

SOC INCIDENT INVESTIGATION REPORT

Wazuh-Based SOC Detection, Monitoring & Response Lab

Environment: Controlled private lab **Evidence basis:** Supplied archive and verified project record

Evidence discipline: Distinguish verified outcomes from setup context, recommendations and unverified tests. No real-world compromise is asserted.

Executive summary

A controlled SOC lab was used to validate centralized Wazuh monitoring, selected custom detections, event-context review, Suricata telemetry forwarding, and a bounded SSH response. Verified outcomes include Rule 5712 SSH correlation with a 60-second firewall-drop response; custom rules 100031, 100033 and 100034; Sysmon/Task Scheduler corroboration; and a custom Suricata ICMP alert. This is a validation report, not a report of an actual compromise.

1. Environment and scope

System	Address / identifier	Function
Ubuntu Docker host	192.168.1.6	Wazuh manager, indexer, dashboard; Docker deployment
Windows 11	192.168.1.7; windows11-agent (ID 001)	Wazuh agent, Sysmon, Windows task auditing
Ubuntu endpoint	192.168.1.8; ubuntu-agent-01 (ID 002)	Wazuh agent, SSH logs, Suricata
Kali Linux	192.168.1.9	Authorized controlled test traffic / invalid SSH attempts

2. Investigation method

For each controlled action, the operator checked the Wazuh alert, inspected source telemetry where available, correlated identifiers/timestamps, and used a benign negative check where documented. Findings below separate observed behavior from interpretation.

3. Findings

3.1 SSH authentication failures and timed response

Kali 192.168.1.9 generated repeated invalid SSH logins to Ubuntu 192.168.1.8. Rule 5712 correlated 8 failures/120 sec at level 10 (T1110). firewall-drop was configured for the matching rule with a 60-second timeout; response logs showed start and removal/end. Detection and timed response were validated in the lab.

3.2 Encoded PowerShell — Rule 100031

Custom level-14 child of 92057; references T1059.001/T1027. Harmless encoded Write-Output test triggered the rule; plain PowerShell negative test did not create a new match. Encoding is a triage signal, not proof of maliciousness.

3.3 Scheduled task — Rule 100033

Windows Event 4698 was collected; built-in 60228 and custom level-12 100033 matched controlled task creation. A separate manual task run was corroborated by Task Scheduler logs and Sysmon cmd.exe PID 9824. Evidence does not show that the task launched encoded PowerShell.

3.4 FIM — Rule 100034

Real-time FIM watched C:\SOC-FIM-Test. Native 550 and custom 100034 (level 12) detected a controlled edit to chain-test.txt with integrity changes; read-only negative check remained quiet. Generic modification alone is not malicious tampering.

3.5 Suricata network telemetry

Suricata 6.0.4 EVE JSON was forwarded to Wazuh. Custom ICMP echo-request alert fired on Kali ping; TCP nc did not match it. Rule 8601 DHCP-related alert was benign. Scan-specific detection was not confirmed.

3.6 Sysmon context and timeline

Sysmon supplied process/parent context. Task creation, encoded PowerShell and FIM were controlled tests in the same exercise, not a proven causal attack chain. Later manual task execution was separately validated.

4. Gaps and disposition

Not validated: custom reconnaissance correlation, Rule 100035, and scan-specific Suricata/Nmap detection. No production compromise, data loss, or unauthorized persistence was established. Disposition: controlled detection/response validation completed for the specific tested patterns; remaining gaps documented.

5. Evidence and handling

Use the figure appendix as an evidence locator, then corroborate with raw alert JSON, response logs, Sysmon/Task Scheduler events, FIM records and Suricata logs. Redact the visible credential in ssh attack.png before any external sharing.

Appendix A — Selected visual evidence

Figures are drawn from the supplied archive. Each caption describes the evidentiary role cautiously; verify exact visible fields against source logs before using a figure to support a specific detection claim. Some setup/alert screenshots are retained as supporting context rather than definitive proof.

Figure 1. Source: 1.png

Wazuh dashboard overview showing the enrolled-agent/monitoring interface. This is platform-state evidence; use the current screenshot contents to confirm exact counts or status.

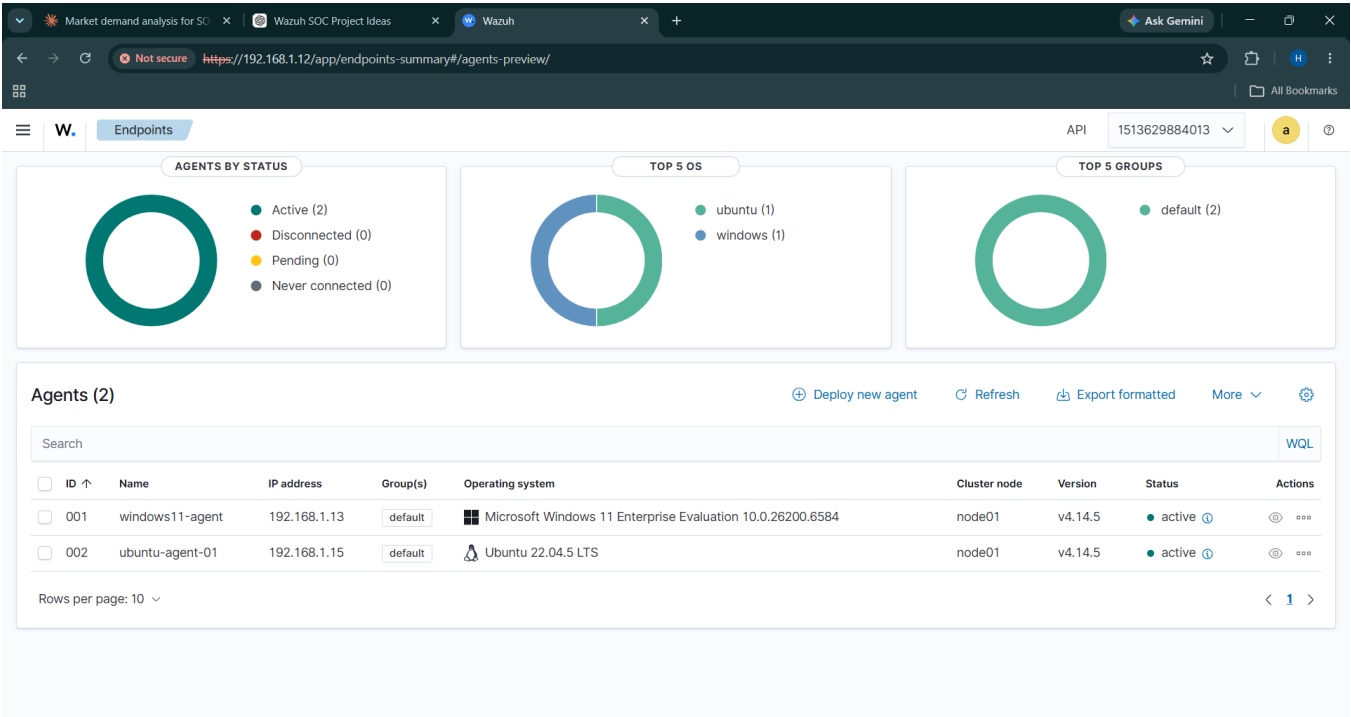
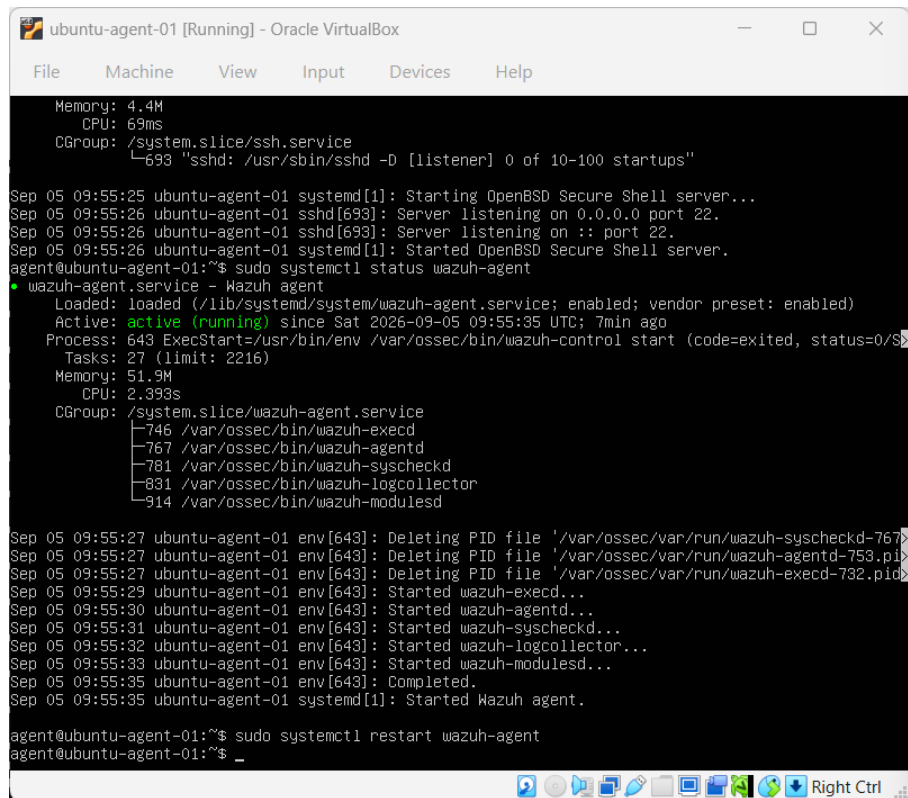


Figure 2. Source: 6.png

Terminal capture from the environment setup sequence. It documents command-line work in the lab; it is not by itself proof that every service is healthy.



```
Memory: 4.4M
CPU: 69ms
CGroup: /system.slice/ssh.service
└─693 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Sep 05 09:55:25 ubuntu-agent-01 systemd[1]: Starting OpenBSD Secure Shell server...
Sep 05 09:55:26 ubuntu-agent-01 sshd[693]: Server listening on 0.0.0.0 port 22.
Sep 05 09:55:26 ubuntu-agent-01 sshd[693]: Server listening on :: port 22.
Sep 05 09:55:26 ubuntu-agent-01 systemd[1]: Started OpenBSD Secure Shell server.
agent@ubuntu-agent-01:~$ sudo systemctl status wazuh-agent
● wazuh-agent.service - Wazuh agent
   Loaded: loaded (/lib/systemd/system/wazuh-agent.service; enabled; vendor preset: enabled)
   Active: active (running) since Sat 2026-09-05 09:55:35 UTC; 7min ago
     Process: 643 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control start (code=exited, status=0/S
   Tasks: 27 (limit: 2216)
   Memory: 51.9M
     CPU: 2.393s
   CGroup: /system.slice/wazuh-agent.service
           └─746 /var/ossec/bin/wazuh-execd
             └─767 /var/ossec/bin/wazuh-agentd
               └─781 /var/ossec/bin/wazuh-syscheckd
                 └─831 /var/ossec/bin/wazuh-logcollector
                   └─914 /var/ossec/bin/wazuh-modulesd

Sep 05 09:55:27 ubuntu-agent-01 env[643]: Deleting PID file '/var/ossec/var/run/wazuh-syscheckd-767
Sep 05 09:55:27 ubuntu-agent-01 env[643]: Deleting PID file '/var/ossec/var/run/wazuh-agentd-753.pi
Sep 05 09:55:27 ubuntu-agent-01 env[643]: Deleting PID file '/var/ossec/var/run/wazuh-execd-732.pid
Sep 05 09:55:29 ubuntu-agent-01 env[643]: Started wazuh-execd...
Sep 05 09:55:30 ubuntu-agent-01 env[643]: Started wazuh-agentd...
Sep 05 09:55:31 ubuntu-agent-01 env[643]: Started wazuh-syscheckd...
Sep 05 09:55:32 ubuntu-agent-01 env[643]: Started wazuh-logcollector...
Sep 05 09:55:33 ubuntu-agent-01 env[643]: Started wazuh-modulesd...
Sep 05 09:55:35 ubuntu-agent-01 env[643]: Completed.
Sep 05 09:55:35 ubuntu-agent-01 systemd[1]: Started Wazuh agent.

agent@ubuntu-agent-01:~$ sudo systemctl restart wazuh-agent
agent@ubuntu-agent-01:~$ _
```

Figure 3. Source: 9.png

Console/configuration evidence from the setup and troubleshooting sequence. Interpret the visible command/output as shown; do not infer unshown successful completion.

Oracle VirtualBox

Input Devices Help

Windows PowerShell

```

Set Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : fe80::1%4
192.168.1.1
[WINDOWS\system32> Get-Service WazuhSvc

Name                DisplayName
-----
WazuhSvc             Wazuh

[WINDOWS\system32> Start-Service WazuhSvc
Service : Service 'Wazuh (WazuhSvc)' cannot be started due to the following error: Cannot open WazuhSvc service
Enter '.':
PS C:\> Get-Service WazuhSvc
CategoryInfo          : OpenError: (System.ServiceProcess.ServiceController:ServiceController) [Start-Service]
Exception              : ServiceCommandException
FullyQualifiedErrorId  : CouldNotStartService,Microsoft.PowerShell.Commands.StartServiceCommand

[WINDOWS\system32> Test-NetConnection 192.168.1.6 -port 1514

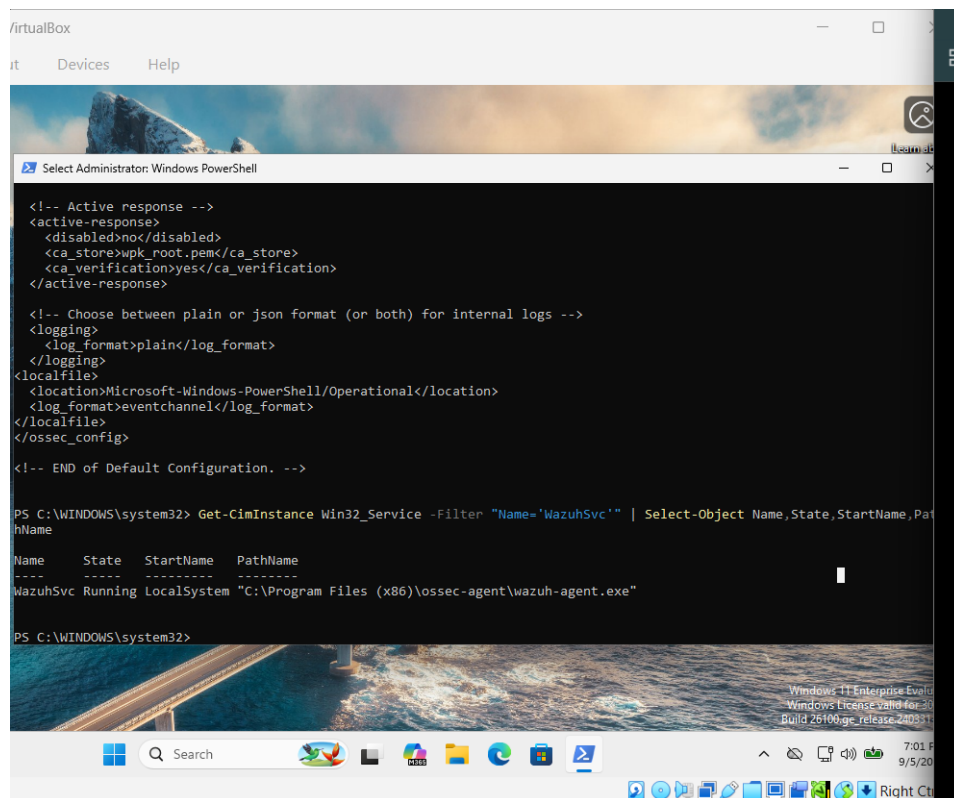
Name                : 192.168.1.6
Address              : 192.168.1.6
Port                 : 1514
InterfaceAlias       : Ethernet
Address              : 192.168.1.7
Succeeded             : True
  
```

Build 26100.ge.rele

Windows taskbar: Search, Task View, File Explorer, Edge, Photos, Mail, Command Prompt

Figure 4. Source: 10.png

Console/configuration evidence from the setup and troubleshooting sequence. Interpret the visible command/output as shown; do not infer unshown successful completion.



```
<!-- Active response -->
<active-response>
  <disabled>no</disabled>
  <ca_store>wpk_root.pem</ca_store>
  <ca_verification>yes</ca_verification>
</active-response>

<!-- Choose between plain or json format (or both) for internal logs -->
<logging>
  <log_format>plain</log_format>
</logging>
<localfile>
  <location>Microsoft-Windows-PowerShell/Operational</location>
  <log_format>eventchannel</log_format>
</localfile>
</ossec_config>

<!-- END of Default Configuration. -->

PS C:\WINDOWS\system32> Get-CimInstance Win32_Service -Filter "Name='WazuhSvc'" | Select-Object Name,State,StartName,PathName
Name      State      StartName PathName
-----
WazuhSvc  Running   LocalSystem "C:\Program Files (x86)\ossec-agent\wazuh-agent.exe"

PS C:\WINDOWS\system32>
```

Figure 5. Source: 13.png

Wazuh dashboard/alert-detail capture from the supplied archive. Use the visible agent, rule, timestamp and event fields to map it to the matching test record; the filename alone does not identify the detection.

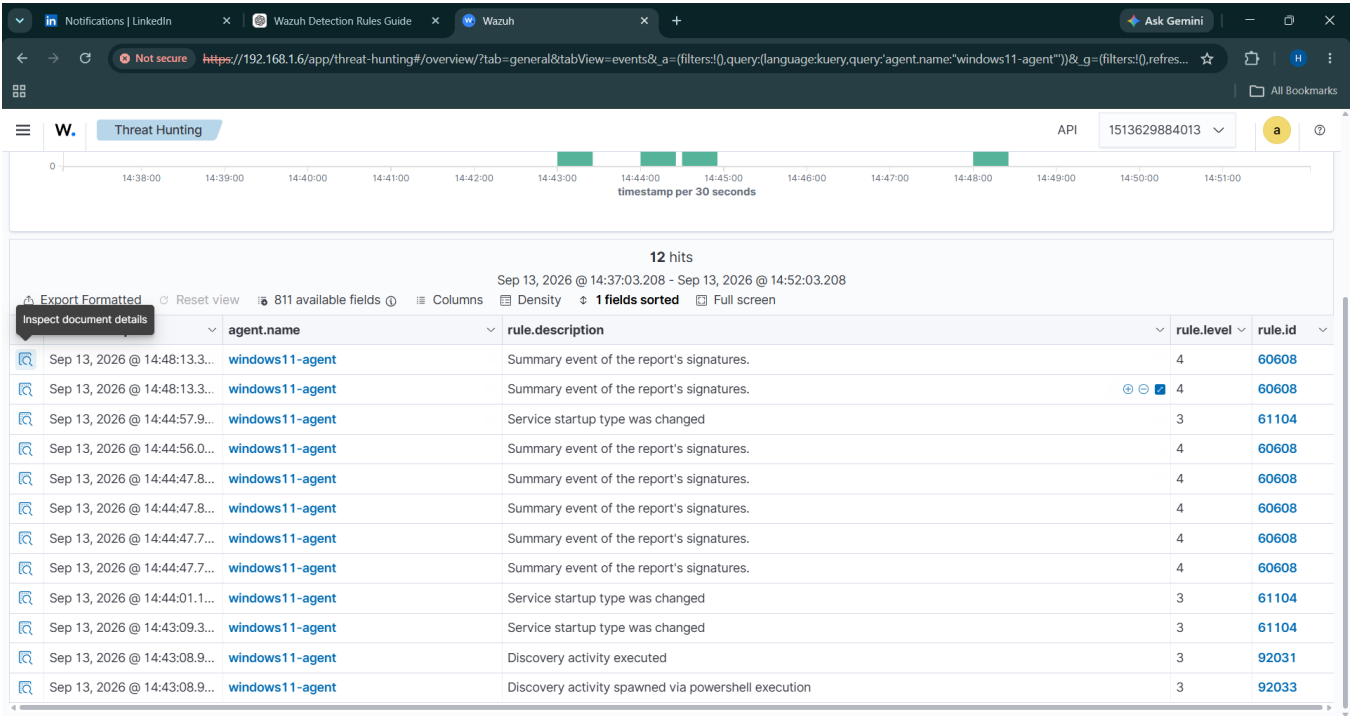


Figure 6. Source: 14.png

Wazuh dashboard/alert-detail capture from the supplied archive. Use the visible agent, rule, timestamp and event fields to map it to the matching test record; the filename alone does not identify the detection.

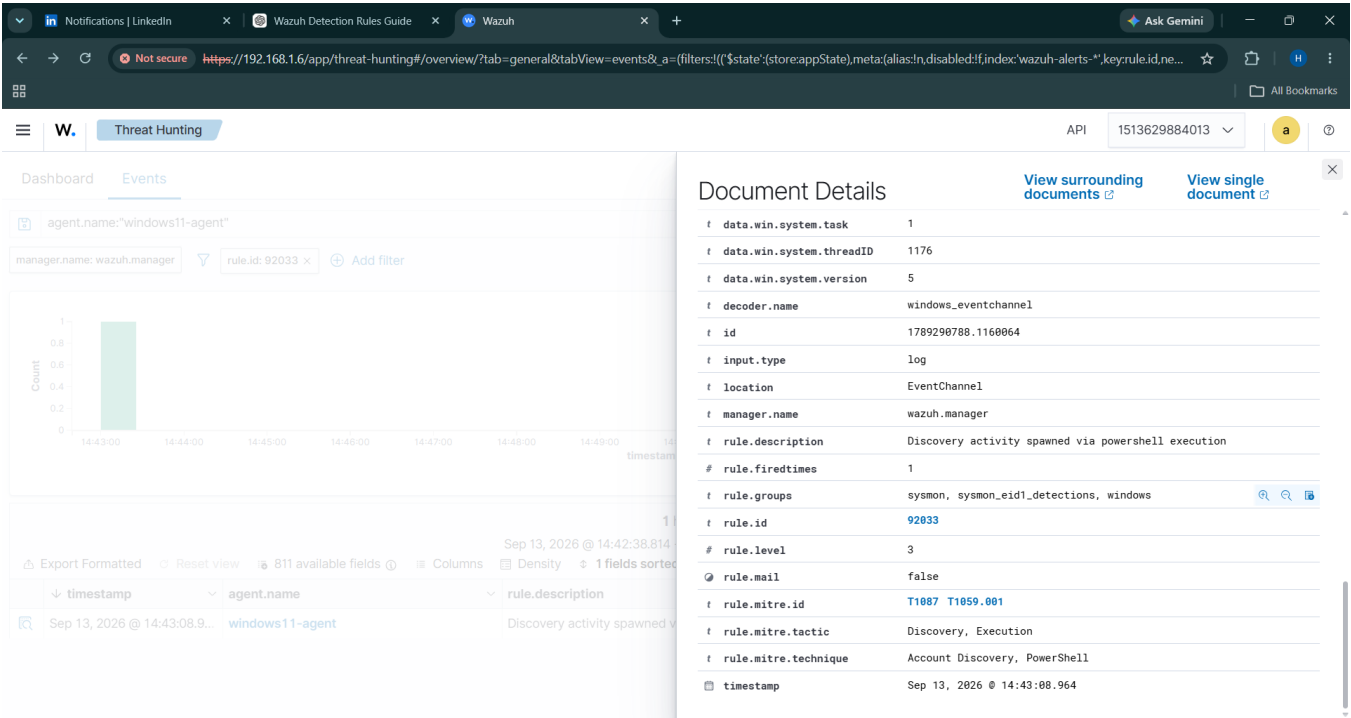


Figure 7. Source: 17.png

Wazuh dashboard/alert-detail capture from the supplied archive. Use the visible agent, rule, timestamp and event fields to map it to the matching test record; the filename alone does not identify the detection.

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Not securehttps://192.168.1.6/app/threat-hunting#/overview/?tab=general&tabView=events&a=(filters:!().query:(language:kuery.query:"agent.name:"windows11-agent"))&_g=(filters:!().refres...

WThreat Hunting

Sep 15, 2026 @ 14:17:08.107 - Sep 15, 2026 @ 14:32:08.107

Export FormattedReset view811 available fieldsColumnsDensity1 fields sortedFull screen

timestamp	agent.name	rule.description	rule.level	rule.id
Sep 15, 2026 @ 14:32:05.9...	windows11-agent	SCA summary: CIS Microsoft Windows 11 Enterprise Benchmark v3.0.0: Score less than 30% (25)	9	19005
Sep 15, 2026 @ 14:31:50.9...	windows11-agent	CIS Microsoft Windows 11 Enterprise Benchmark v3.0.0: Ensure 'Enforce password history' is set to '24 or more ...	9	19014
Sep 15, 2026 @ 14:31:44.4...	windows11-agent	Discovery activity executed	3	92031
Sep 15, 2026 @ 14:31:44.3...	windows11-agent	Discovery activity executed	3	92031
Sep 15, 2026 @ 14:31:44.3...	windows11-agent	Discovery activity executed	3	92031
Sep 15, 2026 @ 14:31:44.3...	windows11-agent	Discovery activity executed	3	92031
Sep 15, 2026 @ 14:31:43.2...	windows11-agent	Powershell process created an executable file in Windows root folder	9	92205
Sep 15, 2026 @ 14:31:41.3...	windows11-agent	Powershell process created an executable file in Windows root folder	9	92205
Sep 15, 2026 @ 14:31:41.2...	windows11-agent	C:\Windows\SysWOW64\SecEdit.exe binary in a suspicious location launched by C:\Windows\SysWOW64\W...	4	92066
Sep 15, 2026 @ 14:31:40.9...	windows11-agent	Powershell script querying system environment variables	4	91816
Sep 15, 2026 @ 14:31:39.9...	windows11-agent	Powershell process created an executable file in Windows root folder	9	92205
Sep 15, 2026 @ 14:31:39.8...	windows11-agent	Powershell was used to delete files or directories	3	92021
Sep 15, 2026 @ 14:31:39.8...	windows11-agent	C:\Windows\SysWOW64\SecEdit.exe binary in a suspicious location launched by C:\Windows\SysWOW64\W...	4	92066
Sep 15, 2026 @ 14:31:38.9...	windows11-agent	Powershell script querying system environment variables	4	91816
Sep 15, 2026 @ 14:31:38.1...	windows11-agent	Powershell process created an executable file in Windows root folder	9	92205

Rows per page: 15< 1 2 3 >

Figure 8. Source: 18.png

Wazuh dashboard/alert-detail capture from the supplied archive. Use the visible agent, rule, timestamp and event fields to map it to the matching test record; the filename alone does not identify the detection.

W.

Threat Hunting

API

1513629884013

a

Sep 15, 2026 @ 14:17:0

Export Formatted

Reset view

811 a

1 fields sorted

Full screen

timestamp	agent.name
Sep 15, 2026 @ 14:32:05.9...	windows11-
Sep 15, 2026 @ 14:31:50.9...	windows11-
Sep 15, 2026 @ 14:31:44.4...	windows11-
Sep 15, 2026 @ 14:31:44.3...	windows11-
Sep 15, 2026 @ 14:31:44.3...	windows11-
Sep 15, 2026 @ 14:31:44.3...	windows11-
Sep 15, 2026 @ 14:31:43.2...	windows11-
Sep 15, 2026 @ 14:31:41.3...	windows11-
Sep 15, 2026 @ 14:31:41.2...	windows11-
Sep 15, 2026 @ 14:31:40.9...	windows11-
Sep 15, 2026 @ 14:31:39.9...	windows11-
Sep 15, 2026 @ 14:31:39.8...	windows11-

Document Details

View surrounding documents

View single document

```
Operating System",
  "parentProcessGuid": "{07af6f40-0978-6aa9-fd04-000000001700}",
  "description": "Net Command",
  "logonGuid": "{07af6f40-35bf-6aa6-e703-000000000000}",
  "parentCommandLine": "net user",
  "processGuid": "{07af6f40-0978-6aa9-ff04-000000001700}",
  "logonId": "0x3e7",
  "parentProcessId": "8544",
  "processId": "8244",
  "currentDirectory": "C:\\\\Program Files (x86)\\\\\\ossec-agent\\\\\\",
  "utcTime": "2026-09-15 09:01:44.671",
  "hashes":
    "MD5=97EF48CB1ED2DFE8B51672B7FAEB1B72, SHA256=0BAAI
    "parentImage":
      "C:\\\\Windows\\\\\\SysWOW64\\\\\\net.exe",
      "company": "Microsoft Corporation",
      "commandLine":
        "C:\\\\WINDOWS\\\\\\system32\\\\\\net1 user",
        "integrityLevel": "System",
        "fileVersion": "10.0.26100.7019 (WinBuild.160101.0000)",
        "user": "NT AUTHORITY\\\\\\SYSTEM",
        "terminalSessionId": "0",
        "parentUser": "NT
```

Figure 9. Source: 19.png

Wazuh dashboard/alert-detail capture from the supplied archive. Use the visible agent, rule, timestamp and event fields to map it to the matching test record; the filename alone does not identify the detection.

W.

Threat Hunting

API

1513629884013

a

Sep 15, 2026 @ 14:17:0

Export Formatted

Reset view

811 a

1 fields sorted

Full screen

timestamp	agent.name
Sep 15, 2026 @ 14:32:05.9...	windows11-
Sep 15, 2026 @ 14:31:50.9...	windows11-
Sep 15, 2026 @ 14:31:44.4...	windows11-
Sep 15, 2026 @ 14:31:44.3...	windows11-
Sep 15, 2026 @ 14:31:44.3...	windows11-
Sep 15, 2026 @ 14:31:44.3...	windows11-
Sep 15, 2026 @ 14:31:43.2...	windows11-
Sep 15, 2026 @ 14:31:41.3...	windows11-
Sep 15, 2026 @ 14:31:41.2...	windows11-
Sep 15, 2026 @ 14:31:40.9...	windows11-
Sep 15, 2026 @ 14:31:39.9...	windows11-
Sep 15, 2026 @ 14:31:39.8...	windows11-

Document Details

View surrounding documents

View single document

Table

JSON

```
{  "_index": "wazuh-alerts-4.x-2026.09.15",  "_id": "x25NpKABDKO-ljSB41h",  "_score": null,  "_source": {    "input": {      "type": "log"    },    "agent": {      "ip": "192.168.1.7",      "name": "windows11-agent",      "id": "001"    },    "manager": {      "name": "wazuh.manager"    },    "data": {      "win": {        "eventdata": {          "originalFileName": "net1.exe",          "image": "C:\\\\Windows\\\\SysWOW64\\\\net1.exe",          "product": "Microsoft Windows Operating System",          "parentProcessGuid": "{07af6f40-0070-6a30-fd04-000000001700}"        }      }    }  }
```

Figure 10. Source: 25.png

Wazuh dashboard/alert-detail capture from the supplied archive. Use the visible agent, rule, timestamp and event fields to map it to the matching test record; the filename alone does not identify the detection.

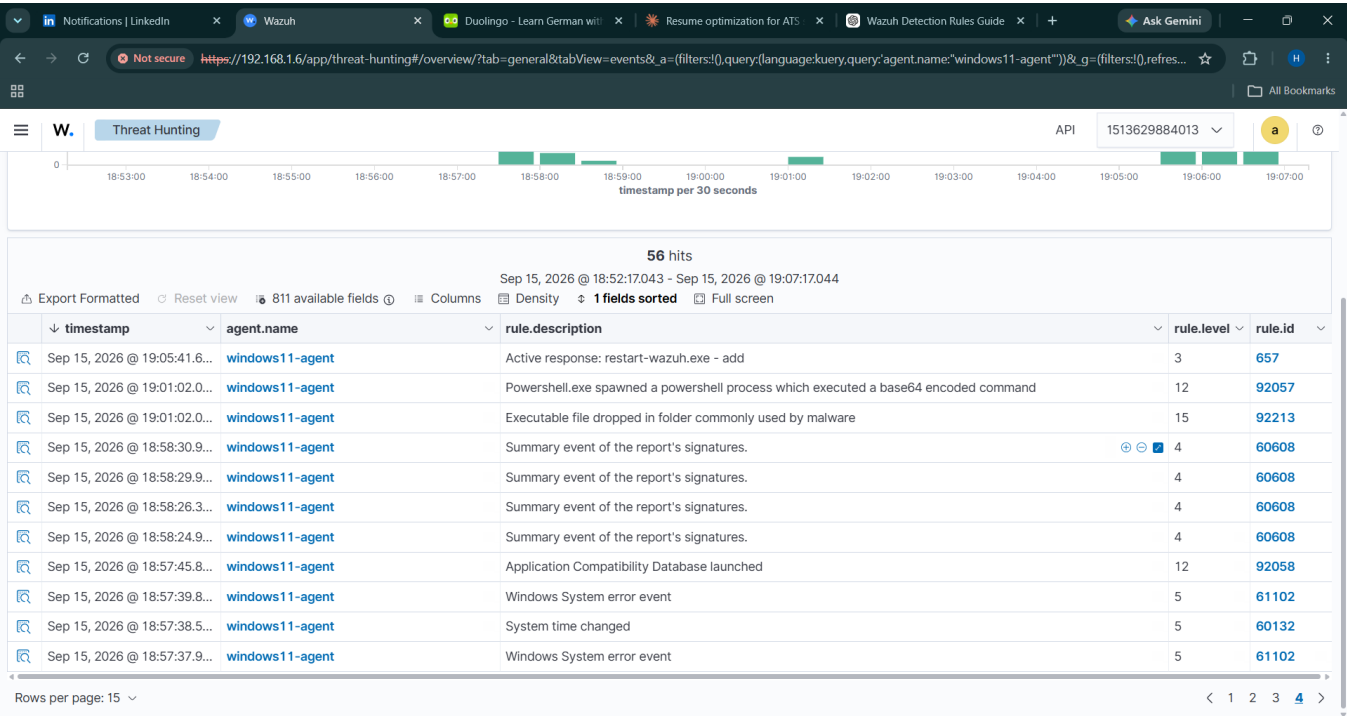


Figure 11. Source: 26.png

Wazuh dashboard/alert-detail capture from the supplied archive. Use the visible agent, rule, timestamp and event fields to map it to the matching test record; the filename alone does not identify the detection.

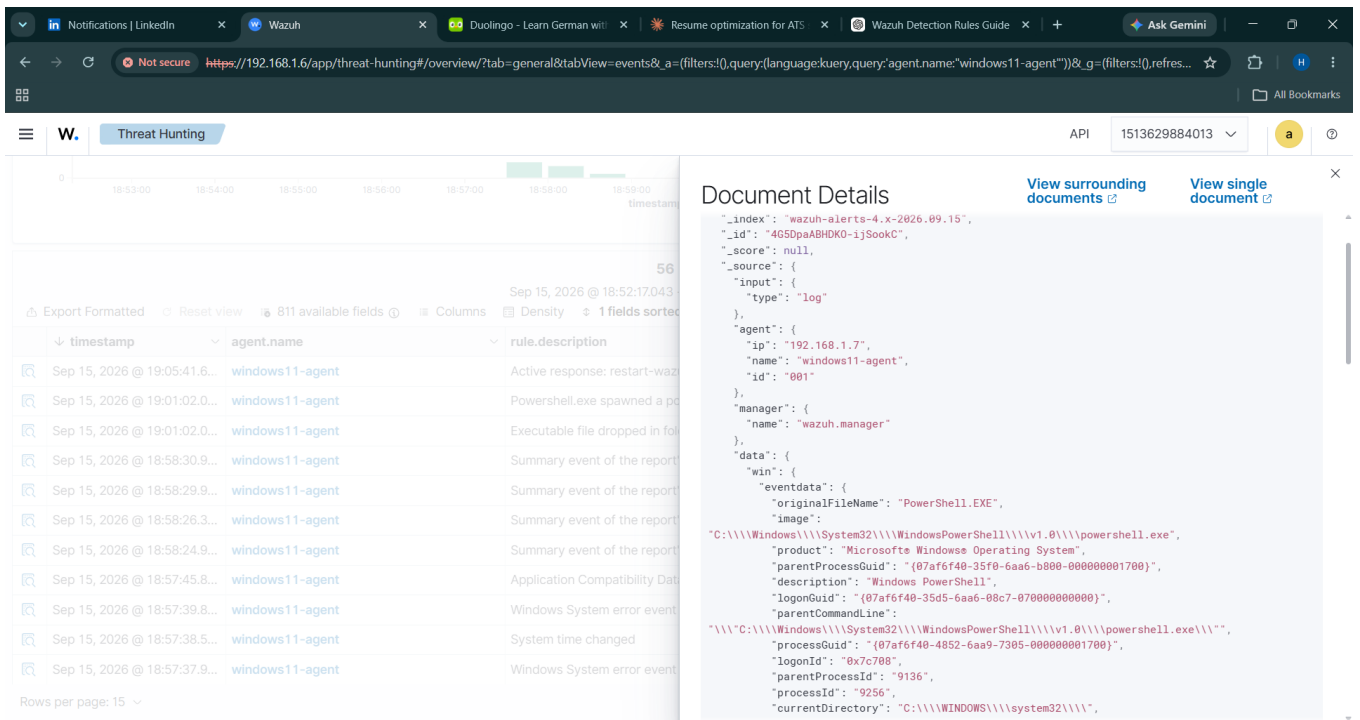


Figure 12. Source: 27.png

Wazuh dashboard/alert-detail capture from the supplied archive. Use the visible agent, rule, timestamp and event fields to map it to the matching test record; the filename alone does not identify the detection.

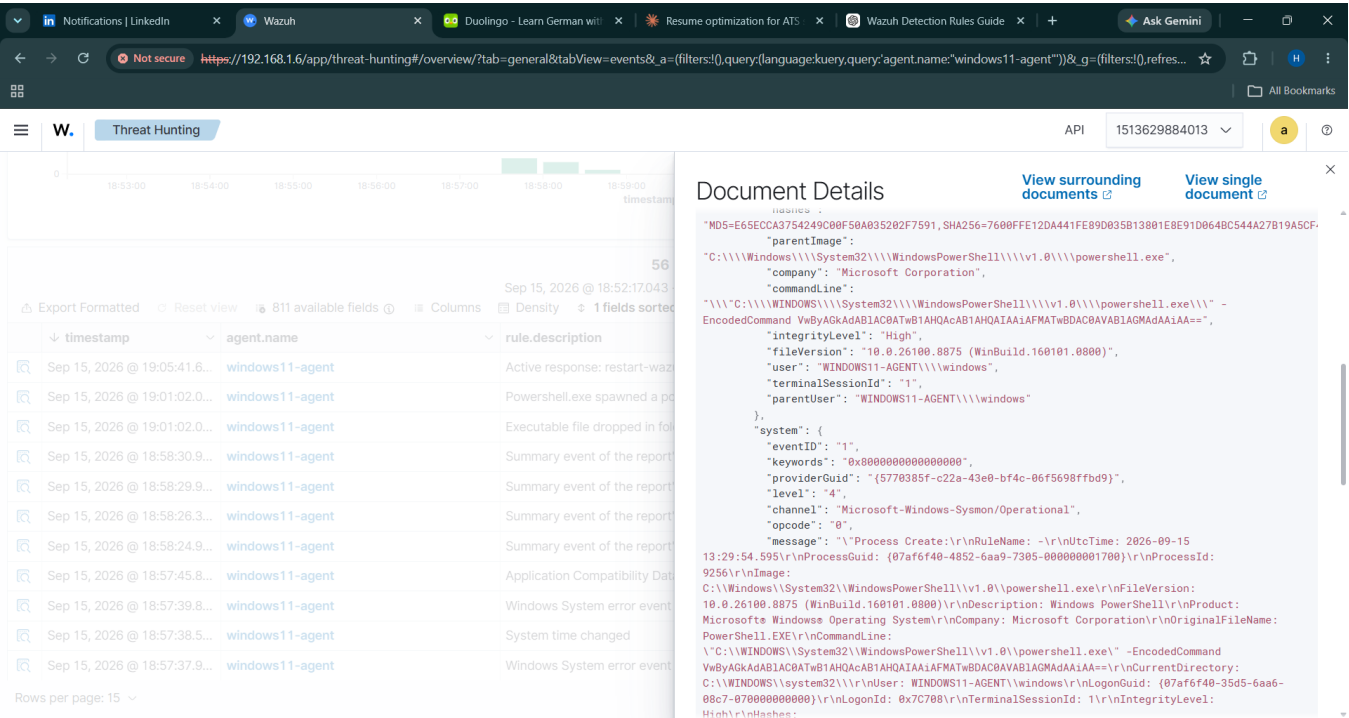


Figure 13. Source: 28.png

Wazuh dashboard/alert-detail capture from the supplied archive. Use the visible agent, rule, timestamp and event fields to map it to the matching test record; the filename alone does not identify the detection.

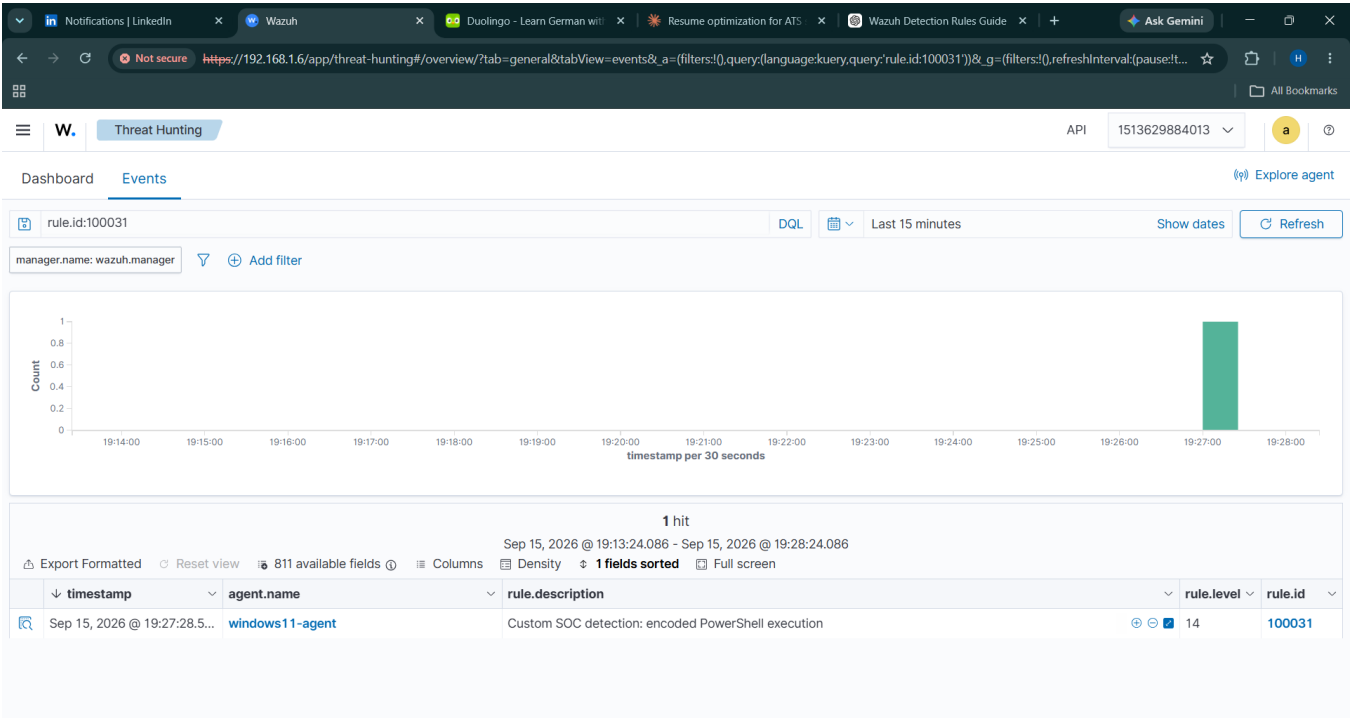


Figure 14. Source: 29.png

Wazuh dashboard/alert-detail capture from the supplied archive. Use the visible agent, rule, timestamp and event fields to map it to the matching test record; the filename alone does not identify the detection.

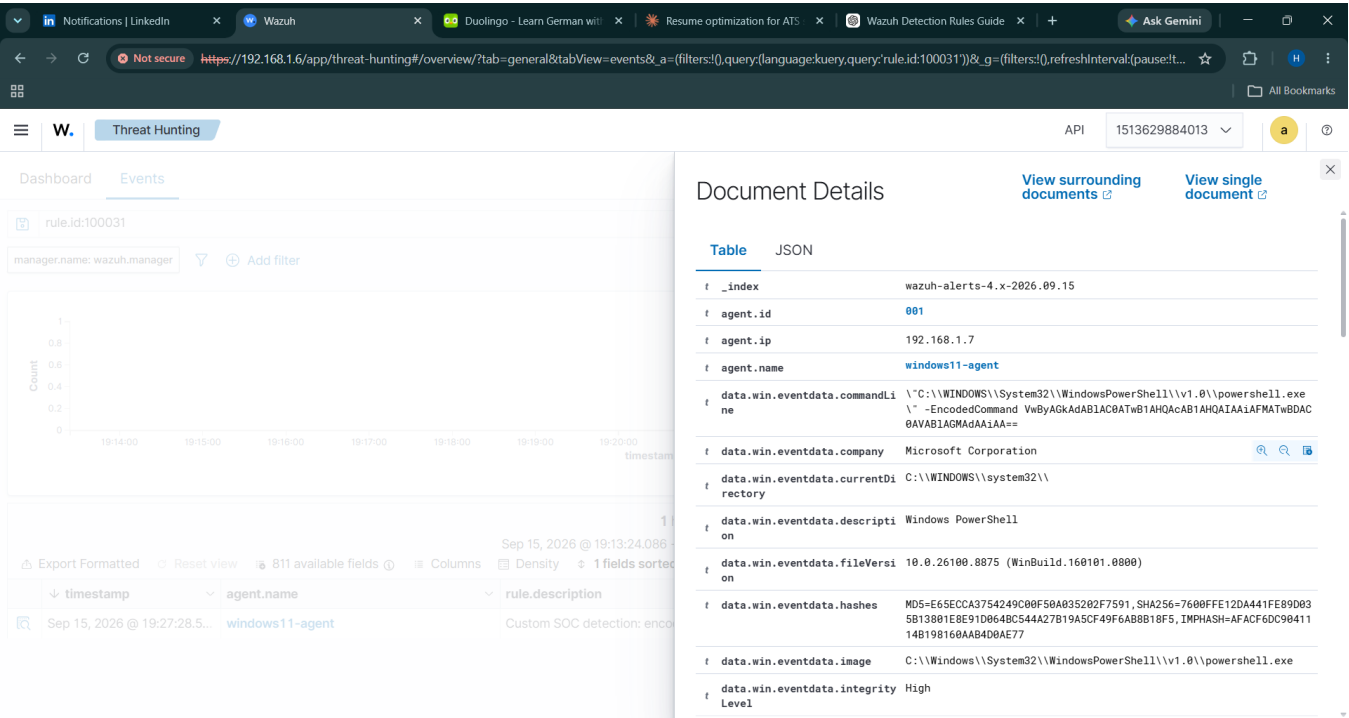


Figure 15. Source: 30.png

Wazuh dashboard/alert-detail capture from the supplied archive. Use the visible agent, rule, timestamp and event fields to map it to the matching test record; the filename alone does not identify the detection.

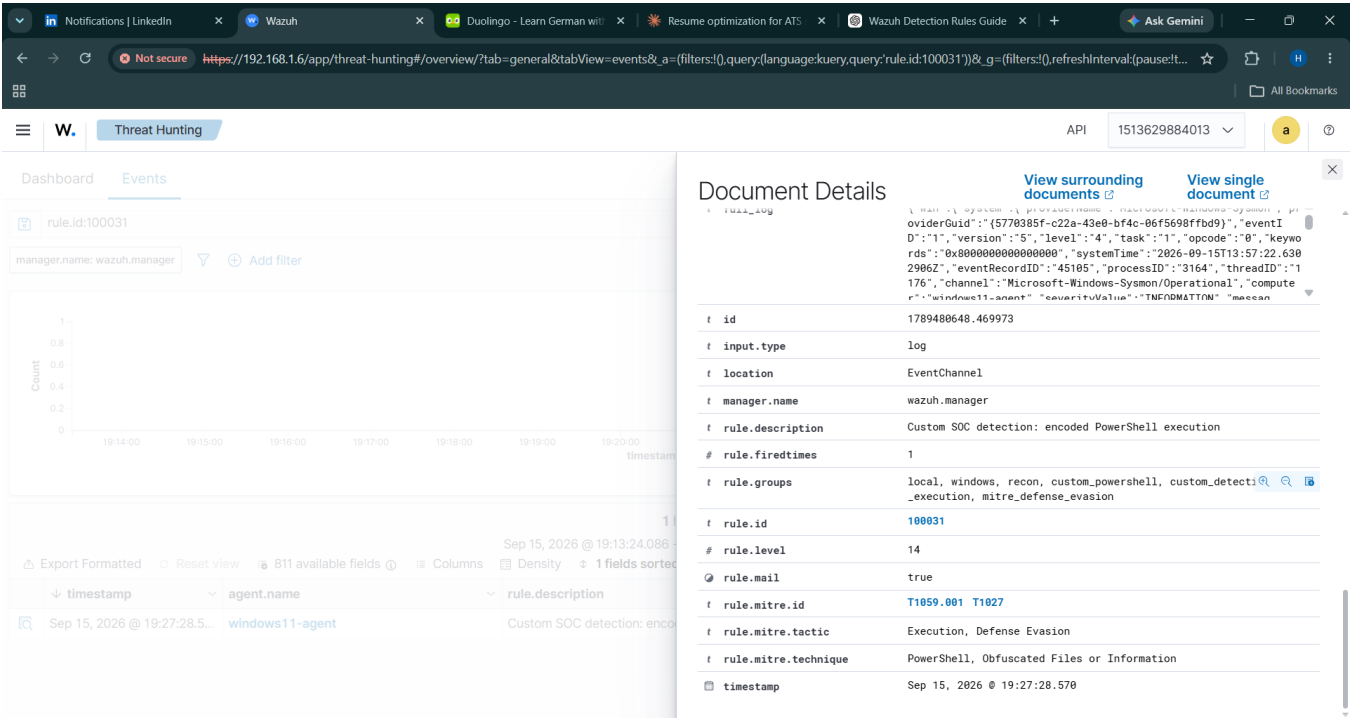


Figure 16. Source: 31.png

Wazuh dashboard/alert-detail capture from the supplied archive. Use the visible agent, rule, timestamp and event fields to map it to the matching test record; the filename alone does not identify the detection.

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Wazuh Detection Rules Guide

Wazuh

Ask Gemini

Not securehttps://192.168.1.6/app/threat-hunting#/overview/?tab=general&tabView=events&a=(filters:!().query:(language:kuery.query:agent.name:"windows11-agent"))&_g=(filters:!().refres...All Bookmarks

W. Threat Hunting

Sep 16, 2026 @ 10:44:38.852 - Sep 16, 2026 @ 10:59:38.852

Export FormattedReset view811 available fieldsColumnsDensity1 fields sortedFull screen

timestamp	agent.name	rule.description	rule.level	rule.id
Sep 16, 2026 @ 10:59:31.0...	windows11-agent	Windows System error event	5	61102
Sep 16, 2026 @ 10:59:31.0...	windows11-agent	Windows System error event	5	61102
Sep 16, 2026 @ 10:59:31.0...	windows11-agent	Windows System error event	5	61102
Sep 16, 2026 @ 10:59:31.0...	windows11-agent	Windows System error event	5	61102
Sep 16, 2026 @ 10:59:27.5...	windows11-agent	Service startup type was changed	3	61104
Sep 16, 2026 @ 10:58:44.7...	windows11-agent	Executable file dropped in folder commonly used by malware	15	92213
Sep 16, 2026 @ 10:57:50.5...	windows11-agent	Logon Failure - Unknown user or bad password	5	60122
Sep 16, 2026 @ 10:57:45.7...	windows11-agent	Executable dropped in Windows root folder	6	92217
Sep 16, 2026 @ 10:57:45.6...	windows11-agent	Executable dropped in Windows root folder	6	92217
Sep 16, 2026 @ 10:57:43.0...	windows11-agent	Executable dropped in Windows root folder	6	92217
Sep 16, 2026 @ 10:57:43.0...	windows11-agent	Executable dropped in Windows root folder	6	92217
Sep 16, 2026 @ 10:57:38.9...	windows11-agent	Executable dropped in Windows root folder	6	92217
Sep 16, 2026 @ 10:57:37.6...	windows11-agent	Executable dropped in Windows root folder	6	92217
Sep 16, 2026 @ 10:57:32.5...	windows11-agent	SCA summary: CIS Microsoft Windows 11 Enterprise Benchmark v3.0.0: Score less than 30% (25)	9	19005
Sep 16, 2026 @ 10:57:30.6...	windows11-agent	Executable dropped in Windows root folder	6	92217

Rows per page: 15

< 1 2 3 4 5 ... 14 >

Figure 17. Source: 32.png

Wazuh dashboard/alert-detail capture from the supplied archive. Use the visible agent, rule, timestamp and event fields to map it to the matching test record; the filename alone does not identify the detection.

Wazuh

Threat Hunting

Sep 16, 2026 @ 10:47:57.984 - Sep 16, 2026 @ 11:02:57.984

Export Formatted Reset view 811 available fields Columns Density 1 fields sorted Full screen

timestamp	agent.name	rule.description	rule.level	rule.id
Sep 16, 2026 @ 11:02:24.4...	windows11-agent	Service startup type was changed	3	61104
Sep 16, 2026 @ 11:00:44.5...	windows11-agent	CIS Microsoft Windows 11 Enterprise Benchmark v3.0.0: Ensure 'Maximum password age' is set to '365 or fewer ...	3	19009
Sep 16, 2026 @ 11:00:44.5...	windows11-agent	CIS Microsoft Windows 11 Enterprise Benchmark v3.0.0: Ensure 'Enforce password history' is set to '24 or more ...	3	19009
Sep 16, 2026 @ 10:59:31.0...	windows11-agent	Windows System error event	5	61102
Sep 16, 2026 @ 10:59:31.0...	windows11-agent	Windows System error event	5	61102
Sep 16, 2026 @ 10:59:31.0...	windows11-agent	Windows System error event	5	61102
Sep 16, 2026 @ 10:59:31.0...	windows11-agent	Windows System error event	5	61102
Sep 16, 2026 @ 10:59:27.5...	windows11-agent	Service startup type was changed	3	61104
Sep 16, 2026 @ 10:58:44.7...	windows11-agent	Executable file dropped in folder commonly used by malware	15	92213
Sep 16, 2026 @ 10:57:50.5...	windows11-agent	Logon Failure - Unknown user or bad password	5	60122
Sep 16, 2026 @ 10:57:45.7...	windows11-agent	Executable dropped in Windows root folder	6	92217
Sep 16, 2026 @ 10:57:45.6...	windows11-agent	Executable dropped in Windows root folder	6	92217
Sep 16, 2026 @ 10:57:43.0...	windows11-agent	Executable dropped in Windows root folder	6	92217
Sep 16, 2026 @ 10:57:43.0...	windows11-agent	Executable dropped in Windows root folder	6	92217
Sep 16, 2026 @ 10:57:38.9...	windows11-agent	Executable dropped in Windows root folder	6	92217

Rows per page: 15

Figure 18. Source: 33.png

Wazuh dashboard/alert-detail capture from the supplied archive. Use the visible agent, rule, timestamp and event fields to map it to the matching test record; the filename alone does not identify the detection.

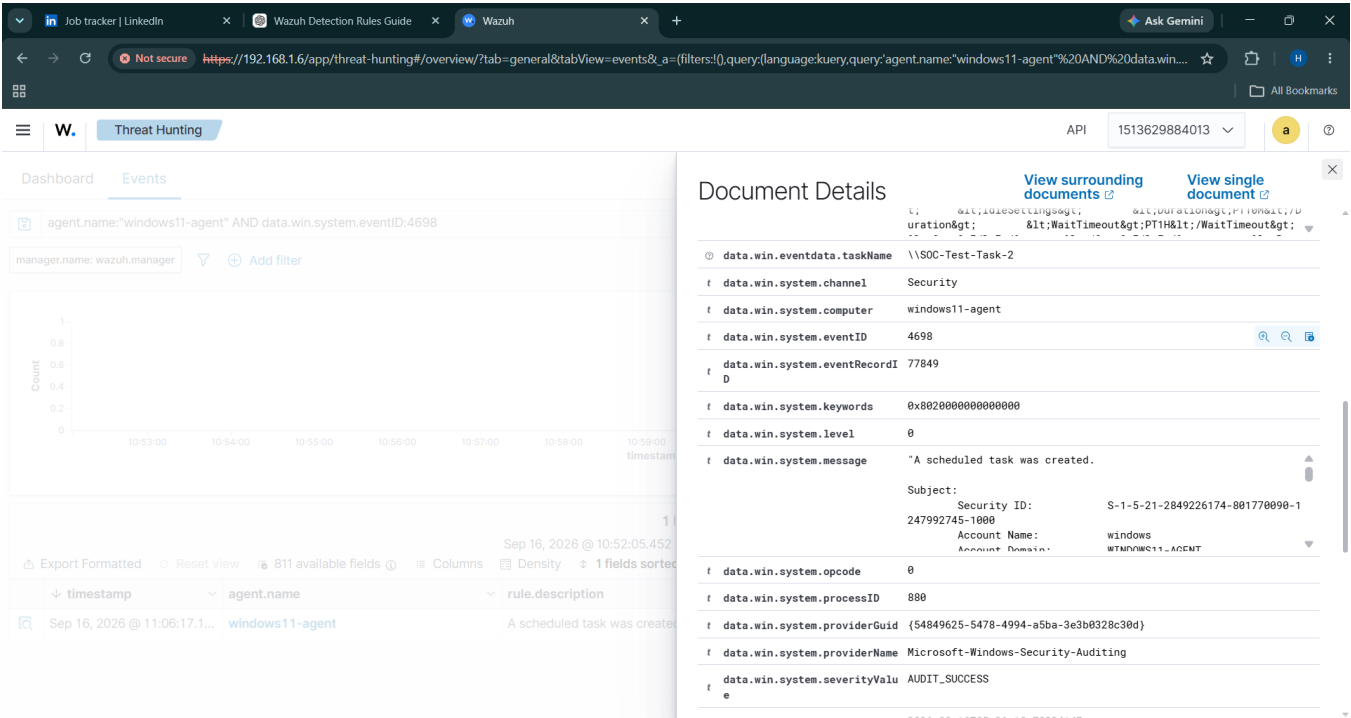


Figure 19. Source: 34.png

Wazuh dashboard/alert-detail capture from the supplied archive. Use the visible agent, rule, timestamp and event fields to map it to the matching test record; the filename alone does not identify the detection.

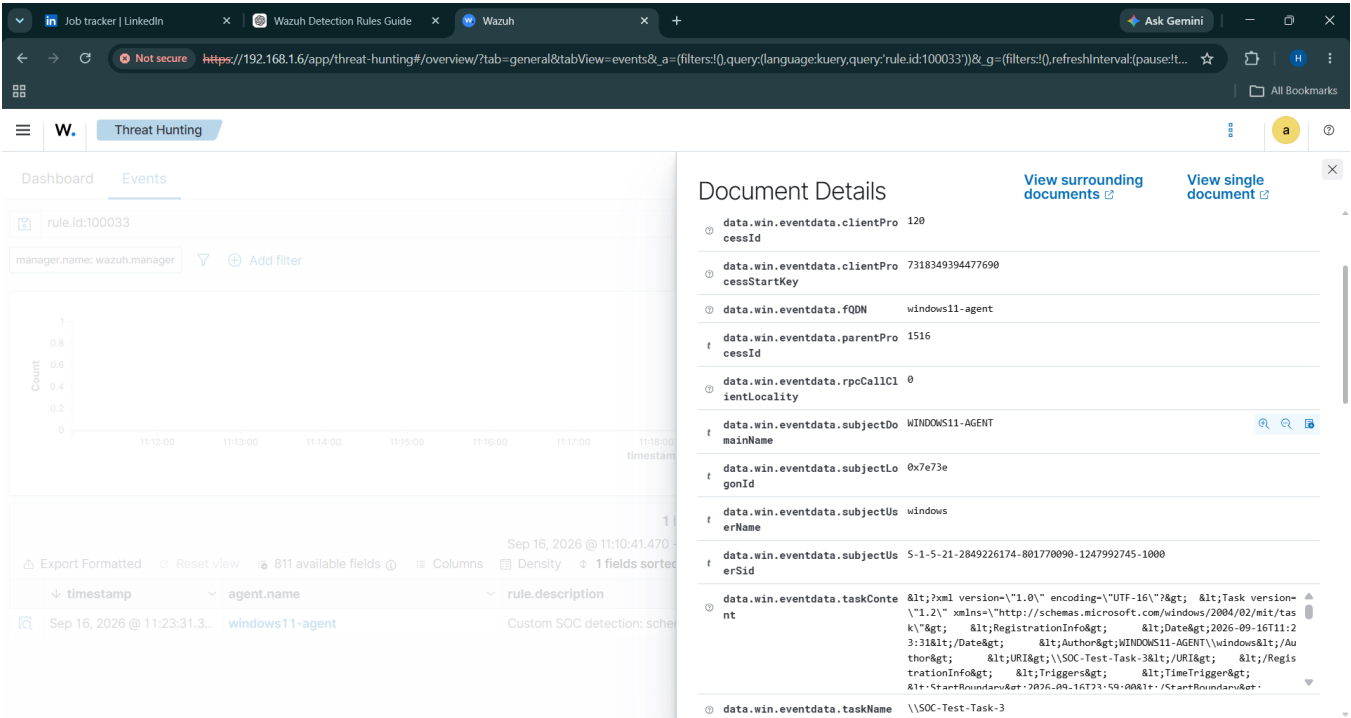


Figure 20. Source: 46.png

Terminal or configuration capture from the lab build/test workflow. It supports implementation context only to the extent that the command and output are legible; correlate with service logs or Wazuh alerts before asserting success.

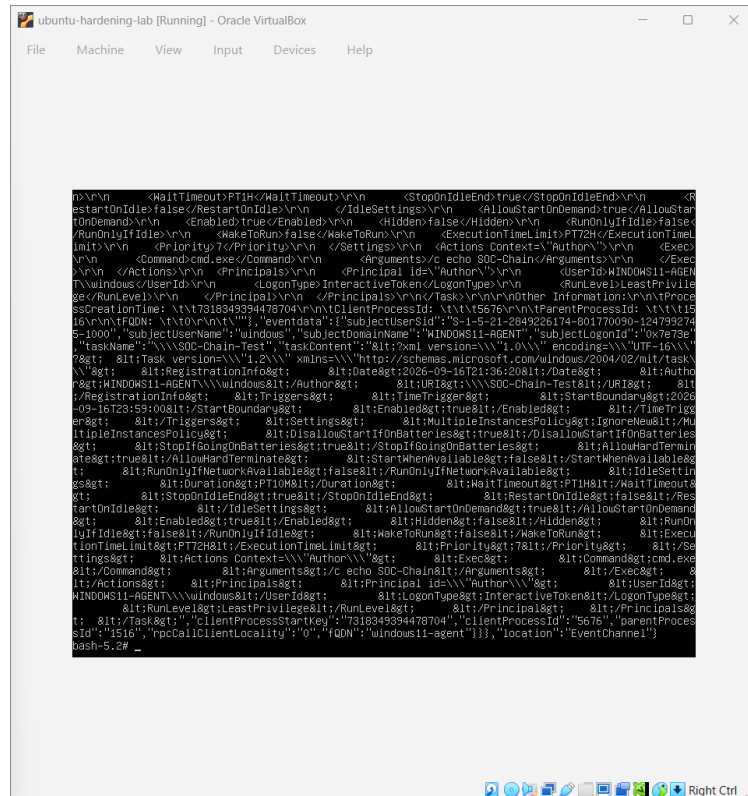


Figure 21. Source: 47.png

Terminal or configuration capture from the lab build/test workflow. It supports implementation context only to the extent that the command and output are legible; correlate with service logs or Wazuh alerts before asserting success.



The screenshot shows a terminal window titled "ubuntu-hardening-lab [Running] - Oracle VirtualBox". The terminal output displays a Windows Event Viewer log entry for a PowerShell process. The log entry is a JSON object with the following fields: "id", "type", "providerName", "providerGuid", "eventID", "version", "level", "task", "opcode", "keywords", "systemTime", "threadID", "channel", "computer", "securityValue", "message", "processCreate", "processRuleName", "processId", "processImage", "processName", "processProduct", "processCompany", "processOriginalFileName", "processCommandLine", "processCurrentDirectory", "processUser", "processIntegrityLevel", "processHashes", "processLogonId", "processSessionId", "processIntegrityLevel", "processHashes", "processLogonId", "processSessionId", "processIntegrityLevel", "processHashes", "processLogonId", "processSessionId". The log entry is a JSON object with the following fields: "id", "type", "providerName", "providerGuid", "eventID", "version", "level", "task", "opcode", "keywords", "systemTime", "threadID", "channel", "computer", "securityValue", "message", "processCreate", "processRuleName", "processId", "processImage", "processName", "processProduct", "processCompany", "processOriginalFileName", "processCommandLine", "processCurrentDirectory", "processUser", "processIntegrityLevel", "processHashes", "processLogonId", "processSessionId", "processIntegrityLevel", "processHashes", "processLogonId", "processSessionId".

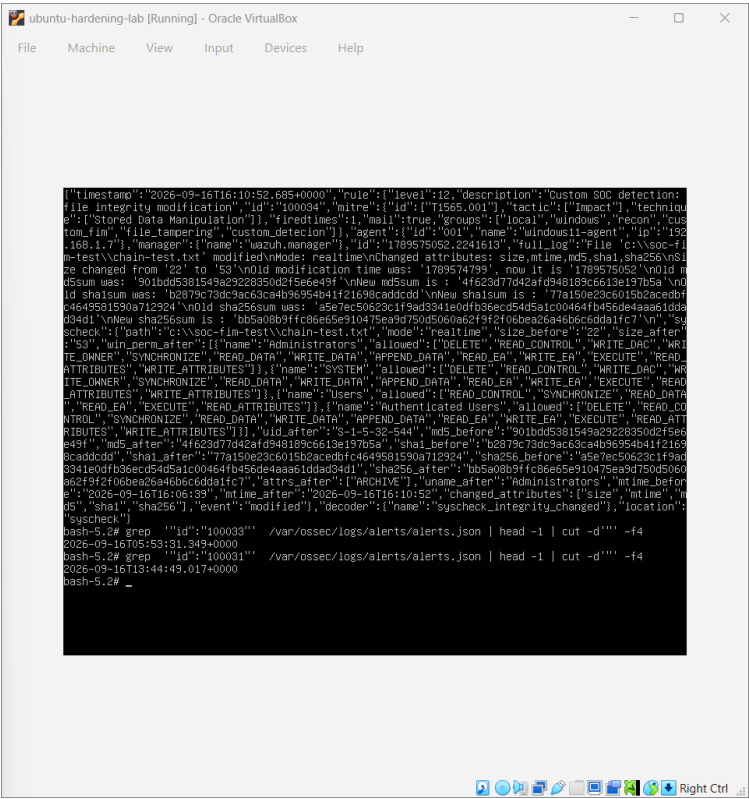
Figure 22. Source: 48.png

Terminal or configuration capture from the lab build/test workflow. It supports implementation context only to the extent that the command and output are legible; correlate with service logs or Wazuh alerts before asserting success.

[illegible]

Figure 23. Source: 50.png

Terminal or configuration capture from the lab build/test workflow. It supports implementation context only to the extent that the command and output are legible; correlate with service logs or Wazuh alerts before asserting success.



```
ubuntu-hardening-lab [Running] - Oracle VirtualBox
File Machine View Input Devices Help

{"timestamp":"2026-09-16T16:10:52.685+0000","rule":{"id":"100034","mitre":{"id":"T1595.001"},"tactic":["Impact"],"technical":{"id":"Stored Data Manipulation"},"firedtimes":1,"mail":{"true","groups":["local"],"windows":"recon","custom_fim":{"file_tampering","custom_detection"},"agent":{"id":"001","name":"windows11-agent","ip":"192.168.1.71"},"manager":{"name":"wazuh.manager"},"id":"1789575062,2241613"},"full_log":{"File":"C:\\soc-fim-test\\chain-test.txt","modified_mtime":"realtime","changed_attributes":{"size","mtime","md5","sha1","sha256"},"changed_from":{"22","to":{"33"},"n0id":{"modification_time":{"was":"1789574799","now":{"is":"1789575062"},"n0id":{"md5sum":{"901bdd5381549a29228350d2f5ee49f"},"nNew":{"md5sum":{"is":"4f623d77d42afd948189c6613e197b5a"},"n0id":{"sha1sum":{"was":"b2879c73dc9ac63ca4b96954b41f21698caddcd"},"nNew":{"sha1sum":{"is":"77a150e23c6015b2acedbf4649581590a712924"},"n0id":{"sha256sum":{"was":"a5e7ec50623c1f9ad3341e0dfb36ecd54d5a1c00464fb456de4aa61ddad34d1"},"nNew":{"sha256sum":{"is":"b05a08b5ffc86e5e910475ea9d750d50a62f9f2106ba2c6a4b6ccddaf7"},"n"},"syscheck":{"path":"C:\\soc-fim-test\\chain-test.txt","mode":"realtime","size_before":"22","size_after":"33","win_perm_after":{"name":"Administrators","allowed":["DELETE","READ_CONTROL","WRITE_DAC","WRITE_OWNER","SYNCHRONIZE","READ_DATA","WRITE_DATA","APPEND_DATA","READ_EA","WRITE_EA","EXECUTE","READ_ATTRIBUTES","WRITE_ATTRIBUTES"]},"name":"Users","allowed":["READ_CONTROL","SYNCHRONIZE","READ_DATA","READ_EA","EXECUTE","READ_ATTRIBUTES"]},"name":"Authenticated Users","allowed":["DELETE","READ_CONTROL","SYNCHRONIZE","READ_DATA","WRITE_DATA","APPEND_DATA","READ_EA","WRITE_EA","EXECUTE","READ_ATTRIBUTES","WRITE_ATTRIBUTES"]},"name":"SYSTEM","allowed":["DELETE","READ_CONTROL","WRITE_DAC","WRITE_OWNER","SYNCHRONIZE","READ_DATA","WRITE_DATA","APPEND_DATA","READ_EA","WRITE_EA","EXECUTE","READ_ATTRIBUTES","WRITE_ATTRIBUTES"]},"uid_after":"S-1-5-32-544","md5_before":{"901bdd5381549a29228350d2f5ee49f"},"md5_after":{"4f623d77d42afd948189c6613e197b5a"},"sha1_before":{"b2879c73dc9ac63ca4b96954b41f21698caddcd"},"sha1_after":{"77a150e23c6015b2acedbf4649581590a712924"},"sha256_before":{"a5e7ec50623c1f9ad3341e0dfb36ecd54d5a1c00464fb456de4aa61ddad34d1"},"sha256_after":{"b05a08b5ffc86e5e910475ea9d750d50a62f9f2106ba2c6a4b6ccddaf7"},"attrs_after":{"ARCHIVE"},"uname_after":{"Administrators"},"mtime_before":{"2026-09-16T16:06:39"},"mtime_after":{"2026-09-16T16:10:52"},"changed_attributes":{"size","mtime"},"ids":{"sha1","sha256"},"event":{"modified"},"decoder":{"name":"syscheck_integrity_changed"},"location":{"syscheck"},"id":"100034"},"id":"100034","2026-09-16T05:53:31.349+0000"}
bash-5.2# grep ""id":"100034"" /var/ossec/logs/alerts/alerts.json | head -1 | cut -d'"' -f4
2026-09-16T05:53:31.349+0000
bash-5.2# grep ""id":"100031"" /var/ossec/logs/alerts/alerts.json | head -1 | cut -d'"' -f4
2026-09-16T13:44:49.017+0000
bash-5.2# _
```

Figure 24. Source: 51.png

Terminal or configuration capture from the lab build/test workflow. It supports implementation context only to the extent that the command and output are legible; correlate with service logs or Wazuh alerts before asserting success.



```
ubuntu-hardening-lab [Running] - Oracle VirtualBox
File Machine View Input Devices Help

{"timestamp":"2026-09-16T16:10:52.685+0000","rule":{"id":"100034","mitre":{"id":"T1595.001"},"tactic":["Impact"],"technical":{"id":"Stored Data Manipulation"},"firedtimes":1,"mail":{"true},"groups":["local"],"windows":{"recon"},"custom":{"file_tampering"},"custom_detection":{"agent":{"id":"001","name":"windows11-agent"},"ip":"192.168.1.71","manager":{"name":"wazuh.manager"},"id":"1789575062,2241613"},"full_log":{"File":"C:\\soc-fil-test\\chain-test.txt","modified\\nmode":"realtime\\nchanged attributes: size,mtime,md5,sha1,sha256\\nsize changed from '22' to '53'\\nold modification time was: 1789574799", "now it is: 1789575052'\\nold modification time was: '901bdd5381549a29228350d2f5ee49f'\\nNew md5sum is: '4f623d77d42afd948189c6613e197b5a'\\nOld sha1sum was: 'b2879c73dc9ac63ca4b96954b41f21698caddcd'\\nNew sha1sum is: '77a150e23c6015b2acedbf34649581590a712924'\\nOld sha256sum was: 'a5e7ec50623c1f9ad3341e0dfb36ecd54d5a1c00464fb456de4aa61ddad34d1'\\nNew sha256sum is: 'b05a0b6b9ffc86e5e910475ea3d750d506062f92f06b926a486cc6ddaf7'\\n"},"syscheck":{"path":"C:\\soc-fil-test\\chain-test.txt","mode":"realtime","size_before":"22","size_after":"53","win_perm_after":{"name":"Administrators","allowed":["DELETE","READ_CONTROL","WRITE_DAC","WRITE_OWNER","SYNCHRONIZE","READ_DATA","WRITE_DATA","APPEND_DATA","READ_EA","WRITE_EA","EXECUTE","READ_ATTRIBUTES","WRITE_ATTRIBUTES"]},"name":"Users","allowed":["READ_CONTROL","SYNCHRONIZE","READ_DATA","READ_EA","EXECUTE","READ_ATTRIBUTES"]},"name":"Authenticated Users","allowed":["DELETE","READ_CONTROL","SYNCHRONIZE","READ_DATA","WRITE_DATA","APPEND_DATA","READ_EA","WRITE_EA","EXECUTE","READ_ATTRIBUTES","WRITE_ATTRIBUTES"]},"name":"SYSTEM","allowed":["DELETE","READ_CONTROL","WRITE_DAC","WRITE_OWNER","SYNCHRONIZE","READ_DATA","WRITE_DATA","APPEND_DATA","READ_EA","WRITE_EA","EXECUTE","READ_ATTRIBUTES","WRITE_ATTRIBUTES"]},"name":"S-1-5-32-544","md5_before":"'901bdd5381549a29228350d2f5ee49f'","md5_after":"'4f623d77d42afd948189c6613e197b5a'","sha1_before":"'b2879c73dc9ac63ca4b96954b41f21698caddcd'","sha1_after":"'77a150e23c6015b2acedbf34649581590a712924'","sha256_before":"'a5e7ec50623c1f9ad34649581590a712924'","sha256_after":"'b05a0b6b9ffc86e5e910475ea3d750d506062f92f06b926a486cc6ddaf7'"},"attributes":{"size","mtime"},"changed_attributes":{"size","mtime"},"location":"C:\\soc-fil-test\\chain-test.txt","event":"modified"},"decoder":{"name":"syscheck_integrity_changed"},"location":"syscheck"}
bash-5.2# grep ''id':'100033'' /var/ossec/logs/alerts/alerts.json | head -1 | cut -d''' -f4
2026-09-16T05:53:31.349+0000
bash-5.2# grep ''id':'100031'' /var/ossec/logs/alerts/alerts.json | head -1 | cut -d''' -f4
2026-09-16T13:44:49.017+0000
bash-5.2# grep ''id':'100033'' /var/ossec/logs/alerts/alerts.json | tail -1 | cut -d''' -f4
2026-09-16T16:06:22.761+0000
bash-5.2#
```

Figure 25. Source: 52.png

Terminal or configuration capture from the lab build/test workflow. It supports implementation context only to the extent that the command and output are legible; correlate with service logs or Wazuh alerts before asserting success.

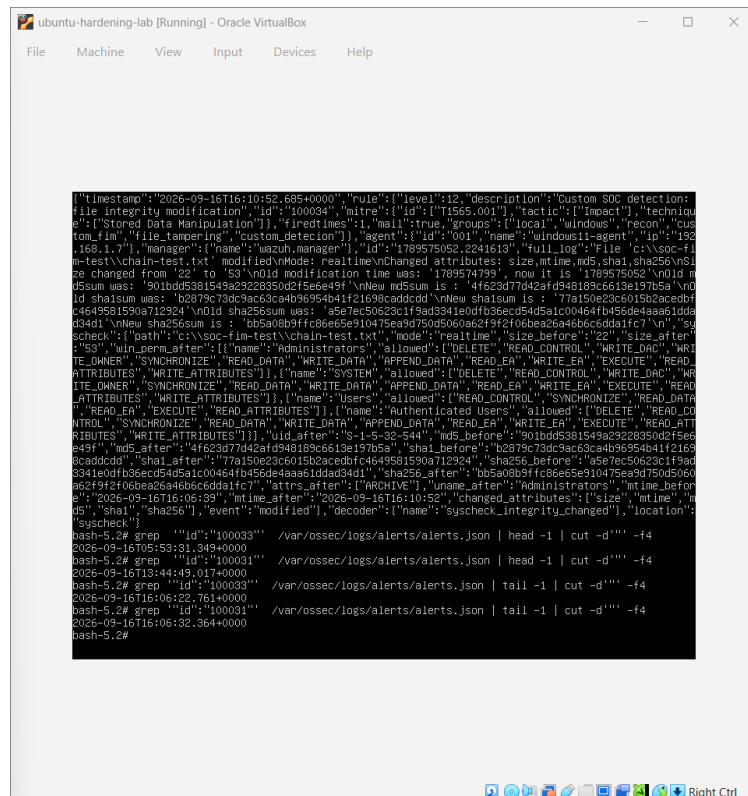
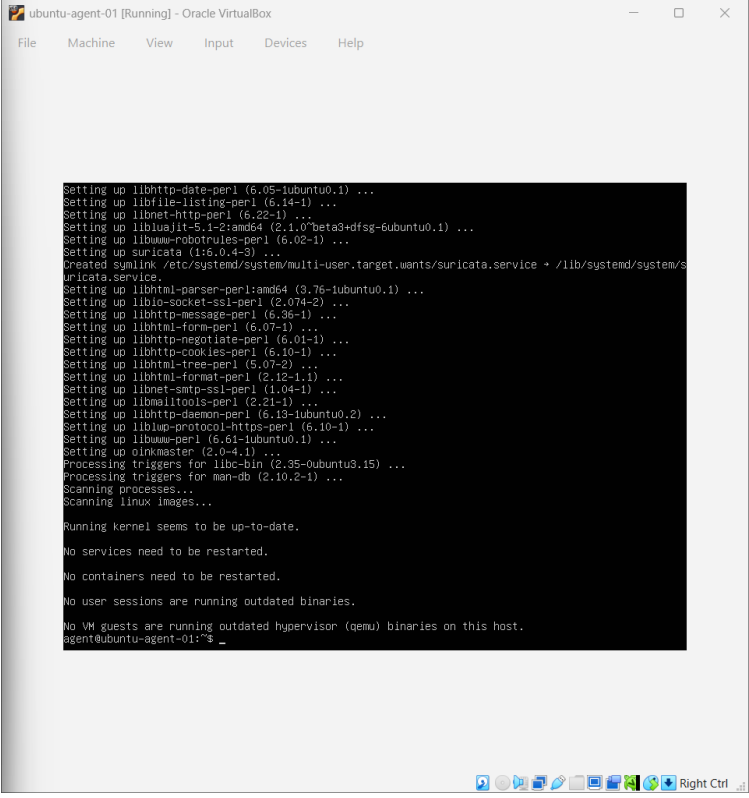


Figure 26. Source: 53.png

Terminal or configuration capture from the lab build/test workflow. It supports implementation context only to the extent that the command and output are legible; correlate with service logs or Wazuh alerts before asserting success.



```
ubuntu-agent-01 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

Setting up libhttp-date-perl (6.05-1ubuntu0.1) ...
Setting up libfile-listing-perl (6.14-1) ...
Setting up libnet-http-perl (6.22-1) ...
Setting up liblua5.1-2:amd64 (2.1.0~beta3+dfsg-6ubuntu0.1) ...
Setting up libwww-robotrules-perl (6.02-1) ...
Setting up suricata (1:6.0.4-3) ...
Created symlink /etc/systemd/system/multi-user.target.wants/suricata.service → /lib/systemd/system/suricata.service.
Setting up libhtml-parser-perl:amd64 (3.76-1ubuntu0.1) ...
Setting up libio-socket-ssl-perl (2.074-2) ...
Setting up libhttp-message-perl (6.36-1) ...
Setting up libhtml-form-perl (6.07-1) ...
Setting up libhttp-negotiate-perl (6.01-1) ...
Setting up libhttp-cookies-perl (6.10-1) ...
Setting up libhtml-tree-perl (5.07+2) ...
Setting up libhtml-format-perl (2.12-1.1) ...
Setting up libnet-smtp-ssl-perl (1.04-1) ...
Setting up libmailtools-perl (2.21-1) ...
Setting up libhttp-daemon-perl (6.13-1ubuntu0.2) ...
Setting up liblua5.1-protocol-http-perl (6.10-1) ...
Setting up libwww-perl (6.61-1ubuntu0.1) ...
Setting up oinkmaster (2.0-4.1) ...
Processing triggers for libc-bin (2.35-0ubuntu3.15) ...
Processing triggers for man-db (2.10.2-1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

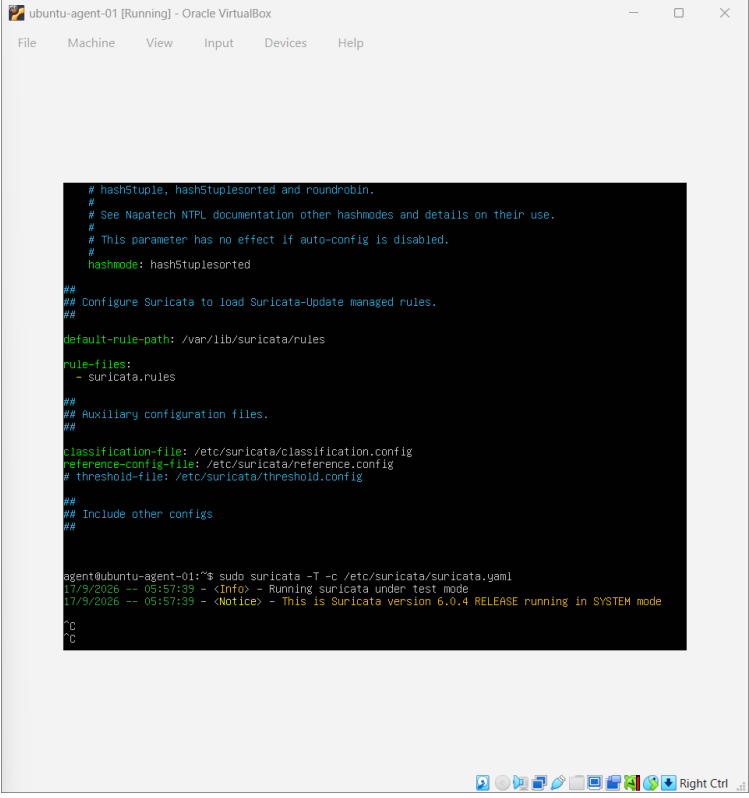
No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
agent@ubuntu-agent-01:~$
```

Figure 27. Source: 60.png

Terminal or configuration capture from the lab build/test workflow. It supports implementation context only to the extent that the command and output are legible; correlate with service logs or Wazuh alerts before asserting success.



```
ubuntu-agent-01 [Running] - Oracle VM VirtualBox
File  Machine  View  Input  Devices  Help

# hashStupie, hashStupiesorted and roundrobin.
#
# See Napatech NTPL documentation other hashmodes and details on their use.
#
# This parameter has no effect if auto-config is disabled.
#
hashmode: hashStupiesorted

##
## Configure Suricata to load Suricata-Update managed rules.
##
default-rule-path: /var/lib/suricata/rules
rule-files:
- suricata.rules

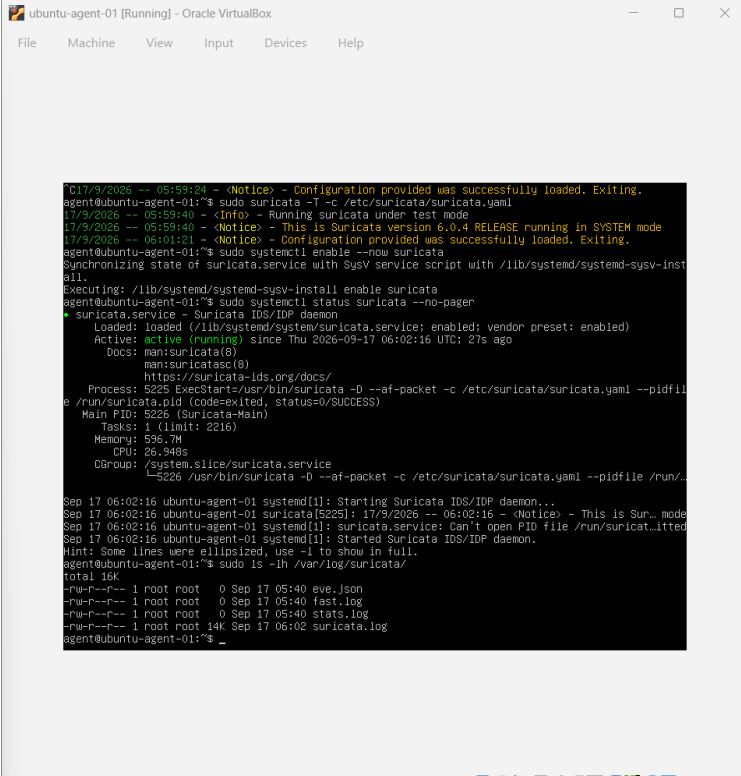
##
## Auxiliary configuration files.
##
classification-file: /etc/suricata/classification.config
reference-config-file: /etc/suricata/reference.config
# threshold-file: /etc/suricata/threshold.config

##
## Include other configs
##

agent@ubuntu-agent-01:~$ sudo suricata -T -c /etc/suricata/suricata.yaml
17/9/2026 -- 05:57:39 - <Info> - Running suricata under test mode
17/9/2026 -- 05:57:39 - <Notice> - This is Suricata version 6.0.4 RELEASE running in SYSTEM mode
^C
^C
```

Figure 28. Source: 61.png

Terminal or configuration capture from the lab build/test workflow. It supports implementation context only to the extent that the command and output are legible; correlate with service logs or Wazuh alerts before asserting success.



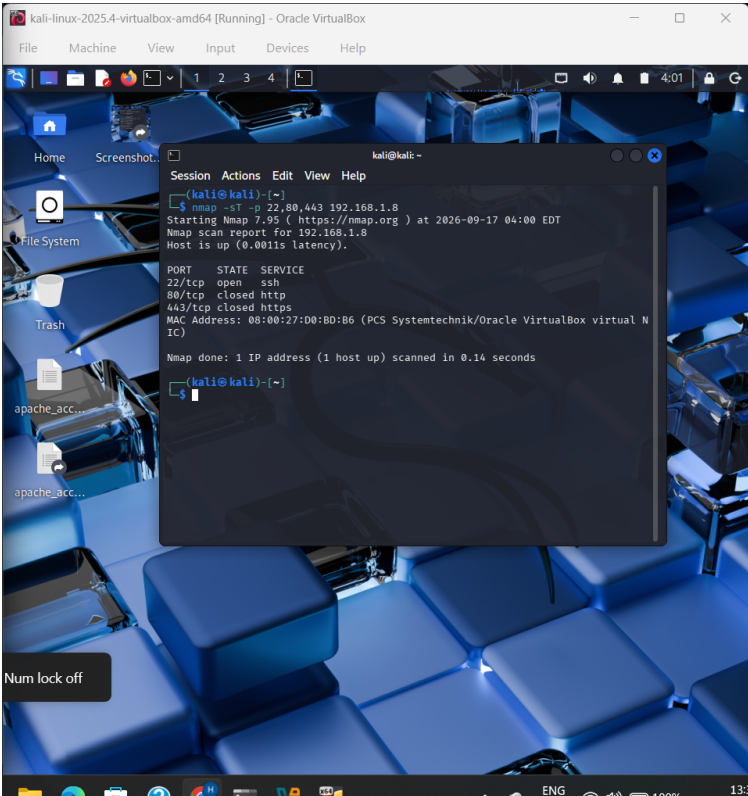
```
ubuntu-agent-01 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

17/9/2026 -- 05:59:04 - <Notice> - Configuration provided was successfully loaded. Exiting.
agent@ubuntu-agent-01:~$ sudo suricata -T -c /etc/suricata/suricata.yaml
17/9/2026 -- 05:59:40 - <Info> - Running suricata under test mode
17/9/2026 -- 05:59:40 - <Notice> - This is Suricata version 6.0.4 RELEASE running in SYSTEM mode
17/9/2026 -- 06:01:21 - <Notice> - Configuration provided was successfully loaded. Exiting.
agent@ubuntu-agent-01:~$ sudo systemctl enable --now suricata
Synchronizing state of suricata.service with SysV service script with /lib/systemd/systemd-sysv-inst
all.
Executing: /lib/systemd/systemd-sysv-install enable suricata
agent@ubuntu-agent-01:~$ sudo systemctl status suricata --no-pager
* suricata.service - Suricata IDS/IDP daemon
   Loaded: loaded (/lib/systemd/system/suricata.service; enabled; vendor preset: enabled)
   Active: active (running) since Thu 2026-09-17 06:02:16 UTC; 27s ago
     Docs: man:suricata(8)
           man:suricata.conf(5)
           https://suricata-ids.org/docs/
   Process: 5225 ExecStart=/usr/bin/suricata -D --af-packet -c /etc/suricata/suricata.yaml --pidfile
e /run/suricata.pid (code=exited, status=0/SUCCESS)
   Main PID: 5225 (Suricata-Main)
     Tasks: 1 (limit: 2216)
    Memory: 596.7M
       CPU: 26.948s
   CGroup: /system.slice/suricata.service
           └─5226 /usr/bin/suricata -D --af-packet -c /etc/suricata/suricata.yaml --pidfile /run/...

Sep 17 06:02:16 ubuntu-agent-01 systemd[1]: Starting Suricata IDS/IDP daemon...
Sep 17 06:02:16 ubuntu-agent-01 suricata[5225]: 17/9/2026 -- 06:02:16 - <Notice> - This is Sur... mode
Sep 17 06:02:16 ubuntu-agent-01 systemd[1]: suricata.service: Can't open PID file /run/suricat...tted
Sep 17 06:02:16 ubuntu-agent-01 systemd[1]: Started Suricata IDS/IDP daemon.
Hint: Some lines were ellipsized, use -l to show in full.
agent@ubuntu-agent-01:~$ sudo ls -lh /var/log/suricata/
total 16K
-rw-r--r-- 1 root root 0 Sep 17 05:40 eve.json
-rw-r--r-- 1 root root 0 Sep 17 05:40 fast.log
-rw-r--r-- 1 root root 0 Sep 17 05:40 stats.log
-rw-r--r-- 1 root root 14K Sep 17 06:02 suricata.log
agent@ubuntu-agent-01:~$ _
```

Figure 29. Source: 62.png

Terminal or configuration capture from the lab build/test workflow. It supports implementation context only to the extent that the command and output are legible; correlate with service logs or Wazuh alerts before asserting success.

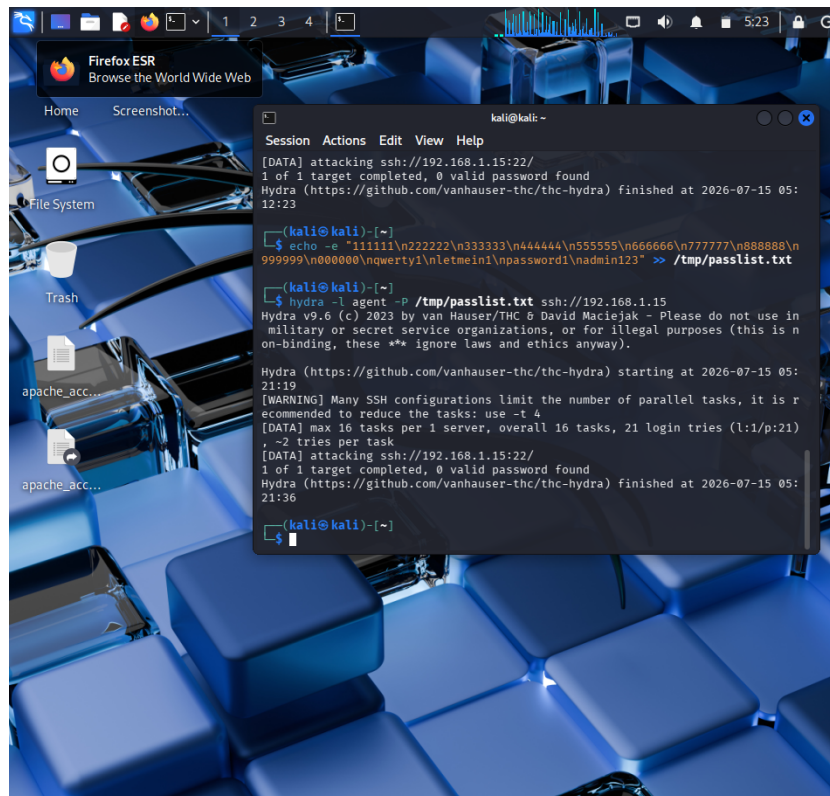


The image shows a screenshot of a Kali Linux virtual machine running in Oracle VM VirtualBox. The desktop background is a blue keyboard. A terminal window is open, displaying the output of an Nmap scan. The terminal title is 'kali@kali: ~'. The command entered is `nmap -sT -p 22,80,443 192.168.1.8`. The output shows the scan results for 192.168.1.8, including the ports 22/tcp (open ssh), 80/tcp (closed http), and 443/tcp (closed https). The scan was completed in 0.14 seconds.

```
kali@kali: ~  
$ nmap -sT -p 22,80,443 192.168.1.8  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-17 04:00 EDT  
Nmap scan report for 192.168.1.8  
Host is up (0.0011s latency).  
  
PORT      STATE SERVICE  
22/tcp    open  ssh  
80/tcp    closed http  
443/tcp   closed https  
MAC Address: 08:00:27:D0:BD:B6 (PCS Systemtechnik/Oracle VirtualBox virtual N IC)  
  
Nmap done: 1 IP address (1 host up) scanned in 0.14 seconds  
kali@kali: ~  
$
```

Figure 30. Source: 70.png

Terminal or configuration capture from the lab build/test workflow. It supports implementation context only to the extent that the command and output are legible; correlate with service logs or Wazuh alerts before asserting success.



The image shows a Kali Linux desktop environment with a blue-themed background. A terminal window is open in the foreground, displaying the output of a Hydra command. The terminal window has a title bar that reads "kali@kali -". The output shows a Hydra attack on an SSH target at 192.168.1.15:22. The attack is completed, but no valid password was found. The terminal also shows the command used to run Hydra and the command to run a password list. The desktop background features a blue keyboard and a Firefox ESR browser window.

```
kali@kali -  
Session Actions Edit View Help  
[DATA] attacking ssh://192.168.1.15:22/  
1 of 1 target completed, 0 valid password found  
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-07-15 05:  
12:23  
  
(kali@kali)-[~]  
$ echo -e "111111\n222222\n333333\n444444\n555555\n666666\n777777\n888888\n999999\n000000\nqwerty1\nletmein\npassword1\nadmin123" >> /tmp/passlist.txt  
  
(kali@kali)-[~]  
$ hydra -l agent -P /tmp/passlist.txt ssh://192.168.1.15  
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in  
military or secret service organizations, or for illegal purposes (this is n  
on-binding, these ** ignore laws and ethics anyway).  
  
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-07-15 05:  
21:19  
[WARNING] Many SSH configurations limit the number of parallel tasks, it is r  
ecommended to reduce the tasks: use -t 4  
[DATA] max 16 tasks per 1 server, overall 16 tasks, 21 login tries (l:1/p:21)  
, ~2 tries per task  
[DATA] attacking ssh://192.168.1.15:22/  
1 of 1 target completed, 0 valid password found  
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-07-15 05:  
21:36  
  
(kali@kali)-[~]  
$
```

Figure 31. Source: 71.png

Terminal or configuration capture from the lab build/test workflow. It supports implementation context only to the extent that the command and output are legible; correlate with service logs or Wazuh alerts before asserting success.

```
* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro

System information as of Wed Sep 16 09:46:49 AM UTC 2026

System load:            0.86
Usage of /:              24.0% of 19.62GB
Memory usage:           14%
Swap usage:              0%
Processes:              109
Users logged in:         0
IPv4 address for enp0s3: 192.168.1.8
IPv6 address for enp0s3: 2401:4900:88e2:d4b7:a00:27ff:fed0:bdb6

* Canonical Workshop gives developers fast, composable, reproducible, and
  secure developer environments that are perfect for agentic workflows.

  https://ubuntu.com/workshop

Expanded Security Maintenance for Applications is not enabled.

1 update can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Sat Sep 12 07:24:50 UTC 2026 on tty1
agent@ubuntu-agent-01:~$ ls -l /var/ossec/active-response/bin/firewall-drop
ls: cannot access '/var/ossec/active-response/bin/firewall-drop': Permission denied
agent@ubuntu-agent-01:~$ sudo ls -l /var/ossec/active-response/bin/firewall-drop
[sudo] password for agent:
-rwxr-x--- 1 root wazuh 21656 Apr 16 12:49 /var/ossec/active-response/bin/firewall-drop
agent@ubuntu-agent-01:~$
```

Figure 32. Source: 74.png

Terminal or configuration capture from the lab build/test workflow. It supports implementation context only to the extent that the command and output are legible; correlate with service logs or Wazuh alerts before asserting success.

```
"srcuser":"wronguser"},"location":"journalid"},"program":"active-response/bin/firewall-drop"}}
2026/09/16 13:06:34 active-response/bin/firewall-drop: Ended
2026/09/16 13:07:35 active-response/bin/firewall-drop: Starting
2026/09/16 13:07:35 active-response/bin/firewall-drop: {"version":1,"origin":{"name":"node01","module":"wazuh-execd"},"command":"delete","parameters":{"extra_args":[],"alert":{"timestamp":"2026-09-16T13:06:34.215+0000"},"rule":{"level":10,"description":"sshd: brute force trying to get access to the system. Non existent user.","id":"5712","mitre":{"id":["T1110"],"tactic":["Credential Access"],"technique":["Brute Force"]},"frequency":8,"firedtimes":1,"mail":false,"groups":["syslog","sshd","authentication_failures"],"gdpr":["IV_35.7.d","IV_32.2"],"hipaa":["164.312.b"],"nist_800_53":["SI.4","AU.14","AC.7"],"pci_dss":["11.4","10.2.4","10.2.5"],"tsc":["CC6.1","CC6.8","CC7.2","CC7.3"]},"agent":{"id":"002","name":"ubuntu-agent-01","ip":"192.168.1.8"},"manager":{"name":"wazuh.manager"},"id":"1789563994.2173849"},"previous_output":"Sep 16 13:06:32 ubuntu-agent-01 sshd[2365]: Invalid user wronguser from 192.168.1.9 port 35146\nSep 16 13:06:32 ubuntu-agent-01 sshd[2363]: Invalid user wronguser from 192.168.1.9 port 35140\nSep 16 13:06:32 ubuntu-agent-01 sshd[2361]: Invalid user wronguser from 192.168.1.9 port 35132\nSep 16 13:06:32 ubuntu-agent-01 sshd[2359]: Invalid user wronguser from 192.168.1.9 port 35126\nSep 16 13:06:31 ubuntu-agent-01 sshd[2357]: Invalid user wronguser from 192.168.1.9 port 35112\nSep 16 13:06:31 ubuntu-agent-01 sshd[2355]: Invalid user wronguser from 192.168.1.9 port 35096\nSep 16 13:06:31 ubuntu-agent-01 sshd[2353]: Invalid user wronguser from 192.168.1.9 port 35090"},"full_log":"Sep 16 13:06:32 ubuntu-agent-01 sshd[2367]: Invalid user wronguser from 192.168.1.9 port 35150"},"predecoder":{"program_name":"sshd","timestamp":"Sep 16 13:06:32","hostname":"ubuntu-agent-01"},"decoder":{"parent":"sshd","name":"sshd"},"data":{"srcip":"192.168.1.9","srcport":"35150"},"srcuser":"wronguser"},"location":"journalid"},"program":"active-response/bin/firewall-drop"}}
2026/09/16 13:07:35 active-response/bin/firewall-drop: Ended
agent@ubuntu-agent-01:~$ [19667.515349] watchdog: BUG: soft lockup - CPU#0 stuck for 4384s! [swapper/0:0]
[21404.965214] watchdog: BUG: soft lockup - CPU#0 stuck for 1617s! [snapd:2518]
[21405.506274] systemd[1]: Failed to start Refresh fwupd metadata and update motd.
[21405.577159] systemd[1]: Failed to start Firmware update daemon.
[23852.543710] e1000 0000:00:03:0 enp0s3: Reset adapter
agent@ubuntu-agent-01:~$ ip -br addr
lo                UNKNOWN      127.0.0.1/8 ::1/128
enp0s3            UP           192.168.1.8/24 metric 100 2401:4900:88e2:d4b7:a00:27ff:fed0:bdb6/64
fe80::a00:27ff:fed0:bdb6/64
agent@ubuntu-agent-01:~$ _
```

Figure 33. Source: 75.png

Terminal or configuration capture from the lab build/test workflow. It supports implementation context only to the extent that the command and output are legible; correlate with service logs or Wazuh alerts before asserting success.

```
agent@ubuntu-agent-01:~$ suricata -V
This is Suricata version 6.0.4 RELEASE
agent@ubuntu-agent-01:~$ sudo grep -n "af-packet" /etc/suricata/suricata.yaml
580:af-packet:
agent@ubuntu-agent-01:~$ sudo sed -n '580,610p' /etc/suricata/suricata.yaml
af-packet:
- interface: eth0
  # Number of receive threads. "auto" uses the number of cores
  #threads: auto
  # Default clusterid. AF_PACKET will load balance packets based on flow.
  cluster-id: 99
  # Default AF_PACKET cluster type. AF_PACKET can load balance per flow or per hash.
  # This is only supported for Linux kernel > 3.1
  # possible value are:
  # * cluster_flow: all packets of a given flow are sent to the same socket
  # * cluster_cpu: all packets treated in kernel by a CPU are sent to the same socket
  # * cluster_qm: all packets linked by network card to a RSS queue are sent to the same
  # socket. Requires at least Linux 3.14.
  # * cluster_ebpf: eBPF file load balancing. See doc/userguide/capture-hardware/ebpf-xdp.rst for
  # more info.
  # Recommended modes are cluster_flow on most boxes and cluster_cpu or cluster_qm on system
  # with capture card using RSS (requires cpu affinity tuning and system IRQ tuning)
  cluster-type: cluster_flow
  # In some fragmentation cases, the hash can not be computed. If "defrag" is set
  # to yes, the kernel will do the needed defragmentation before sending the packets.
  defrag: yes
  # To use the ring feature of AF_PACKET, set 'use-mmap' to yes
  #use-mmap: yes
  # Lock memory map to avoid it being swapped. Be careful that over
  # subscribing could lock your system
  #mmap-locked: yes
  # Use tpacket_v3 capture mode, only active if use-mmap is true
  # Don't use it in IPS or TAP mode as it causes severe latency
  #tpacket-v3: yes
  # Ring size will be computed with respect to "max-pending-packets" and number
  # of threads. You can set manually the ring size in number of packets by setting
agent@ubuntu-agent-01:~$
```

Figure 34. Source: 77.png

Terminal or configuration capture from the lab build/test workflow. It supports implementation context only to the extent that the command and output are legible; correlate with service logs or Wazuh alerts before asserting success.

```
CGroup: /system.slice/suricata.service
└─5226 /usr/bin/suricata -D --af-packet -c /etc/suricata/suricata.yaml --pidfile /run/...

Sep 17 06:02:16 ubuntu-agent-01 systemd[1]: Starting Suricata IDS/IDP daemon...
Sep 17 06:02:16 ubuntu-agent-01 suricata[5225]: 17/9/2026 -- 06:02:16 - <Notice> - This is Sur... mode
Sep 17 06:02:16 ubuntu-agent-01 systemd[1]: suricata.service: Can't open PID file /run/suricat...itted
Sep 17 06:02:16 ubuntu-agent-01 systemd[1]: Started Suricata IDS/IDP daemon.
Hint: Some lines were ellipsized, use -l to show in full.
agent@ubuntu-agent-01:~$ sudo ls -lh /var/log/suricata/
total 16K
-rw-r--r-- 1 root root  0 Sep 17 05:40 eve.json
-rw-r--r-- 1 root root  0 Sep 17 05:40 fast.log
-rw-r--r-- 1 root root  0 Sep 17 05:40 stats.log
-rw-r--r-- 1 root root 14K Sep 17 06:02 suricata.log
agent@ubuntu-agent-01:~$ sudo grep -n -A 20 "eve-log:" /etc/suricata/suricata.yaml
84: - eve-log:
85:     enabled: yes
86:     filetype: regular #regular|syslog|unix_dgram|unix_stream|redis
87:     filename: eve.json
88:     # Enable for multi-threaded eve.json output; output files are amended with
89:     # with an identifier, e.g., eve.9.json
90:     #threaded: false
91:     #prefix: "@cee: " # prefix to prepend to each log entry
92:     # the following are valid when type: syslog above
93:     #identity: "suricata"
94:     #facility: local5
95:     #level: Info ## possible levels: Emergency, Alert, Critical,
96:     ## Error, Warning, Notice, Info, Debug
97:     #ethernet: no # log ethernet header in events when available
98:     #redis:
99:     #   server: 127.0.0.1
100:    #   port: 6379
101:    #   async: true ## if redis replies are read asynchronously
102:    #   mode: list ## possible values: list|lpush (default), rpush, channel|publish
103:    #           ## lpush and rpush are using a Redis list. "list" is an alias for lpush
104:    #           ## publish is using a Redis channel. "channel" is an alias for publish
agent@ubuntu-agent-01:~$
```

Figure 35. Source: 79.png

Terminal or configuration capture from the lab build/test workflow. It supports implementation context only to the extent that the command and output are legible; correlate with service logs or Wazuh alerts before asserting success.

```
p":0,"dns_tcp":0,"nfs_tcp":0,"ntp":3,"ftp-data":0,"tftp":0,"ikev2":0,"krb5_tcp":0,"dhcp":24,"snmp":0
,"sip":0,"rfb":0,"mqtt":0,"rdp":0,"dcerpc_udp":0,"dns_udp":8,"nfs_udp":0,"krb5_udp":0},"expectations
":0},"http":{"memuse":0,"memcap":0},"ftp":{"memuse":0,"memcap":0},"file_store":{"open_files":0}}}}
{"timestamp":"2026-09-17T07:58:04.546462+0000","flow_id":1274113264411867,"in_iface":"enp0s3","event
_type":"flow","src_ip":"192.168.1.5","src_port":64197,"dest_ip":"224.0.0.252","dest_port":5355,"prot
o":"UDP","app_proto":"failed","flow":{"pkts_toserver":2,"pkts_toclient":0,"bytes_toserver":130,"byte
s_toclient":0,"start":"2026-09-17T07:53:43.746715+0000","end":"2026-09-17T07:53:44.158776+0000","age
":1,"state":"new","reason":"timeout","alerted":false}}
agent@ubuntu-agent-01:~$ sudo grep -i suricata /var/ossec/logs/ossec.log | tail -n 10
2026/09/17 07:57:11 wazuh-logcollector: INFO: (1950): Analyzing file: '/var/log/suricata/eve.json'.
agent@ubuntu-agent-01:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 100
0
    link/ether 08:00:27:d0:bd:b6 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.8/24 metric 100 brd 192.168.1.255 scope global dynamic enp0s3
        valid_lft 85984sec preferred_lft 85984sec
    inet6 2401:4900:88e2:d4b7:a00:27ff:fed0:bdb6/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 85984sec preferred_lft 85984sec
    inet6 fe80::a00:27ff:fed0:bdb6/64 scope link
        valid_lft forever preferred_lft forever
agent@ubuntu-agent-01:~$ sudo grep "event_type\":\"alert\"" /var/log/suricata/eve.json | tail -n 10
{"timestamp":"2026-09-17T08:00:37.220840+0000","flow_id":2024882164087061,"in_iface":"enp0s3","event
_type":"alert","src_ip":"0.0.0.0","src_port":68,"dest_ip":"255.255.255.255","dest_port":67,"proto":"
UDP","alert":{"action":"allowed","gid":1,"signature_id":2022973,"rev":1,"signature":"ET POLICY Possi
ble Kali Linux hostname in DHCP Request Packet","category":"Potential Corporate Privacy Violation","
severity":1,"metadata":{"attack_target":["Client_Endpoint"],"confidence":["Medium"],"created_at":["2
016_07_18"],"deployment":["Perimeter"],"performance_impact":["Low"],"reviewed_at":["2024_04_19"],"si
gnature_severity":["Informational"],"updated_at":["2019_07_26"]}},"app_proto":"dhcp","flow":{"pkts_t
oserver":2,"pkts_toclient":0,"bytes_toserver":675,"bytes_toclient":0,"start":"2026-09-17T08:00:26.66
9973+0000"}}}
agent@ubuntu-agent-01:~$
```

Figure 36. Source: 81.png

Terminal or configuration capture from the lab build/test workflow. It supports implementation context only to the extent that the command and output are legible; correlate with service logs or Wazuh alerts before asserting success.

```
- local.rules
##
## Auxiliary configuration files.
##
classification-file: /etc/suricata/classification.config
reference-config-file: /etc/suricata/reference.config
# threshold-file: /etc/suricata/threshold.config

##
## Include other configs
##

# Includes: Files included here will be handled as if they were in-lined
# in this configuration file. Files with relative pathnames will be

agent@ubuntu-agent-01:~$ sudo grep '"event_type":"alert"' /var/log/suricata/eve.json | tail -n 5
{"timestamp":"2026-09-17T08:00:37.220840+0000","flow_id":2024882164087061,"in_iface":"enp0s3","event_type":"alert","src_ip":"0.0.0.0","src_port":68,"dest_ip":"255.255.255.255","dest_port":67,"proto":"UDP","alert":{"action":"allowed","gid":1,"signature_id":2022973,"rev":1,"signature":"ET POLICY Possible Kali Linux hostname in DHCP Request Packet","category":"Potential Corporate Privacy Violation","severity":1,"metadata":{"attack_target":["Client_Endpoint"],"confidence":["Medium"],"created_at":["2016_07_18"],"deployment":["Perimeter"],"performance_impact":["Low"],"reviewed_at":["2024_04_19"],"signature_severity":["Informational"],"updated_at":["2019_07_26"]},"app_proto":"dhcp","flow":{"pkts_toserver":2,"pkts_toclient":0,"bytes_toserver":675,"bytes_toclient":0,"start":"2026-09-17T08:00:26.669973+0000"}}}
{"timestamp":"2026-09-17T08:13:33.894286+0000","flow_id":2024882215691598,"in_iface":"enp0s3","event_type":"alert","src_ip":"0.0.0.0","src_port":68,"dest_ip":"255.255.255.255","dest_port":67,"proto":"UDP","alert":{"action":"allowed","gid":1,"signature_id":2022973,"rev":1,"signature":"ET POLICY Possible Kali Linux hostname in DHCP Request Packet","category":"Potential Corporate Privacy Violation","severity":1,"metadata":{"attack_target":["Client_Endpoint"],"confidence":["Medium"],"created_at":["2016_07_18"],"deployment":["Perimeter"],"performance_impact":["Low"],"reviewed_at":["2024_04_19"],"signature_severity":["Informational"],"updated_at":["2019_07_26"]},"app_proto":"dhcp","flow":{"pkts_toserver":1,"pkts_toclient":0,"bytes_toserver":330,"bytes_toclient":0,"start":"2026-09-17T08:13:33.894286+0000"}}}
agent@ubuntu-agent-01:~$
```

Figure 37. Source: 82.png

Terminal or configuration capture from the lab build/test workflow. It supports implementation context only to the extent that the command and output are legible; correlate with service logs or Wazuh alerts before asserting success.

```
{
  "timestamp": "2026-09-17T08:33:58.238498+0000",
  "flow_id": "285051162753474",
  "in_iface": "enp0s3",
  "event_type": "flow",
  "src_ip": "192.168.1.9",
  "src_port": 44613,
  "dest_ip": "192.168.1.8",
  "dest_port": 51,
  "proto": "TCP",
  "flow": {
    "pkts_to_server": 1,
    "pkts_to_client": 1,
    "bytes_to_server": 60,
    "bytes_to_client": 54,
    "start": "2026-09-17T08:31:08.227778+0000",
    "end": "2026-09-17T08:31:08.227814+0000",
    "age": 0,
    "state": "closed",
    "reason": "timeout",
    "alerted": false,
    "tcp": {
      "tcp_flags": "16",
      "tcp_flags_ts": "02",
      "tcp_flags_tc": "14",
      "syn": true,
      "rst": true,
      "ack": true,
      "state": "closed"
    }
  }
}

{
  "timestamp": "2026-09-17T08:34:00.240898+0000",
  "flow_id": "427261825506196",
  "in_iface": "enp0s3",
  "event_type": "flow",
  "src_ip": "192.168.1.9",
  "src_port": 44613,
  "dest_ip": "192.168.1.8",
  "dest_port": 17,
  "proto": "TCP",
  "flow": {
    "pkts_to_server": 1,
    "pkts_to_client": 1,
    "bytes_to_server": 60,
    "bytes_to_client": 54,
    "start": "2026-09-17T08:31:17.056212+0000",
    "end": "2026-09-17T08:31:17.056236+0000",
    "age": 0,
    "state": "closed",
    "reason": "timeout",
    "alerted": false,
    "tcp": {
      "tcp_flags": "16",
      "tcp_flags_ts": "02",
      "tcp_flags_tc": "14",
      "syn": true,
      "rst": true,
      "ack": true,
      "state": "closed"
    }
  }
}

{
  "timestamp": "2026-09-17T08:34:04.242608+0000",
  "flow_id": "1977745018812942",
  "in_iface": "enp0s3",
  "event_type": "flow",
  "src_ip": "192.168.1.9",
  "src_port": 44613,
  "dest_ip": "192.168.1.8",
  "dest_port": 5,
  "proto": "TCP",
  "flow": {
    "pkts_to_server": 1,
    "pkts_to_client": 1,
    "bytes_to_server": 60,
    "bytes_to_client": 54,
    "start": "2026-09-17T08:31:09.031246+0000",
    "end": "2026-09-17T08:31:09.031281+0000",
    "age": 0,
    "state": "closed",
    "reason": "timeout",
    "alerted": false,
    "tcp": {
      "tcp_flags": "16",
      "tcp_flags_ts": "02",
      "tcp_flags_tc": "14",
      "syn": true,
      "rst": true,
      "ack": true,
      "state": "closed"
    }
  }
}

{
  "timestamp": "2026-09-17T08:34:10.247366+0000",
  "flow_id": "855521603157134",
  "in_iface": "enp0s3",
  "event_type": "flow",
  "src_ip": "192.168.1.9",
  "src_port": 44613,
  "dest_ip": "192.168.1.8",
  "dest_port": 80,
  "proto": "TCP",
  "flow": {
    "pkts_to_server": 1,
    "pkts_to_client": 1,
    "bytes_to_server": 60,
    "bytes_to_client": 54,
    "start": "2026-09-17T08:30:56.195726+0000",
    "end": "2026-09-17T08:30:56.195751+0000",
    "age": 0,
    "state": "closed",
    "reason": "timeout",
    "alerted": false,
    "tcp": {
      "tcp_flags": "16",
      "tcp_flags_ts": "02",
      "tcp_flags_tc": "14",
      "syn": true,
      "rst": true,
      "ack": true,
      "state": "closed"
    }
  }
}

{
  "timestamp": "2026-09-17T08:34:14.252225+0000",
  "flow_id": "1561012932224992",
  "in_iface": "enp0s3",
  "event_type": "flow",
  "src_ip": "192.168.1.9",
  "src_port": 44613,
  "dest_ip": "192.168.1.8",
  "dest_port": 1,
  "proto": "TCP",
  "flow": {
    "pkts_to_server": 1,
    "pkts_to_client": 1,
    "bytes_to_server": 60,
    "bytes_to_client": 54,
    "start": "2026-09-17T08:31:12.239584+0000",
    "end": "2026-09-17T08:31:12.239620+0000",
    "age": 0,
    "state": "closed",
    "reason": "timeout",
    "alerted": false,
    "tcp": {
      "tcp_flags": "16",
      "tcp_flags_ts": "02",
      "tcp_flags_tc": "14",
      "syn": true,
      "rst": true,
      "ack": true,
      "state": "closed"
    }
  }
}

{
  "timestamp": "2026-09-17T08:34:14.252282+0000",
  "flow_id": "153848501460442",
  "in_iface": "enp0s3",
  "event_type": "flow",
  "src_ip": "192.168.1.9",
  "src_port": 44613,
  "dest_ip": "192.168.1.8",
  "dest_port": 42,
  "proto": "TCP",
  "flow": {
    "pkts_to_server": 1,
    "pkts_to_client": 1,
    "bytes_to_server": 60,
    "bytes_to_client": 54,
    "start": "2026-09-17T08:31:03.417242+0000",
    "end": "2026-09-17T08:31:03.417264+0000",
    "age": 0,
    "state": "closed",
    "reason": "timeout",
    "alerted": false,
    "tcp": {
      "tcp_flags": "16",
      "tcp_flags_ts": "02",
      "tcp_flags_tc": "14",
      "syn": true,
      "rst": true,
      "ack": true,
      "state": "closed"
    }
  }
}

agent@ubuntu-agent-01:~$ _
```

Figure 38. Source: 83.png

Terminal or configuration capture from the lab build/test workflow. It supports implementation context only to the extent that the command and output are legible; correlate with service logs or Wazuh alerts before asserting success.

[illegible]

Figure 39. Source: 84.png

Terminal or configuration capture from the lab build/test workflow. It supports implementation context only to the extent that the command and output are legible; correlate with service logs or Wazuh alerts before asserting success.

```
alert tcp $EXTERNAL_NET any -> $HOME_NET 21 (msg:"ET SCAN Multiple FTP Administrator Login Attempts
from Single Source - Possible Brute Force Attempt"; flow:established,to_server; content:"USER "; noc
ase; depth:5; content:"administrator"; within:25; nocase; threshold: type threshold, track by_src, c
ount 5, seconds 60; classtype:attempted-recon; sid:2010643; rev:3; metadata:created_at 2010_07_30, c
onfidence Medium, signature_severity Informational, updated_at 2019_07_26;)
alert tcp any any -> any 21 (msg:"ET SCAN Grim's Ping ftp scanning tool"; flow:to_server,established
; content:"PASS "; content:"gpuser@home.com"; within:18; reference:url,archives.neohapsis.com/archiv
es/snort/2002-04/0448.html; reference:url,grimsping.cjb.net; classtype:network-scan; sid:2007802; re
v:4; metadata:created_at 2010_07_30, confidence Medium, signature_severity Informational, updated_at
2019_07_26;)
alert icmp $EXTERNAL_NET any -> $HOME_NET any (msg:"ET SCAN ICMP PING IPTools"; itype: 8; icode: 0;
content:"|A7 A7 A7 A7 A7 A7 A7 A7 A7 A7 A7 A7 A7 A7 A7 A7 A7 A7 A7 A7 A7 A7 A7 A7 A7 A7 A7 A7 A7
A7 A7 A7 A7 A7 A7 A7 A7 A7 A7 A7 A7 A7 A7 A7 A7 A7 A7 A7 A7 A7 A7 A7 A7 A7 A7 A7 A7 A7 A7 A7 A7
A7|"; depth: 64; reference:url,www.ks-soft.net/ip-tools.eng; reference:url,www.ks-soft.net/ip-tools.e
ng/index.htm; classtype:misc-activity; sid:2000575; rev:7; metadata:created_at 2010_07_30, confidenc
e High, signature_severity Informational, updated_at 2019_07_26;)
alert udp $EXTERNAL_NET any -> $HOME_NET 1434 (msg:"ET SCAN NNG MS02-039 Exploit False Positive Gene
rator - May Conceal A Genuine Attack"; content:"nng Snort (Snort)"; offset:90; threshold:type thresh
old, track by_dst, count 4, seconds 15; reference:url,packetstormsecurity.nl/filedesc/nng-4.13r-publ
ic.rar.html; classtype:misc-activity; sid:2008560; rev:2; metadata:created_at 2010_07_30, confidence
Medium, signature_severity Major, updated_at 2019_07_26;)
alert tcp $EXTERNAL_NET any -> $SQL_SERVERS 3306 (msg:"ET SCAN MYSQL 4.0 brute force root login atte
mpt"; flow:to_server,established; content:"|01|"; offset:3; depth:4; content:"root|00|"; nocase; dis
tance:5; within:5; threshold:type both,track by_src,count 5,seconds 60; reference:url,www.redferni.u
klinux.net/mysql/MySQL-323.html; classtype:protocol-command-decode; sid:2001906; rev:6; metadata:cre
ated_at 2010_07_30, confidence Medium, signature_severity Informational, updated_at 2019_07_26;)
alert tcp $EXTERNAL_NET any -> $SQL_SERVERS 3306 (msg:"ET SCAN MYSQL 4.1 brute force root login atte
mpt"; flow:to_server,established; content:"|01|"; offset:3; depth:4; content:"root|00|"; nocase; dis
tance:32; within:5; threshold:type both,track by_src,count 5,seconds 60; reference:url,www.redferni.
uklinux.net/mysql/MySQL-Protocol.html; classtype:protocol-command-decode; sid:2002842; rev:4; metada
ta:created_at 2010_07_30, confidence Medium, signature_severity Informational, updated_at 2019_07_26
;)
agent@ubuntu-agent-01:~$ sudo grep -n -E '^[[:space:]]*HOME_NET|^[[:space:]]*EXTERNAL_NET' /etc/suric
cata/suricata.yaml
15:   HOME_NET: "[192.168.0.0/16,10.0.0.0/8,172.16.0.0/12]"
21:   EXTERNAL_NET: "!$HOME_NET"
agent@ubuntu-agent-01:~$
```

Figure 40. Source: 85.png

Terminal or configuration capture from the lab build/test workflow. It supports implementation context only to the extent that the command and output are legible; correlate with service logs or Wazuh alerts before asserting success.

[illegible]

Figure 41. Source: 87.png

Terminal or configuration capture from the lab build/test workflow. It supports implementation context only to the extent that the command and output are legible; correlate with service logs or Wazuh alerts before asserting success.

```
##
## Step 1: Inform Suricata about your network
##
vars:
# more specific is better for alert accuracy and performance
address-groups:
  HOME_NET: "[192.168.1.8/32]"
  #HOME_NET: "[192.168.0.0/16]"
  #HOME_NET: "[10.0.0.0/8]"
  #HOME_NET: "[172.16.0.0/12]"
  #HOME_NET: "any"

  EXTERNAL_NET: "!$HOME_NET"
  #EXTERNAL_NET: "any"

  HTTP_SERVERS: "$HOME_NET"
  SMTP_SERVERS: "$HOME_NET"
  SQL_SERVERS: "$HOME_NET"
  DNS_SERVERS: "$HOME_NET"
  TELNET_SERVERS: "$HOME_NET"
  AIM_SERVERS: "$EXTERNAL_NET"
  DC_SERVERS: "$HOME_NET"
  DNP3_SERVER: "$HOME_NET"
  DNP3_CLIENT: "$HOME_NET"
  MODBUS_CLIENT: "$HOME_NET"

agent@ubuntu-agent-01:~$ sudo tail -n 15 /var/log/suricata/fast.log
09/17/2026-08:00:37.220840  [**] [1:2022973:1] ET POLICY Possible Kali Linux hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy Violation] [Priority: 1] {UDP} 0.0.0.0:68
-> 255.255.255.255:67
09/17/2026-08:13:33.894286  [**] [1:2022973:1] ET POLICY Possible Kali Linux hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy Violation] [Priority: 1] {UDP} 0.0.0.0:68
-> 255.255.255.255:67
agent@ubuntu-agent-01:~$
```

Figure 42. Source: 89.png

Terminal or configuration capture from the lab build/test workflow. It supports implementation context only to the extent that the command and output are legible; correlate with service logs or Wazuh alerts before asserting success.

```
EXTERNAL_NET: "!$HOME_NET"
#EXTERNAL_NET: "any"

HTTP_SERVERS: "$HOME_NET"
SMTP_SERVERS: "$HOME_NET"
SQL_SERVERS: "$HOME_NET"
DNS_SERVERS: "$HOME_NET"
TELNET_SERVERS: "$HOME_NET"
AIM_SERVERS: "$EXTERNAL_NET"
DC_SERVERS: "$HOME_NET"
DNP3_SERVER: "$HOME_NET"
DNP3_CLIENT: "$HOME_NET"
MODBUS_CLIENT: "$HOME_NET"

agent@ubuntu-agent-01:~$ sudo tail -n 15 /var/log/suricata/fast.log
09/17/2026-08:00:37.220840  [**] [1:2022973:1] ET POLICY Possible Kali Linux hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy Violation] [Priority: 1] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
09/17/2026-08:13:33.894286  [**] [1:2022973:1] ET POLICY Possible Kali Linux hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy Violation] [Priority: 1] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
agent@ubuntu-agent-01:~$ sudo tail -n 15 /var/log/suricata/fast.log
09/17/2026-08:00:37.220840  [**] [1:2022973:1] ET POLICY Possible Kali Linux hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy Violation] [Priority: 1] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
09/17/2026-08:13:33.894286  [**] [1:2022973:1] ET POLICY Possible Kali Linux hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy Violation] [Priority: 1] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
agent@ubuntu-agent-01:~$ sudo tail -n 20 /var/log/suricata/fast.log
09/17/2026-08:00:37.220840  [**] [1:2022973:1] ET POLICY Possible Kali Linux hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy Violation] [Priority: 1] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
09/17/2026-08:13:33.894286  [**] [1:2022973:1] ET POLICY Possible Kali Linux hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy Violation] [Priority: 1] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
agent@ubuntu-agent-01:~$ _
```

Figure 43. Source: 90.png

Terminal or configuration capture from the lab build/test workflow. It supports implementation context only to the extent that the command and output are legible; correlate with service logs or Wazuh alerts before asserting success.

```
TELNET_SERVERS: "$HOME_NET"
AIM_SERVERS: "$EXTERNAL_NET"
DC_SERVERS: "$HOME_NET"
DNP3_SERVER: "$HOME_NET"
DNP3_CLIENT: "$HOME_NET"
MODBUS_CLIENT: "$HOME_NET"

agent@ubuntu-agent-01:~$ sudo tail -n 15 /var/log/suricata/fast.log
09/17/2026-08:00:37.220840  [**] [1:2022973:1] ET POLICY Possible Kali Linux hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy Violation] [Priority: 1] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
09/17/2026-08:13:33.894286  [**] [1:2022973:1] ET POLICY Possible Kali Linux hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy Violation] [Priority: 1] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
agent@ubuntu-agent-01:~$ sudo tail -n 15 /var/log/suricata/fast.log
09/17/2026-08:00:37.220840  [**] [1:2022973:1] ET POLICY Possible Kali Linux hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy Violation] [Priority: 1] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
09/17/2026-08:13:33.894286  [**] [1:2022973:1] ET POLICY Possible Kali Linux hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy Violation] [Priority: 1] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
agent@ubuntu-agent-01:~$ sudo tail -n 20 /var/log/suricata/fast.log
09/17/2026-08:00:37.220840  [**] [1:2022973:1] ET POLICY Possible Kali Linux hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy Violation] [Priority: 1] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
09/17/2026-08:13:33.894286  [**] [1:2022973:1] ET POLICY Possible Kali Linux hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy Violation] [Priority: 1] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
agent@ubuntu-agent-01:~$ sudo tail -n 20 /var/log/suricata/fast.log
09/17/2026-08:00:37.220840  [**] [1:2022973:1] ET POLICY Possible Kali Linux hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy Violation] [Priority: 1] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
09/17/2026-08:13:33.894286  [**] [1:2022973:1] ET POLICY Possible Kali Linux hostname in DHCP Request Packet [**] [Classification: Potential Corporate Privacy Violation] [Priority: 1] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
agent@ubuntu-agent-01:~$ _
```

Figure 44. Source: 91.png

Terminal or configuration capture from the lab build/test workflow. It supports implementation context only to the extent that the command and output are legible; correlate with service logs or Wazuh alerts before asserting success.

```
{
  "timestamp": "2026-09-17T08:55:43.086551+0000",
  "flow_id": "2176486126520758",
  "in_iface": "enp0s3",
  "event_type": "flow",
  "src_ip": "192.168.1.9",
  "src_port": 59439,
  "dest_ip": "192.168.1.8",
  "dest_port": 846,
  "proto": "TCP",
  "flow": {
    "pkts_to_server": 1,
    "pkts_to_client": 1,
    "bytes_to_server": 60,
    "bytes_to_client": 54,
    "start": "2026-09-17T08:53:02.395702+0000",
    "end": "2026-09-17T08:53:02.395849+0000",
    "age": 0,
    "state": "closed",
    "reason": "timeout",
    "alerted": false,
    "tcp": {
      "tcp_flags": "16",
      "tcp_flags_ts": "02",
      "tcp_flags_tc": "14",
      "syn": true,
      "rst": true,
      "ack": true,
      "state": "closed"
    }
  }
}

{
  "timestamp": "2026-09-17T08:55:43.086554+0000",
  "flow_id": "1895157174668636",
  "in_iface": "enp0s3",
  "event_type": "flow",
  "src_ip": "192.168.1.9",
  "src_port": 42311,
  "dest_ip": "192.168.1.8",
  "dest_port": 29,
  "proto": "TCP",
  "flow": {
    "pkts_to_server": 1,
    "pkts_to_client": 1,
    "bytes_to_server": 60,
    "bytes_to_client": 54,
    "start": "2026-09-17T08:52:00.232796+0000",
    "end": "2026-09-17T08:52:00.232817+0000",
    "age": 0,
    "state": "closed",
    "reason": "timeout",
    "alerted": false,
    "tcp": {
      "tcp_flags": "16",
      "tcp_flags_ts": "02",
      "tcp_flags_tc": "14",
      "syn": true,
      "rst": true,
      "ack": true,
      "state": "closed"
    }
  }
}

{
  "timestamp": "2026-09-17T08:55:43.086558+0000",
  "flow_id": "1191482625321999",
  "in_iface": "enp0s3",
  "event_type": "flow",
  "src_ip": "192.168.1.9",
  "src_port": 49349,
  "dest_ip": "192.168.1.8",
  "dest_port": 155,
  "proto": "TCP",
  "flow": {
    "pkts_to_server": 1,
    "pkts_to_client": 1,
    "bytes_to_server": 60,
    "bytes_to_client": 54,
    "start": "2026-09-17T08:53:55.224271+0000",
    "end": "2026-09-17T08:53:55.224274+0000",
    "age": 0,
    "state": "closed",
    "reason": "timeout",
    "alerted": false,
    "tcp": {
      "tcp_flags": "16",
      "tcp_flags_ts": "02",
      "tcp_flags_tc": "14",
      "syn": true,
      "rst": true,
      "ack": true,
      "state": "closed"
    }
  }
}

{
  "timestamp": "2026-09-17T08:55:43.086561+0000",
  "flow_id": "910050598259659",
  "in_iface": "enp0s3",
  "event_type": "flow",
  "src_ip": "192.168.1.9",
  "src_port": 49349,
  "dest_ip": "192.168.1.8",
  "dest_port": 137,
  "proto": "TCP",
  "flow": {
    "pkts_to_server": 1,
    "pkts_to_client": 1,
    "bytes_to_server": 60,
    "bytes_to_client": 54,
    "start": "2026-09-17T08:53:55.330699+0000",
    "end": "2026-09-17T08:53:55.330745+0000",
    "age": 0,
    "state": "closed",
    "reason": "timeout",
    "alerted": false,
    "tcp": {
      "tcp_flags": "16",
      "tcp_flags_ts": "02",
      "tcp_flags_tc": "14",
      "syn": true,
      "rst": true,
      "ack": true,
      "state": "closed"
    }
  }
}

{
  "timestamp": "2026-09-17T08:55:43.086565+0000",
  "flow_id": "910059180695797",
  "in_iface": "enp0s3",
  "event_type": "flow",
  "src_ip": "192.168.1.9",
  "src_port": 42311,
  "dest_ip": "192.168.1.8",
  "dest_port": 54,
  "proto": "TCP",
  "flow": {
    "pkts_to_server": 1,
    "pkts_to_client": 1,
    "bytes_to_server": 60,
    "bytes_to_client": 54,
    "start": "2026-09-17T08:52:00.237813+0000",
    "end": "2026-09-17T08:52:00.237943+0000",
    "age": 0,
    "state": "closed",
    "reason": "timeout",
    "alerted": false,
    "tcp": {
      "tcp_flags": "16",
      "tcp_flags_ts": "02",
      "tcp_flags_tc": "14",
      "syn": true,
      "rst": true,
      "ack": true,
      "state": "closed"
    }
  }
}

{
  "timestamp": "2026-09-17T08:55:43.086569+0000",
  "flow_id": "628592793922999",
  "in_iface": "enp0s3",
  "event_type": "flow",
  "src_ip": "192.168.1.9",
  "src_port": 42311,
  "dest_ip": "192.168.1.8",
  "dest_port": 391,
  "proto": "TCP",
  "flow": {
    "pkts_to_server": 1,
    "pkts_to_client": 1,
    "bytes_to_server": 60,
    "bytes_to_client": 54,
    "start": "2026-09-17T08:52:00.241079+0000",
    "end": "2026-09-17T08:52:00.241377+0000",
    "age": 0,
    "state": "closed",
    "reason": "timeout",
    "alerted": false,
    "tcp": {
      "tcp_flags": "16",
      "tcp_flags_ts": "02",
      "tcp_flags_tc": "14",
      "syn": true,
      "rst": true,
      "ack": true,
      "state": "closed"
    }
  }
}

agent@ubuntu-agent-01:~$
```

Figure 45. Source: 109.png

Selected evidence screenshot previously retained for the project. Refer to the visible event/rule/endpoint identifiers and cross-check against raw logs; avoid claiming causation from proximity alone.

```
bash-5.2# cp /var/ossec/etc/rules/local_rules.xml /var/ossec/etc/rules/local_rules.xml.bak
bash-5.2# ls -l /var/ossec/etc/rules/local_rules.xml*
-rw-rw---- 1 wazuh wazuh 1760 Jul 24 05:21 /var/ossec/etc/rules/local_rules.xml
-rw-r----- 1 root root 1760 Sep 13 09:34 /var/ossec/etc/rules/local_rules.xml.bak
bash-5.2# python3 --version
Python 3.9.25
bash-5.2# cp /var/ossec/etc/rules/local_rules.xml /var/ossec/etc/rules/local_rules.xml.bak
bash-5.2# echo BACKUP_DONE
BACKUP_DONE
bash-5.2# grep -R -n -B 5 -A 15 'rule id="92031"' /var/ossec/ruleset/rules/
/var/ossec/ruleset/rules/0800-sysmon_id_1.xml-335-      <mitre>
/var/ossec/ruleset/rules/0800-sysmon_id_1.xml-336-        <id>T1033</id>
/var/ossec/ruleset/rules/0800-sysmon_id_1.xml-337-      </mitre>
/var/ossec/ruleset/rules/0800-sysmon_id_1.xml-338-    </rule>
/var/ossec/ruleset/rules/0800-sysmon_id_1.xml-339-
/var/ossec/ruleset/rules/0800-sysmon_id_1.xml:340:    <rule id="92031" level="3">
/var/ossec/ruleset/rules/0800-sysmon_id_1.xml-341-      <if_group>sysmon_event1</if_group>
/var/ossec/ruleset/rules/0800-sysmon_id_1.xml-342-      <field name="win.eventdata.originalFileName" type="
type="pcrc2">(?) (net\.EXE|net1\.EXE) </field>
/var/ossec/ruleset/rules/0800-sysmon_id_1.xml-343-      <options>no_full_log</options>
/var/ossec/ruleset/rules/0800-sysmon_id_1.xml-344-      <description>Discovery activity executed</desc
ription>
/var/ossec/ruleset/rules/0800-sysmon_id_1.xml-345-      <mitre>
/var/ossec/ruleset/rules/0800-sysmon_id_1.xml-346-        <id>T1087</id>
/var/ossec/ruleset/rules/0800-sysmon_id_1.xml-347-      </mitre>
/var/ossec/ruleset/rules/0800-sysmon_id_1.xml-348-    </rule>
/var/ossec/ruleset/rules/0800-sysmon_id_1.xml-349-
/var/ossec/ruleset/rules/0800-sysmon_id_1.xml:350:    <rule id="92032" level="3">
/var/ossec/ruleset/rules/0800-sysmon_id_1.xml-351-      <if_sid>92031, 61603</if_sid>
/var/ossec/ruleset/rules/0800-sysmon_id_1.xml-352-      <field name="win.eventdata.parentImage" type="
pcrc2">(?) cmd\.EXE</field>
/var/ossec/ruleset/rules/0800-sysmon_id_1.xml-353-      <field name="win.eventdata.parentCommandLine"
type="pcrc2">(?) \\s\\C\\s</field>
/var/ossec/ruleset/rules/0800-sysmon_id_1.xml-354-      <options>no_full_log</options>
/var/ossec/ruleset/rules/0800-sysmon_id_1.xml-355-      <description>Suspicious Windows cmd shell exec
ution</description>
bash-5.2#
```

Figure 46. Source: 112.png

Selected evidence screenshot previously retained for the project. Refer to the visible event/rule/endpoint identifiers and cross-check against raw logs; avoid claiming causation from proximity alone.

```
</group>
<group name="local, windows, recon,">
  <rule id="100030" level="9" frequency="3" timeframe="120">
    <if_matched_sid>100029</if_matched_sid>
    <field name="win.eventdata.commandLine" type="pcre2">(?(i)(whoami|net1? user|net1? group|systeminfo|tasklist|nltest|ipconfig /all)</field>

    <description>Multiple reconnaissance commands executed in short window</description>
    <mitre><id>T1082</id><id>T1087</id></mitre>
    <group>recon, mitre_discovery,</group>
  </rule>

  <rule id="100031" level="12">
bash-5.2# python3 -c "import xml.etree.ElementTree as ET; ET.parse('/var/ossec/etc/rules/local_rules.xml'); print('XML_OK')"
File "<string>", line 1
    import xml.etree.ElementTree as ET; ET.parse('/var/ossec/etc/rules/local_rules.xml'); print('XML_OK')
^
SyntaxError: EOL while scanning string literal
bash-5.2# cp /var/ossec/etc/rules/local_rules.xml.bak /var/ossec/etc/rules/local_rules.xml
bash-5.2# grep -n '100029\|100030' /var/ossec/etc/rules/local_rules.xml
21:  <rule id="100030" level="9" frequency="3" timeframe="120">
bash-5.2# cat >> /var/ossec/etc/rules/local_rules.xml
<rule id="100029" level="3">
<if_group>sysmon_event1</if_group>
<field name="win.eventdata.commandLine" type="pcre2">(?(i)(whoami|systeminfo|tasklist|net(\.exe)?\s+user|net(\.exe)?\s+group|ipconfig(\.exe)?\s+/all)</field>
<description>Reconnaissance command executed on Windows endpoint</description>
<mitre>
<id>T1082</id>
<id>T1087</id>
</mitre>
<group>recon, mitre_discovery,</group>
</rule>
bash-5.2#
```

Figure 47. Source: 120.png

Selected evidence screenshot previously retained for the project. Refer to the visible event/rule/endpoint identifiers and cross-check against raw logs; avoid claiming causation from proximity alone.

```
ubuntu@ubuntu-server:~/wazuh-docker/single-node$ docker exec -it single-node-wazuh.manager-1 bash
bash-5.2# cp /var/ossec/etc/rules/local_rules.xml /var/ossec/etc/rules/local_rules.xml.bak2
bash-5.2# python3
Python 3.9.25 (main, Apr 17 2026, 00:00:00)
[GCC 11.5.0 20240719 (Red Hat 11.5.0-5)] on linux
Type "help", "copyright", "credits" or "license()" for more information.
>>> p="/var/ossec/etc/rules/local_rules.xml"
>>> s=open(p).read()
>>> a=s.index('<rule id="100031"')
>>> b=s.index('</rule>',a)+7
>>> new='<rule id="100031" level="14">
...     <if_sid>92057</if_sid>
...     <description>Custom SOC detection: encoded PowerShell execution</description>
...     <mitre>
...         <id>T1059.001</id>
...         <id>T1027</id>
...     </mitre>
...     <group>custom_powershell,custom_detection,mitre_execution,mitre_defense_evasion,</group>
... </rule>'
>>> open(p,"w").write(s[:a]+new+s[b:])
1692
>>> exit()
bash-5.2# grep -n -A 10 -B 2 '<rule id="100031"' /var/ossec/etc/rules/local_rules.xml
28- </rule>
29-
30: <rule id="100031" level="14">
31:   <if_sid>92057</if_sid>
32:   <description>Custom SOC detection: encoded PowerShell execution</description>
33:   <mitre>
34:     <id>T1059.001</id>
35:     <id>T1027</id>
36:   </mitre>
37:   <group>custom_powershell,custom_detection,mitre_execution,mitre_defense_evasion,</group>
38: </rule>
39-
40: <rule id="100032" level="12">
bash-5.2# _
```

Figure 48. Source: 122.png

Selected evidence screenshot previously retained for the project. Refer to the visible event/rule/endpoint identifiers and cross-check against raw logs; avoid claiming causation from proximity alone.

```
/var/ossec/ruleset/rules/0392-fortimail_rules.xml-415-    <field name="msg">System has been changed
to \w+ mode</field>
/var/ossec/ruleset/rules/0392-fortimail_rules.xml-416-    <description>FortiMail: An administrator o
r LCD user changed the mode to gateway, server, or transparent mode using the CLI, web-based manager
or LCD.</description>
/var/ossec/ruleset/rules/0392-fortimail_rules.xml-417-    </rule>
/var/ossec/ruleset/rules/0392-fortimail_rules.xml-418-
/var/ossec/ruleset/rules/0392-fortimail_rules.xml-419-    <rule id="44706" level="3">
/var/ossec/ruleset/rules/0392-fortimail_rules.xml-420-        <if_sid>44642</if_sid>
/var/ossec/ruleset/rules/0392-fortimail_rules.xml-421-        <field name="subtype">update</field>
/var/ossec/ruleset/rules/0392-fortimail_rules.xml-422-        <description>FortiMail: The FortiMail unit
updated the following FortiGuard services.</description>
/var/ossec/ruleset/rules/0392-fortimail_rules.xml-423-    </rule>
/var/ossec/ruleset/rules/0392-fortimail_rules.xml-424-
/var/ossec/ruleset/rules/0392-fortimail_rules.xml-425-    <rule id="44707" level="3">
/var/ossec/ruleset/rules/0392-fortimail_rules.xml-426-        <field name="subtype">imap</field>
bash-5.2# grep -R -n -B 5 -A 10 'eventID.*4698' /var/ossec/ruleset/rules/
/var/ossec/ruleset/rules/0580-win-security_rules.xml-1322-    </mitre>
/var/ossec/ruleset/rules/0580-win-security_rules.xml-1323-    </rule>
/var/ossec/ruleset/rules/0580-win-security_rules.xml-1324-
/var/ossec/ruleset/rules/0580-win-security_rules.xml-1325-    <rule id="60228" level="4">
/var/ossec/ruleset/rules/0580-win-security_rules.xml-1326-        <if_sid>60103</if_sid>
/var/ossec/ruleset/rules/0580-win-security_rules.xml-1327-        <field name="win.system.eventID">^4698
$</field>
/var/ossec/ruleset/rules/0580-win-security_rules.xml-1328-        <description>A scheduled task was crea
ted</description>
/var/ossec/ruleset/rules/0580-win-security_rules.xml-1329-        <options>no_full_log</options>
/var/ossec/ruleset/rules/0580-win-security_rules.xml-1330-        <mitre>
/var/ossec/ruleset/rules/0580-win-security_rules.xml-1331-            <id>T1053</id>
/var/ossec/ruleset/rules/0580-win-security_rules.xml-1332-        </mitre>
/var/ossec/ruleset/rules/0580-win-security_rules.xml-1333-    </rule>
/var/ossec/ruleset/rules/0580-win-security_rules.xml-1334-
/var/ossec/ruleset/rules/0580-win-security_rules.xml-1335-    <rule id="60229" level="4">
/var/ossec/ruleset/rules/0580-win-security_rules.xml-1336-        <if_sid>60103</if_sid>
/var/ossec/ruleset/rules/0580-win-security_rules.xml-1337-        <field name="win.system.eventID">^5136
$</field>
bash-5.2# _
```

Figure 49. Source: 124.png

Selected evidence screenshot previously retained for the project. Refer to the visible event/rule/endpoint identifiers and cross-check against raw logs; avoid claiming causation from proximity alone.

```
<rule id="60229" level="4">
bash-5.2# cp /var/ossec/etc/rules/local_rules.xml /var/ossec/etc/rules/local_rules.xml.bak3
bash-5.2# python3
Python 3.9.25 (main, Apr 17 2026, 00:00:00)
[GCC 11.5.0 20240719 (Red Hat 11.5.0-5)] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> p="/var/ossec/etc/rules/local_rules.xml"
>>> s=open(p).read()
>>> a=s.index('<rule id="100032">')
>>> new='''<rule id="100033" level="12">
...     <if_sid>60228</if_sid>
...     <description>Custom SOC detection: scheduled task persistence</description>
...     <mitre>
...         <id>T1053.005</id>
...     </mitre>
...     <group>custom_persistence,scheduled_task,mitre_persistence,</group>
... </rule>
... '''
>>> open(p,"w").write(s[:a]+new+s[a:])
1967
>>> exit()
bash-5.2# grep -n -A 10 -B 2 '<rule id="100033">' /var/ossec/etc/rules/local_rules.xml
38-</rule>
39-
40: <rule id="100033" level="12">
41-     <if_sid>60228</if_sid>
42-     <description>Custom SOC detection: scheduled task persistence</description>
43-     <mitre>
44-         <id>T1053.005</id>
45-     </mitre>
46-     <group>custom_persistence,scheduled_task,mitre_persistence,</group>
47-</rule>
48-
49-<rule id="100032" level="12">
50-     <if_group>sysmon_event1</if_group>
bash-5.2# _
```

Figure 50. Source: 130.png

Selected evidence screenshot previously retained for the project. Refer to the visible event/rule/endpoint identifiers and cross-check against raw logs; avoid claiming causation from proximity alone.

```
</rule>
<rule id="100034" level="12">
  <if_sid>550</if_sid>
  <description>Custom SOC detection: file integrity modification</description>
  <mitre><id>T1565.001</id></mitre>
  <group>custom_fim,file_tampering,custom_detection,</group>
</rule>
</group>
bash-5.2# /var/ossec/bin/wazuh-analysisd -t
2026/09/16 08:33:33 wazuh-analysisd: WARNING: (7612): Rule ID '100034' is duplicated. Only the first
occurrence will be considered.
bash-5.2# cp /var/ossec/etc/rules/local_rules.xml.bak4 /var/ossec/etc/rules/local_rules.xml
bash-5.2# grep '100034' /var/ossec/etc/rules/local_rules.xml
bash-5.2# /var/ossec/bin/wazuh-analysisd -t
bash-5.2# cp /var/ossec/etc/rules/local_rules.xml /var/ossec/etc/rules/local_rules.xml.bak7
bash-5.2# sed -i '5d' /var/ossec/etc/rules/local_rules.xml
bash-5.2# E=$(printf '\075')
bash-5.2# printf 'rule id%"100034" level%"12">\n    <if_sid>550</if_sid>\n        <description>Custo
m SOC detection: file integrity modification</description>\n        <mitre><id>T1565.001</id></mitre>\n
    <group>custom_fim,file_tampering,custom_detection,</group>\n</rule>\n</group>\n' "$E" "$E" >> /var
/ossec/etc/rules/local_rules.xml
bash-5.2# grep -n '100034' /var/ossec/etc/rules/local_rules.xml
56:<rule id="100034" level="12">
bash-5.2# /var/ossec/bin/wazuh-analysisd -t
bash-5.2# exit
exit
ubuntu@ubuntu-server:~/wazuh-docker/single-node$ docker restart single-node-wazuh.manager-1
single-node-wazuh.manager-1
ubuntu@ubuntu-server:~/wazuh-docker/single-node$ docker exec -it single-node-wazuh.manager-1 bash
bash-5.2# grep -A5 '<command>' /var/ossec/etc/ossec.conf | grep -A4 firewall-drop
    <name>firewall-drop</name>
    <executable>firewall-drop</executable>
    <timeout_allowed>yes</timeout_allowed>
  </command>

  <command>
bash-5.2#
```

Figure 51. Source: 140.png

Selected evidence screenshot previously retained for the project. Refer to the visible event/rule/endpoint identifiers and cross-check against raw logs; avoid claiming causation from proximity alone.

```

        <location>local</location>\
        <rules_id>5712</rules_id>\
        <timeout>'"$E"'60</timeout>\
    </active-response>' /var/ossec/etc/ossec.conf
bash-5.2# sed -n '220,235p' /var/ossec/etc/ossec.conf
    <executable>netsh.exe</executable>
    <timeout_allowed>yes</timeout_allowed>
</command>

    <active-response>
        <disabled>no</disabled>
        <command>firewall-drop</command>
        <location>local</location>
        <rules_id>5712</rules_id>
        <timeout>=60</timeout>
    </active-response>
    <timeout>=60</timeout>
</active-response>

<!-- Log analysis -->
<localfile>
bash-5.2# cp /var/ossec/etc/ossec.conf.ar-backup3 /var/ossec/etc/ossec.conf
bash-5.2# /var/ossec/bin/wazuh-analysisd -t
bash-5.2# tail -12 /var/ossec/etc/ossec.conf
    <disabled>yes</disabled>
</cluster>

</ossec_config>

<ossec_config>
    <localfile>
        <log_format>syslog</log_format>
        <location>/var/ossec/logs/active-responses.log</location>
    </localfile>

</ossec_config>
bash-5.2#
```

Figure 52. Source: 142.png

Selected evidence screenshot previously retained for the project. Refer to the visible event/rule/endpoint identifiers and cross-check against raw logs; avoid claiming causation from proximity alone.

```
2026-09-16T09:44:14.360Z      INFO      memlog/store.go:124      Finished loading transaction log fil
e for '/var/lib/filebeat/registry/filebeat'. Active transaction id=2055
2026-09-16T09:44:14.361Z      INFO      [registrar]      registrar/registrar.go:109      States Load
d from registrar: 1
2026-09-16T09:44:14.361Z      INFO      [crawler]      beater/crawler.go:71      Loading Inputs: 1
2026-09-16T09:44:14.364Z      INFO      log/input.go:157      Configured paths: [/var/ossec/logs/a
lerts/alerts.json]
2026-09-16T09:44:14.364Z      INFO      [crawler]      beater/crawler.go:141      Starting input (ID:
9132358592892857476)
2026-09-16T09:44:14.364Z      INFO      [crawler]      beater/crawler.go:108      Loading and starting
Inputs completed. Enabled inputs: 1
2026-09-16T09:44:14.365Z      INFO      log/harvester.go:302      Harvester started for file: /var/osse
c/logs/alerts/alerts.json
2026-09-16T09:44:15.365Z      INFO      [publisher_pipeline_output]      pipeline/output.go:143 Conn
ecting to backoff(elasticsearch(https://wazuh.indexer:9200))
2026-09-16T09:44:15.366Z      INFO      [publisher]      pipeline/retry.go:219      retryer: send unwait
signal to consumer
2026-09-16T09:44:15.366Z      INFO      [publisher]      pipeline/retry.go:223      done
2026-09-16T09:44:15.385Z      INFO      [esclientleg]      eslegclient/connection.go:314      Attempting t
o connect to Elasticsearch version 7.10.2
2026-09-16T09:44:15.387Z      INFO      [esclientleg]      eslegclient/connection.go:314      Attempting t
o connect to Elasticsearch version 7.10.2
2026-09-16T09:44:15.388Z      INFO      template/load.go:183      Existing template will be overwritte
n, as overwrite is enabled.
2026-09-16T09:44:15.389Z      INFO      template/load.go:117      Try loading template wazuh to Elast
icsearch
2026-09-16T09:44:15.490Z      INFO      template/load.go:109      template with name 'wazuh' loaded.
2026-09-16T09:44:15.490Z      INFO      [index-management]      idxmgmt/std.go:298      Loaded index
template.
2026-09-16T09:44:15.493Z      INFO      [publisher_pipeline_output]      pipeline/output.go:151 Conn
ection to backoff(elasticsearch(https://wazuh.indexer:9200)) established
2026/09/16 09:44:16 sca: INFO: Evaluation finished for policy '/var/ossec/ruleset/sca/cis_amazon_lin
ux_2023.yml'
2026/09/16 09:44:16 sca: INFO: Security Configuration Assessment scan finished. Duration: 5 seconds.
2026/09/16 09:44:23 wazuh-syscheckd: INFO: netstat not available. Skipping port check.
2026/09/16 09:44:28 rootcheck: INFO: Ending rootcheck scan.
ubuntu@ubuntu-server:~/wazuh-docker/single-node$ _
```

Figure 53. Source: 144.png

Selected evidence screenshot previously retained for the project. Refer to the visible event/rule/endpoint identifiers and cross-check against raw logs; avoid claiming causation from proximity alone.

```
zuh-states-inventory-interfaces-wazuh.manager.
2026/09/16 09:44:12 indexer-connector: INFO: IndexerConnector initialized successfully for index: wa
zuh-states-inventory-networks-wazuh.manager.
2026/09/16 09:44:12 indexer-connector: INFO: IndexerConnector initialized successfully for index: wa
zuh-states-inventory-users-wazuh.manager.
2026/09/16 09:44:12 wazuh-modulesd:vulnerability-scanner: INFO: Vulnerability scanner module started
.
2026/09/16 09:44:12 indexer-connector: INFO: IndexerConnector initialized successfully for index: wa
zuh-states-inventory-groups-wazuh.manager.
2026/09/16 09:44:12 logger-helper: INFO: InventoryHarvesterFacade module started.
2026/09/16 09:44:12 indexer-connector: INFO: IndexerConnector initialized successfully for index: wa
zuh-states-inventory-browser-extensions-wazuh.manager.
2026/09/16 09:44:12 indexer-connector: INFO: IndexerConnector initialized successfully for index: wa
zuh-states-inventory-services-wazuh.manager.
2026/09/16 09:44:12 wazuh-modulesd:syscollector: INFO: Evaluation finished.
2026/09/16 09:44:16 sca: INFO: Evaluation finished for policy '/var/ossec/ruleset/sca/cis_amazon_lin
ux_2023.yml'
2026/09/16 09:44:16 sca: INFO: Security Configuration Assessment scan finished. Duration: 5 seconds.
2026/09/16 09:44:23 wazuh-syscheckd: INFO: netstat not available. Skipping port check.
2026/09/16 09:44:28 rootcheck: INFO: Ending rootcheck scan.
2026/09/16 10:59:50 wazuh-modulesd:syscollector: INFO: Starting evaluation.
2026/09/16 13:04:35 wazuh-modulesd:content-updater: WARNING: Offset processing failed. Triggered a s
napshot download.
2026/09/16 13:04:35 wazuh-modulesd:content-updater: WARNING: Couldn't run full content download: Err
or -1 from server: Could not resolve hostname - Response body: .
2026/09/16 13:04:35 wazuh-modulesd:content-updater: ERROR: Action for 'vulnerability_feed_manager' f
ailed: Error -1 from server: Could not resolve hostname - Response body: .
2026/09/16 13:04:43 wazuh-modulesd:syscollector: INFO: Evaluation finished.
2026/09/16 13:04:44 indexer-connector: WARNING: Failed to sync agent '002': No available server
2026/09/16 13:04:44 indexer-connector: WARNING: Failed to sync agent '002': No available server
2026/09/16 13:04:44 indexer-connector: WARNING: Failed to sync agent '002': No available server
2026/09/16 13:04:44 indexer-connector: WARNING: Failed to sync agent '002': No available server
2026/09/16 13:21:20 wazuh-analysisd: ERROR: (1226): Error reading XML file 'etc/rules/local_rules.xml'
1': XMLERR: Element 'if_matched_sid' not closed. (line 64).
2026/09/16 13:21:20 wazuh-analysisd: CRITICAL: (1220): Error loading the rules: 'etc/rules/local_rul
es.xml'.
bash-5.2# _
```

Figure 54. Source: 146.png

Selected evidence screenshot previously retained for the project. Refer to the visible event/rule/endpoint identifiers and cross-check against raw logs; avoid claiming causation from proximity alone.

```
<rule id="100035" level="14" frequency="3" timeframe="120">\
  <if_matched_sid>100034</if_matched_sid>\
  <same_field>syscheck.path</same_field>\
  <description>Custom SOC correlation:repeated file tampering on the same file</description>
n>\
  <mitre>\
    <id>T1565.001</id>\
  </mitre>\
  <group>custom_correlation,file_tampering,custom_detection,mitre_impact,</group>\
</rule>' /var/ossec/etc/rules/local_rules.xml
sed: -e expression #1, char 6: extra characters after command
bash-5.2# cp /var/ossec/etc/rules/local_rules.xml.bak8 /var/ossec/etc/rules/local_rules.xml
bash-5.2# /var/ossec/bin/wazuh-analysisd -t
bash-5.2# nl -ba /var/ossec/etc/rules/local_rules.xml | tail -20
43     <mitre>
44       <id>T1053.005</id>
45     </mitre>
46     <group>custom_persistence,scheduled_task,mitre_persistence,</group>
47   </rule>
48
49   <rule id="100032" level="12">
50     <if_group>sysmon_event1</if_group>
51     <field name="win.eventdata.commandLine" type="pcr2">(?)certutil.*(-urlcache|-decode|-e
ncode)</field>
52     <description>Suspicious certutil usage - possible LOLBin file download or encoding abuse
</description>
53     <mitre><id>T1105</id><id>T1218.011</id></mitre>
54     <group>lolbin,mitre_defense_evasion,</group>
55   </rule>
56   <rule id="100034" level="12">
57     <if_sid>550</if_sid>
58     <description>Custom SOC detection: file integrity modification</description>
59     <mitre><id>T1565.001</id></mitre>
60     <group>custom_fim,file_tampering,custom_detecion,</group>
61   </rule>
62 </group>
bash-5.2#
```

Figure 55. Source: 152.png

Selected evidence screenshot previously retained for the project. Refer to the visible event/rule/endpoint identifiers and cross-check against raw logs; avoid claiming causation from proximity alone.

```
519E017C,SHA256=8BB6FA8C283B4D92120B1EF249A9B311B0F804D4CABBE9981159976C8BE76A5E,IMPHASH=2F87CE4617F593D7148B739BBF06D059",\parentProcessGuid": "{07af6f40-290c-6aaa-eb01-000000001a00}",\parentProcessId": "1516",\parentImage": "C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe",\parentCommandLine": "C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe -EncodedCommand VuByAGkAdABlAC0ATwB1AHQACAB1AHQAIAAIAFMATwBDAC0AVAB1AHMAdAAIAA=",\currentDirectory": "C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe",\parentUser": "WINDOWS11-AGENT\\windows\\",\logonGuid": "{07af6f40-2825-6aaa-3ee7-070000000000}",\logonId": "0x7e73e",\terminalSessionId": "1",\integrityLevel": "High",\hashes": "MD5=61F649A6FDC374A6BF36C860519E017C,SHA256=8BB6FA8C283B4D92120B1EF249A9B311B0F804D4CABBE9981159976C8BE76A5E,IMPHASH=2F87CE4617F593D7148B739BBF06D059",\parentProcessGuid": "{07af6f40-290c-6aaa-eb01-000000001a00}",\parentProcessId": "1516",\parentImage": "C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe",\parentCommandLine": "C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe -EncodedCommand VuByAGkAdABlAC0ATwB1AHQACAB1AHQAIAAIAFMATwBDAC0AVAB1AHMAdAAIAA=",\currentDirectory": "C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe",\parentUser": "WINDOWS11-AGENT\\windows\\",\location": "EventChannel"}
bash-5.2# _
```

Figure 56. Source: 157.png

Selected evidence screenshot previously retained for the project. Refer to the visible event/rule/endpoint identifiers and cross-check against raw logs; avoid claiming causation from proximity alone.

```
519E017C,SHA256=8BB6FA8C283B4D92120B1EF249A9B311B0F804D4CABBE9981159976C8BE76A5E,IMPHASH=2F87CE4617F593D7148B739BBF06D059",\parentProcessGuid": "{07af6f40-290c-6aaa-eb01-000000001a00}",\parentProcessId": "1516",\parentImage": "C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe",\parentCommandLine": "C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe -EncodedCommand VuByAGkAdABlAC0ATwB1AHQACAB1AHQAIAAIAFMATwBDAC0AVAB1AHMAdAAIAA=",\currentDirectory": "C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe",\parentUser": "WINDOWS11-AGENT\\windows\\",\logonGuid": "{07af6f40-2825-6aaa-3ee7-070000000000}",\logonId": "0x7e73e",\terminalSessionId": "1",\integrityLevel": "High",\hashes": "MD5=61F649A6FDC374A6BF36C860519E017C,SHA256=8BB6FA8C283B4D92120B1EF249A9B311B0F804D4CABBE9981159976C8BE76A5E,IMPHASH=2F87CE4617F593D7148B739BBF06D059",\parentProcessGuid": "{07af6f40-290c-6aaa-eb01-000000001a00}",\parentProcessId": "1516",\parentImage": "C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe",\parentCommandLine": "C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe -EncodedCommand VuByAGkAdABlAC0ATwB1AHQACAB1AHQAIAAIAFMATwBDAC0AVAB1AHMAdAAIAA=",\currentDirectory": "C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe",\parentUser": "WINDOWS11-AGENT\\windows\\",\location": "EventChannel"}
bash-5.2#
```

Figure 57. Source: 158.png

Selected evidence screenshot previously retained for the project. Refer to the visible event/rule/endpoint identifiers and cross-check against raw logs; avoid claiming causation from proximity alone.

```
tion>\r\n      <WaitTimeout>PT1H</WaitTimeout>\r\n      <StopOnIdleEnd>true</StopOnIdleEnd>\r\n      <RestartOnIdle>false</RestartOnIdle>\r\n      </IdleSettings>\r\n      <AllowStartOnDemand>true</AllowStartOnDemand>\r\n      <Enabled>true</Enabled>\r\n      <Hidden>false</Hidden>\r\n      <RunOnlyIfIdle>false</RunOnlyIfIdle>\r\n      <WakeToRun>false</WakeToRun>\r\n      <ExecutionTimeLimit>PT72H</ExecutionTimeLimit>\r\n      <Priority>7</Priority>\r\n      </Settings>\r\n      <Actions Context=\\\"Author\\\">\r\n      <Exec>\r\n      <Command>cmd.exe</Command>\r\n      <Arguments>/c echo SOC-Test</Arguments>\r\n      </Exec>\r\n      </Actions>\r\n      <Principals>\r\n      <Principal id=\\\"Author\\\">\r\n      <UserId>WINDOWS11-AGENT\\windows</UserId>\r\n      <LogonType>InteractiveToken</LogonType>\r\n      <RunLevel>LeastPrivilege</RunLevel>\r\n      </Principal>\r\n      </Principals>\r\n      </Task>\r\n      <Other Information>\r\n      <ProcessCreationTime> \t\t\t7318349394477690\r\n      <ClientProcessId> \t\t\t120\r\n      <ParentProcessId> \t\t\t1516\r\n      <FQDN> \t\t\t0\r\n      <eventdata>:{\"subjectUserId\":\"S-1-5-21-2849226174-801770090-1247992745-1000\",\"subjectUserName\":\"windows\",\"subjectDomainName\":\"WINDOWS11-AGENT\",\"subjectLogonId\":\"0x7e73e\",\"taskName\":\"\\\\\\\\\\\\SOC-Test-Task-3\",\"taskContent\":\"&lt;?xml version=\\\"1.0\\\" encoding=\\\"UTF-16\\\"&gt;&lt;Task version=\\\"1.2\\\" xmlns=\\\"http://schemas.microsoft.com/windows/2004/02/mit/task\\\"&gt;&lt;RegistrationInfo&gt;&lt;Date&gt;2026-09-16T11:23:31&lt;/Date&gt;&lt;Author&gt;WINDOWS11-AGENT\\\\\\\\windows&lt;/Author&gt;&lt;URI&gt;&lt;/URI&gt;&lt;/RegistrationInfo&gt;&lt;Triggers&gt;&lt;TimeTrigger&gt;&lt;StartBoundary&gt;2026-09-16T23:59:00&lt;/StartBoundary&gt;&lt;Enabled&gt;true&lt;/Enabled&gt;&lt;TimeTrigger&gt;&lt;/Triggers&gt;&lt;Settings&gt;&lt;MultipleInstancesPolicy&gt;IgnoreNew&lt;/MultipleInstancesPolicy&gt;&lt;DisallowStartIfOnBatteries&gt;true&lt;/DisallowStartIfOnBatteries&gt;&lt;StopIfGoingOnBatteries&gt;true&lt;/StopIfGoingOnBatteries&gt;&lt;AllowHardTerminate&gt;true&lt;/AllowHardTerminate&gt;&lt;StartWhenAvailable&gt;false&lt;/StartWhenAvailable&gt;&lt;RunOnlyIfNetworkAvailable&gt;false&lt;/RunOnlyIfNetworkAvailable&gt;&lt;IdleSettings&gt;&lt;Duration&gt;PT10M&lt;/Duration&gt;&lt;WaitTimeout&gt;PT1H&lt;/WaitTimeout&gt;&lt;StopOnIdleEnd&gt;true&lt;/StopOnIdleEnd&gt;&lt;RestartOnIdle&gt;false&lt;/RestartOnIdle&gt;&lt;/IdleSettings&gt;&lt;AllowStartOnDemand&gt;true&lt;/AllowStartOnDemand&gt;&lt;Enabled&gt;true&lt;/Enabled&gt;&lt;Hidden&gt;false&lt;/Hidden&gt;&lt;RunOnlyIfIdle&gt;false&lt;/RunOnlyIfIdle&gt;&lt;WakeToRun&gt;false&lt;/WakeToRun&gt;&lt;ExecutionTimeLimit&gt;PT72H&lt;/ExecutionTimeLimit&gt;&lt;Priority&gt;7&lt;/Priority&gt;&lt;/Settings&gt;&lt;Actions Context=\\\"Author\\\"&gt;&lt;Exec&gt;&lt;Command&gt;cmd.exe&lt;/Command&gt;&lt;Arguments&gt;/c echo SOC-Test&lt;/Arguments&gt;&lt;/Exec&gt;&lt;/Actions&gt;&lt;Principals&gt;&lt;Principal id=\\\"Author\\\"&gt;&lt;UserId&gt;WINDOWS11-AGENT\\\\\\\\windows&lt;/UserId&gt;&lt;LogonType&gt;InteractiveToken&lt;/LogonType&gt;&lt;RunLevel&gt;LeastPrivilege&lt;/RunLevel&gt;&lt;/Principal&gt;&lt;/Principals&gt;&lt;/Task&gt;&lt;clientProcessStartKey>\"7318349394477690\",\"clientProcessId>\"120\",\"parentProcessId>\"1516\",\"rpcCallClientLocality>\"0\",\"fQDN>\"windows11-agent\"}}&lt;/location>\"EventChannel\"}
```

bash-5.2# _

Figure 58. Source: 159.png

Selected evidence screenshot previously retained for the project. Refer to the visible event/rule/endpoint identifiers and cross-check against raw logs; avoid claiming causation from proximity alone.

```
nOnlyIfIdle&gt;;false&lt;/RunOnlyIfIdle&gt;      &lt;/WakeToRun&gt;;false&lt;/WakeToRun&gt;      &lt;/Ex
ecutionTimeLimit&gt;;PT72H&lt;/ExecutionTimeLimit&gt;      &lt;/Priority&gt;;7&lt;/Priority&gt;      &lt;
/Settings&gt;      &lt;/Actions Context=\\\"Author\\\"&gt;      &lt;/Exec&gt;      &lt;/Command&gt;;cmd.
exe&lt;/Command&gt;      &lt;/Arguments&gt;;c echo SOC-Test&lt;/Arguments&gt;      &lt;/Exec&gt;
&lt;/Actions&gt;      &lt;/Principals&gt;      &lt;/Principal id=\\\"Author\\\"&gt;      &lt;/UserId&g
t;WINDOWS11-AGENT\\\"\\windows&lt;/UserId&gt;      &lt;/LogonType&gt;;InteractiveToken&lt;/LogonType&g
t;      &lt;/RunLevel&gt;;LeastPrivilege&lt;/RunLevel&gt;      &lt;/Principal&gt;      &lt;/Principals
&gt;      &lt;/Task&gt;;\"clientProcessStartKey\":\"7318349394477690\",\"clientProcessId\":\"120\",\"parentProce
ssId\":\"1516\",\"rpcCallClientLocality\":\"0\",\"fqdn\":\"windows11-agent\"}},\"location\":\"EventChannel1}
bash-5.2# grep \"id\":\"100034\" /var/ossec/logs/alerts/alerts.json | head -1
{\"timestamp\":\"2026-09-16T08:43:25.243+0000\",\"rule\":{\"level\":12,\"description\":\"Custom SOC detection:
file integrity modification\",\"id\":\"100034\",\"mitre\":{\"id\":\"T1565.001\"},\"tactic\":[\"Impact\"],\"techniqu
e\":[\"Stored Data Manipulation\"]},\"firedtimes\":1,\"mail\":true,\"groups\":[\"local\",\"windows\",\"recon\",\"cus
tom_fim\",\"file_tampering\",\"custom_detecion\"]},\"agent\":{\"id\":\"001\",\"name\":\"windows11-agent\",\"ip\":\"192
.168.1.7\"},\"manager\":{\"name\":\"wazuh.manager\"},\"id\":\"1789548205.1948222\",\"full_log\":{\"file\":\"c:\\soc-fil
m-test\\realtime-test.txt\" modified\\nMode: realtime\\nChanged attributes: mtime,md5,sha1,sha256\\nOld
modification time was: '1789542015', now it is '1789548129'\\nOld md5sum was: '2c8bdac1b0aa61bcca33ef
812b1781d8'\\nNew md5sum is : '195c4b274bcb29f5a22f6c70e21ba4e2'\\nOld sha1sum was: '288ea90ca5aca3f82
0220311fbb63e5ca0af4f40'\\nNew sha1sum is : 'a0ccc29686114b69ab2f30d92723e643b12ea2ec'\\nOld sha256sum
was: 'ae9a572d0ae6af5d81c33b299e74656fa834fe5eb3a222f5b131ba55de0d94ac'\\nNew sha256sum is : '80865d
2e442819f25a098e15263c3e417a97b4569a2545f0ad1707aee7357b27'\\n\",\"syscheck\":{\"path\":\"c:\\soc-fim-test\\
\\realtime-test.txt\", \"mode\":\"realtime\", \"size_after\":\"40\", \"win_perm_after\":{\"name\":\"Administrators\",
\"allowed\":[\"DELETE\",\"READ_CONTROL\",\"WRITE_DAC\",\"WRITE_OWNER\",\"SYNCHRONIZE\",\"READ_DATA\",\"WRITE_DATA\",
\"APPEND_DATA\",\"READ_EA\",\"WRITE_EA\",\"EXECUTE\",\"READ_ATTRIBUTES\",\"WRITE_ATTRIBUTES\"]}, {\"name\":\"SYSTEM\",
\"allowed\":[\"DELETE\",\"READ_CONTROL\",\"WRITE_DAC\",\"WRITE_OWNER\",\"SYNCHRONIZE\",\"READ_DATA\",\"WRITE_DATA\",
\"APPEND_DATA\",\"READ_EA\",\"WRITE_EA\",\"EXECUTE\",\"READ_ATTRIBUTES\",\"WRITE_ATTRIBUTES\"]}, {\"name\":\"Users\",
\"allowed\":[\"READ_CONTROL\",\"SYNCHRONIZE\",\"READ_DATA\",\"READ_EA\",\"EXECUTE\",\"READ_ATTRIBUTES\"]}, {\"name\":
\"Authenticated Users\", \"allowed\":[\"DELETE\",\"READ_CONTROL\",\"SYNCHRONIZE\",\"READ_DATA\",\"WRITE_DATA\", \"APP
END_DATA\",\"READ_EA\",\"WRITE_EA\",\"EXECUTE\",\"READ_ATTRIBUTES\",\"WRITE_ATTRIBUTES\"]}}, \"uid_after\":\"S-1-5-
32-544\", \"md5_before\":\"2c8bdac1b0aa61bcca33ef812b1781d8\", \"md5_after\":\"195c4b274bcb29f5a22f6c70e21ba4e
2\", \"sha1_before\":\"288ea90ca5aca3f820220311fbb63e5ca0af4f40\", \"sha1_after\":\"a0ccc29686114b69ab2f30d927
23e643b12ea2ec\", \"sha256_before\":\"ae9a572d0ae6af5d81c33b299e74656fa834fe5eb3a222f5b131ba55de0d94ac\",
\"sha256_after\":\"80865d2e442819f25a098e15263c3e417a97b4569a2545f0ad1707aee7357b27\", \"attrs_after\":[\"ARC
HIVE\"], \"uname_after\":\"Administrators\", \"mtime_before\":\"2026-09-16T07:00:15\", \"mtime_after\":\"2026-09-16
T08:42:09\", \"changed_attributes\":[\"mtime\", \"md5\", \"sha1\", \"sha256\"], \"event\":\"modified\"}, \"decoder\":{\"name
\":\"syscheck_integrity_changed\"}, \"location\":\"syscheck\"}
bash-5.2# _
```

Figure 59. Source: 160.png

Selected evidence screenshot previously retained for the project. Refer to the visible event/rule/endpoint identifiers and cross-check against raw logs; avoid claiming causation from proximity alone.

```
END_DATA", "READ_EA", "WRITE_EA", "EXECUTE", "READ_ATTRIBUTES", "WRITE_ATTRIBUTES"]}], "uid_after": "S-1-5-32-544", "md5_before": "2c8bdac1b0aa61bcca33ef812b1781d8", "md5_after": "195c4b274bcb29f5a22f6c70e21ba4e2", "sha1_before": "288ea90ca5aca3f820220311fbb63e5ca0af4f40", "sha1_after": "a0ccc29686114b69ab2f30d92723e643b12ea2ec", "sha256_before": "ae9a572d0ae6af5d81c33b299e74656fa834fe5eb3a222f5b131ba55de0d94ac", "sha256_after": "80865d2e442819f25a098e15263c3e417a97b4569a2545f0ad1707aee7357b27", "attrs_after": ["ARCHIVE"], "uname_after": "Administrators", "mtime_before": "2026-09-16T07:00:15", "mtime_after": "2026-09-16T08:42:09", "changed_attributes": ["mtime", "md5", "sha1", "sha256"], "event": "modified", "decoder": {"name": "syscheck_integrity_changed"}, "location": "syscheck"}
bash-5.2# grep "id": "100034" /var/ossec/logs/alerts/alerts.json | tail -1
{"timestamp": "2026-09-16T16:10:52.685+0000", "rule": {"level": 12, "description": "Custom SOC detection: file integrity modification", "id": "100034", "mitre": {"id": "T1565.001"}, "tactic": ["Impact"], "technique": ["Stored Data Manipulation"]}, "firedtimes": 1, "mail": true, "groups": ["local", "windows", "recon", "custom_fim", "file_tampering", "custom_deteccion"]}, "agent": {"id": "001", "name": "windows11-agent", "ip": "192.168.1.7"}, "manager": {"name": "wazuh.manager", "id": "1789575052.2241613", "full_log": "File 'c:\\soc-fim-test\\chain-test.txt' modified\\nMode: realtime\\nChanged attributes: size, mtime, md5, sha1, sha256\\nSize changed from '22' to '53'\\nOld modification time was: '1789574799', now it is '1789575052'\\nOld md5sum was: '901bdd5381549a29228350d2f5e6e49f'\\nNew md5sum is: '4f623d77d42afd948189c6613e197b5a'\\nOld sha1sum was: 'b2879c73dc9ac63ca4b96954b41f21698caddcdd'\\nNew sha1sum is: '77a150e23c6015b2acedbfc4649581590a712924'\\nOld sha256sum was: 'a5e7ec50623c1f9ad3341e0dfb36ecd54d5a1c00464fb456de4aaa61dda334d1'\\nNew sha256sum is: 'bb5a08b9ffc86e65e910475ea9d750d5060a62f9f2f06bea26a46b6c6dda1fc7'\\n", "syscheck": {"path": "c:\\soc-fim-test\\chain-test.txt", "mode": "realtime", "size_before": "22", "size_after": "53", "win_perm_after": [{"name": "Administrators", "allowed": ["DELETE", "READ_CONTROL", "WRITE_DAC", "WRITE_OWNER", "SYNCHRONIZE", "READ_DATA", "WRITE_DATA", "APPEND_DATA", "READ_EA", "WRITE_EA", "EXECUTE", "READ_ATTRIBUTES", "WRITE_ATTRIBUTES"]}, {"name": "Users", "allowed": ["READ_CONTROL", "SYNCHRONIZE", "READ_DATA", "READ_EA", "EXECUTE", "READ_ATTRIBUTES"]}, {"name": "Authenticated Users", "allowed": ["DELETE", "READ_CONTROL", "SYNCHRONIZE", "READ_DATA", "WRITE_DATA", "APPEND_DATA", "READ_EA", "WRITE_EA", "EXECUTE", "READ_ATTRIBUTES", "WRITE_ATTRIBUTES"]}], "uid_after": "S-1-5-32-544", "md5_before": "901bdd5381549a29228350d2f5e6e49f", "md5_after": "4f623d77d42afd948189c6613e197b5a", "sha1_before": "b2879c73dc9ac63ca4b96954b41f21698caddcdd", "sha1_after": "77a150e23c6015b2acedbfc4649581590a712924", "sha256_before": "a5e7ec50623c1f9ad3341e0dfb36ecd54d5a1c00464fb456de4aaa61dda334d1", "sha256_after": "bb5a08b9ffc86e65e910475ea9d750d5060a62f9f2f06bea26a46b6c6dda1fc7", "attrs_after": ["ARCHIVE"], "uname_after": "Administrators", "mtime_before": "2026-09-16T16:06:39", "mtime_after": "2026-09-16T16:10:52", "changed_attributes": ["size", "mtime", "md5", "sha1", "sha256"], "event": "modified", "decoder": {"name": "syscheck_integrity_changed"}, "location": "syscheck"}
bash-5.2# _
```

Figure 60. Source: 161.png

Selected evidence screenshot previously retained for the project. Refer to the visible event/rule/endpoint identifiers and cross-check against raw logs; avoid claiming causation from proximity alone.

```
976C8BE76A5E, IMPHASH=2F87CE4617F593D7148B739BBF06D059\", \"parentProcessGuid\": \"\\{07af6f40-290c-6aaa-eb01-000000001a00}\\\", \"parentProcessId\": \"1516\", \"parentImage\": \"C:\\\\Windows\\\\System32\\\\WindowsPowerShell\\\\v1.0\\\\powershell.exe\", \"parentCommandLine\": \"C:\\\\Windows\\\\System32\\\\WindowsPowerShell\\\\v1.0\\\\powershell.exe\", \"parentUser\": \"\\\"WINDOWS11-AGENT\\\\windows\\\\\"\", \"decoder\": {\"name\": \"windows_eventchannel\", \"data\": {\"win\": {\"system\": {\"providerName\": \"Microsoft-Windows-Sysmon\", \"providerGuid\": \"{5770385f-c22a-43e0-bf4c-06f5698ffbd9}\", \"eventID\": \"1\", \"version\": \"5\", \"level\": \"4\", \"task\": \"1\", \"opcode\": \"0\", \"keywords\": \"0x8000000000000000\", \"systemTime\": \"2026-09-16T16:06:29.79886122\", \"eventRecordID\": \"47277\", \"processID\": \"3436\", \"threadID\": \"1328\", \"channel\": \"Microsoft-Windows-Sysmon/Operational\", \"computer\": \"windows11-agent\", \"severityValue\": \"INFORMATION\", \"message\": \"\\\"Process Create:\\\\\\\\RuleName: -\\\\\\\\UtcTime: 2026-09-16 16:06:29.786\\\\\\\\ProcessGuid: {07af6f40-be85-6aaa-7106-000000001a00}\\\\\\\\ProcessId: 9768\\\\\\\\Image: C:\\\\Windows\\\\System32\\\\WindowsPowerShell\\\\v1.0\\\\powershell.exe\\\\\\\\FileVersion: 10.0.26100.9278 (WinBuild.160101.0800)\\\\\\\\Description: Windows PowerShell\\\\\\\\Product: Microsoft@ Windows@ Operating System\\\\\\\\Company: Microsoft Corporation\\\\\\\\OriginalFileName: PowerShell.EXE\\\\\\\\CommandLine: \\\\\"C:\\\\WINDOWS\\\\System32\\\\WindowsPowerShell\\\\v1.0\\\\powershell.exe\\\\\\\\ -EncodedCommand VwByAGKAdABlAC0ATwB1AHQAcAB1AHQAIAAIAFMATwBDAC0AQwBoAGEAaQBuaCOAVAB1AHMAAAIAA==\\\\\\\\CurrentDirectory: C:\\\\WINDOWS\\\\system32\\\\\\\\User: WINDOWS11-AGENT\\\\windows\\\\LogonGuid: {07af6f40-2825-6aaa-3ee7-070000000000}\\\\\\\\LogonId: 0x7E73E\\\\\\\\TerminalSessionId: 1\\\\\\\\IntegrityLevel: High\\\\\\\\Hashes: MD5=61F649A6FDC374A6BF36C860519E017C, SHA256=8BB6FA8C283B4D92120B1EF249A9B311B0F804D4CABBE9981159976C8BE76A5E, IMPHASH=2F87CE4617F593D7148B739BBF06D059\\\\\\\\ParentProcessGuid: {07af6f40-290c-6aaa-eb01-000000001a00}\\\\\\\\ParentProcessId: 1516\\\\\\\\ParentImage: C:\\\\Windows\\\\System32\\\\WindowsPowerShell\\\\v1.0\\\\powershell.exe\\\\\\\\ParentCommandLine: \\\\\"C:\\\\Windows\\\\System32\\\\WindowsPowerShell\\\\v1.0\\\\powershell.exe\\\\\\\\\" \\\\"\\\\\\\\ParentUser: WINDOWS11-AGENT\\\\windows\\\\\", \"eventdata\": {\"utcTime\": \"2026-09-16 16:06:29.786\", \"processGuid\": \"{07af6f40-be85-6aaa-7106-000000001a00}\", \"processId\": \"9768\", \"image\": \"C:\\\\Windows\\\\System32\\\\WindowsPowerShell\\\\v1.0\\\\powershell.exe\", \"fileVersion\": \"10.0.26100.9278 (WinBuild.160101.0800)\", \"description\": \"Windows PowerShell\", \"product\": \"Microsoft@ Windows@ Operating System\", \"company\": \"Microsoft Corporation\", \"originalFileName\": \"PowerShell.EXE\", \"commandLine\": \"\\\"C:\\\\WINDOWS\\\\System32\\\\WindowsPowerShell\\\\v1.0\\\\powershell.exe\\\\\\\\ -EncodedCommand VwByAGKAdABlAC0ATwB1AHQAcAB1AHQAIAAIAFMATwBDAC0AQwBoAGEAaQBuaCOAVAB1AHMAAAIAA==\\\\\\\\currentDirectory: C:\\\\WINDOWS\\\\system32\\\\\\\\, \"user\": \"WINDOWS11-AGENT\\\\windows\", \"logonGuid\": \"{07af6f40-2825-6aaa-3ee7-070000000000}\", \"logonId\": \"0x7e73e\", \"terminalSessionId\": \"1\", \"integrityLevel\": \"High\", \"hashes\": \"MD5=61F649A6FDC374A6BF36C860519E017C, SHA256=8BB6FA8C283B4D92120B1EF249A9B311B0F804D4CABBE9981159976C8BE76A5E, IMPHASH=2F87CE4617F593D7148B739BBF06D059\", \"parentProcessGuid\": \"{07af6f40-290c-6aaa-eb01-000000001a00}\", \"parentProcessId\": \"1516\", \"parentImage\": \"C:\\\\Windows\\\\System32\\\\WindowsPowerShell\\\\v1.0\\\\powershell.exe\", \"parentCommandLine\": \"\\\"C:\\\\Windows\\\\System32\\\\WindowsPowerShell\\\\v1.0\\\\powershell.exe\\\\\\\\\", \"parentUser\": \"WINDOWS11-AGENT\\\\windows\"}}, \"location\": \"EventChannel\"}
bash-5.2# _
```

Figure 61. Source: 162.png

Selected evidence screenshot previously retained for the project. Refer to the visible event/rule/endpoint identifiers and cross-check against raw logs; avoid claiming causation from proximity alone.

```
n>\r\n      <WaitTimeout>PT1H</WaitTimeout>\r\n      <StopOnIdleEnd>true</StopOnIdleEnd>\r\n      <RestartOnIdle>false</RestartOnIdle>\r\n      </IdleSettings>\r\n      <AllowStartOnDemand>true</AllowStartOnDemand>\r\n      <Enabled>true</Enabled>\r\n      <Hidden>false</Hidden>\r\n      <RunOnlyIfIdle>false</RunOnlyIfIdle>\r\n      <WakeToRun>false</WakeToRun>\r\n      <ExecutionTimeLimit>PT72H</ExecutionTimeLimit>\r\n      <Priority>7</Priority>\r\n      </Settings>\r\n      <Actions Context=\\\"Author\\\">\r\n      <Command>cmd.exe</Command>\r\n      <Arguments>/c echo SOC-Chain</Arguments>\r\n      </Exec>\r\n      </Actions>\r\n      <Principals>\r\n      <Principal id=\\\"Author\\\">\r\n      <UserId>WINDOWS11-AGENT\\windows</UserId>\r\n      <LogonType>InteractiveToken</LogonType>\r\n      <RunLevel>LeastPrivilege</RunLevel>\r\n      </Principal>\r\n      </Principals>\r\n      </Task>\r\n      <Other Information>\r\n      <CreationTime> \t\t7318349394478704\r\n      <ClientProcessId> \t\t\t5676\r\n      <ParentProcessId> \t\t\t1516\r\n      <FQDN> \t\t\t0\r\n      <eventdata>{\r\n      <subjectUserSid>\"S-1-5-21-2849226174-801770090-124799274-5-1000\",\r\n      <subjectUserName>\"windows\",\r\n      <subjectDomainName>\"WINDOWS11-AGENT\",\r\n      <subjectLogonId>\"0x7e73e\",\r\n      <taskName>\"\\\\\\\\SOC-Chain-Test\",\r\n      <taskContent>\"&lt;?xml version=\\\"1.0\\\" encoding=\\\"UTF-16\\\"&gt;&lt;Task version=\\\"1.2\\\" xmlns=\\\"http://schemas.microsoft.com/windows/2004/02/mit/task\\\"&gt;&lt;RegistrationInfo&gt;&lt;Date&gt;2026-09-16T21:36:20&lt;/Date&gt;&lt;Author&gt;WINDOWS11-AGENT\\\\\\\\windows&lt;/Author&gt;&lt;URI&gt;\\\\\\\\\\\\\\\\SOC-Chain-Test&lt;/URI&gt;&lt;/RegistrationInfo&gt;&lt;Triggers&gt;&lt;TimeTrigger&gt;&lt;StartBoundary&gt;2026-09-16T23:59:00&lt;/StartBoundary&gt;&lt;Enabled&gt;true&lt;/Enabled&gt;&lt;/TimeTrigger&gt;&lt;/Triggers&gt;&lt;Settings&gt;&lt;MultipleInstancesPolicy&gt;IgnoreNew&lt;/MultipleInstancesPolicy&gt;&lt;DisallowStartIfOnBatteries&gt;true&lt;/DisallowStartIfOnBatteries&gt;&lt;StopIfGoingOnBatteries&gt;true&lt;/StopIfGoingOnBatteries&gt;&lt;AllowHardTerminate&gt;true&lt;/AllowHardTerminate&gt;&lt;StartWhenAvailable&gt;false&lt;/StartWhenAvailable&gt;&lt;RunOnlyIfNetworkAvailable&gt;false&lt;/RunOnlyIfNetworkAvailable&gt;&lt;IdleSettings&gt;&lt;Duration&gt;PT10M&lt;/Duration&gt;&lt;WaitTimeout&gt;PT1H&lt;/WaitTimeout&gt;&lt;StopOnIdleEnd&gt;true&lt;/StopOnIdleEnd&gt;&lt;RestartOnIdle&gt;false&lt;/RestartOnIdle&gt;&lt;/IdleSettings&gt;&lt;AllowStartOnDemand&gt;true&lt;/AllowStartOnDemand&gt;&lt;Enabled&gt;true&lt;/Enabled&gt;&lt;Hidden&gt;false&lt;/Hidden&gt;&lt;RunOnlyIfIdle&gt;false&lt;/RunOnlyIfIdle&gt;&lt;WakeToRun&gt;false&lt;/WakeToRun&gt;&lt;ExecutionTimeLimit&gt;PT72H&lt;/ExecutionTimeLimit&gt;&lt;Priority&gt;7&lt;/Priority&gt;&lt;/Settings&gt;&lt;Actions Context=\\\"Author\\\"&gt;&lt;Exec&gt;&lt;Command&gt;cmd.exe&lt;/Command&gt;&lt;Arguments&gt;/c echo SOC-Chain&lt;/Arguments&gt;&lt;/Exec&gt;&lt;/Actions&gt;&lt;Principals&gt;&lt;Principal id=\\\"Author\\\"&gt;&lt;UserId&gt;WINDOWS11-AGENT\\\\\\\\windows&lt;/UserId&gt;&lt;LogonType&gt;InteractiveToken&lt;/LogonType&gt;&lt;RunLevel&gt;LeastPrivilege&lt;/RunLevel&gt;&lt;/Principal&gt;&lt;/Principals&gt;&lt;/Task&gt;\",\r\n      <clientProcessStartKey>\"7318349394478704\",\r\n      <clientProcessId>\"5676\",\r\n      <parentProcessId>\"1516\",\r\n      <rpcCallClientLocality>\"0\",\r\n      <FQDN>\"windows11-agent\"&lt;/FQDN>&lt;/eventdata>\r\n    }\r\n  }\r\n  bash-5.2#
```

Figure 62. Source: 166.png

Selected evidence screenshot previously retained for the project. Refer to the visible event/rule/endpoint identifiers and cross-check against raw logs; avoid claiming causation from proximity alone.

```
<Interval>PT15M</Interval> <StopAtDurationEnd>false</StopAtDurationEnd> <Repetition> </TimeTrigger> </Triggers> <Actions>
Context=\\\"Author\\\"> <ComHandler> <ClassId>{AF28889E-B0FD-494C-9FB0-D3
B058A96B34}</ClassId> <Data> <CDATA[CreativeId:18=128000000001615609;Courtesy
OverrideRuleName:4=none;]> </Data> <ComHandler> </Actions> <Regi
strationInfo> <URI>\\\"SoftLanding\\\"S-1-5-21-2849226174-801770090-1247992745-1000\\\"
SoftLandingDeferralTask-{43babb81-1103-479d-8cc8-f8961c0fe444}</URI> <SecurityDescrip
tor>D:P(A;FA;;;SY)(A;CI;0x80010000;;;WD)(A;FA;;;S-1-5-21-2849226174-801770090-1247992745-1000)&
lt;/SecurityDescriptor> </RegistrationInfo> </Task>\",
  \"clientProcessStartKey\": \"7318349394478742\",
  \"clientProcessId\": \"5832\",
  \"parentProcessId\": \"1008\",
  \"rpcCallClientLocality\": \"0\",
  \"fqdn\": \"windows11-agent\"
}
}
}
}
ubuntu@ubuntuuser:~/wazuh-docker/single-node$ docker exec single-node-wazuh.manager-1 sh -c \"grep
-E '100031|100033|100034' /var/ossec/logs/alerts/alerts.json\" | jq -r '[.timestamp, .agent.name, .ru
le.id, .rule.description, .data.win.eventdata.taskName] | @tsv'
2026-09-17T13:37:51.337+0000 windows11-agent 100033 Custom SOC detection: scheduled task persist
ence \\\"SoftLanding\\\"S-1-5-21-2849226174-801770090-1247992745-1000\\\"SoftLandingTriggerTask-1
28000000001615609-render-{78c2b1be-0f94-4775-8acd-2679b0cc435d}
2026-09-17T13:37:51.518+0000 windows11-agent 100033 Custom SOC detection: scheduled task persist
ence \\\"SoftLanding\\\"S-1-5-21-2849226174-801770090-1247992745-1000\\\"SoftLandingTriggerTask-1
28000000001615609-render-{0347648b-7490-4d12-9dfc-96765c2f965e}
2026-09-17T13:37:51.758+0000 windows11-agent 100033 Custom SOC detection: scheduled task persist
ence \\\"SoftLanding\\\"S-1-5-21-2849226174-801770090-1247992745-1000\\\"SoftLandingDeferralTask-
{6e10c2be-4c32-4dc1-820a-0e0083957098}
2026-09-17T13:38:03.744+0000 windows11-agent 100033 Custom SOC detection: scheduled task persist
ence \\\"SoftLanding\\\"S-1-5-21-2849226174-801770090-1247992745-1000\\\"SoftLandingTriggerTask-1
28000000001615609-render-{d748808b-f2f7-45cd-84b2-b1dc4df639f8}
2026-09-17T13:38:03.849+0000 windows11-agent 100033 Custom SOC detection: scheduled task persist
ence \\\"SoftLanding\\\"S-1-5-21-2849226174-801770090-1247992745-1000\\\"SoftLandingDeferralTask-
{43babb81-1103-479d-8cc8-f8961c0fe444}
ubuntu@ubuntuuser:~/wazuh-docker/single-node$
```

Figure 63. Source: 167.png

Selected evidence screenshot previously retained for the project. Refer to the visible event/rule/endpoint identifiers and cross-check against raw logs; avoid claiming causation from proximity alone.

```
[229650.685073] rcu: #0-...! (1 ticks this GP) idle=95f/0/0x1 softirq=3680504/3680504 fqs=0
[229650.690566] rcu: rcu_sched kthread timer wakeup didn't happen for 39497 jiffies! g4984313 f0x0 R
CU_GP_WAIT_FQS(5) ->state=0x402
[229650.692107] rcu: #Possible timer handling issue on cpu=0 timer-softirq=3756447
[229650.692771] rcu: rcu_sched kthread starved for 39498 jiffies! g4984313 f0x0 RCU_GP_WAIT_FQS(5) -
>state=0x402 ->cpu=0
[229650.694147] rcu: #Unless rcu_sched kthread gets sufficient CPU time, OOM is now expected behavio
r.
[229650.695686] rcu: RCU grace-period kthread stack dump:
[229650.712674] rcu: Stack dump where RCU GP kthread last ran:
[229650.727096] rcu: INFO: rcu_sched self-detected stall on CPU
[229650.728599] rcu: #0-...! (2 ticks this GP) idle=95f/0/0x1 softirq=3680504/3680504 fqs=0
[229650.730139] rcu: rcu_sched kthread starved for 1139025 jiffies! g4984313 f0x0 RCU_GP_WAIT_FQS(5)
->state=0x0 ->cpu=0
[229650.733372] rcu: #Unless rcu_sched kthread gets sufficient CPU time, OOM is now expected behavio
r.
[229650.736554] rcu: RCU grace-period kthread stack dump:
[229650.739451] rcu: Stack dump where RCU GP kthread last ran:
[229650.744532] systemd[1]: systemd-networkd.service: Watchdog timeout (limit 3min)!
[229650.745795] systemd[1]: snapd.service: Watchdog timeout (limit 5min)!
[229650.796187] systemd[1]: systemd-udevd.service: Watchdog timeout (limit 3min)!
[229650.797193] systemd[1]: systemd-logind.service: Watchdog timeout (limit 3min)!

ubuntu@ubuntu-server:~/wazuh-docker/single-node$ docker exec single-node-wazuh.manager-1 sh -c "grep
-A 8 -B 2 '100033' /var/ossec/etc/rules/local_rules.xml"
</rule>

<rule id="100033" level="12">
  <if_sid>60228</if_sid>
  <description>Custom SOC detection: scheduled task persistence</description>
  <mitre>
    <id>T1053.005</id>
  </mitre>
  <group>custom_persistence,scheduled_task,mitre_persistence,</group>
</rule>

ubuntu@ubuntu-server:~/wazuh-docker/single-node$
```

Figure 64. Source: 168.png

Selected evidence screenshot previously retained for the project. Refer to the visible event/rule/endpoint identifiers and cross-check against raw logs; avoid claiming causation from proximity alone.

```
>state=0x402 ->cpu=0
[229650.694147] rcu: #Unless rcu_sched kthread gets sufficient CPU time, OOM is now expected behavior.
[229650.695686] rcu: RCU grace-period kthread stack dump:
[229650.712674] rcu: Stack dump where RCU GP kthread last ran:
[229650.727096] rcu: INFO: rcu_sched self-detected stall on CPU
[229650.728599] rcu: #0-...! (2 ticks this GP) idle=95f/0/0x1 softirq=3680504/3680504 fqs=0
[229650.730139] rcu: rcu_sched kthread starved for 1139025 jiffies! g4984313 f0x0 RCU_GP_WAIT_FQS(5)
->state=0x0 ->cpu=0
[229650.733372] rcu: #Unless rcu_sched kthread gets sufficient CPU time, OOM is now expected behavior.
[229650.736554] rcu: RCU grace-period kthread stack dump:
[229650.739451] rcu: Stack dump where RCU GP kthread last ran:
[229650.744532] systemd[1]: systemd-networkd.service: Watchdog timeout (limit 3min)!
[229650.745795] systemd[1]: snapd.service: Watchdog timeout (limit 5min)!
[229650.796187] systemd[1]: systemd-udevd.service: Watchdog timeout (limit 3min)!
[229650.797193] systemd[1]: systemd-logind.service: Watchdog timeout (limit 3min)!

ubuntu@ubuntu-server:~/wazuh-docker/single-node$ docker exec single-node-wazuh.manager-1 sh -c "grep
-A 8 -B 2 '100033' /var/ossec/etc/rules/local_rules.xml"
</rule>

  <rule id="100033" level="12">
    <if_sid>60228</if_sid>
    <description>Custom SOC detection: scheduled task persistence</description>
    <mitre>
      <id>T1053.005</id>
    </mitre>
    <group>custom_persistence,scheduled_task,mitre_persistence,</group>
  </rule>

ubuntu@ubuntu-server:~/wazuh-docker/single-node$ docker exec single-node-wazuh.manager-1 cp /var/osse
c/etc/rules/local_rules.xml /var/ossec/etc/rules/local_rules.xml.pre100033
ubuntu@ubuntu-server:~/wazuh-docker/single-node$ docker exec single-node-wazuh.manager-1 ls -l /var/o
ssec/etc/rules/local_rules.xml.pre100033
-rw-r----- 1 root root 2212 Sep 17 15:53 /var/ossec/etc/rules/local_rules.xml.pre100033
ubuntu@ubuntu-server:~/wazuh-docker/single-node$
```

Figure 65. Source: 169.png

Selected evidence screenshot previously retained for the project. Refer to the visible event/rule/endpoint identifiers and cross-check against raw logs; avoid claiming causation from proximity alone.

```
GNU nano 6.2 /home/ubuntu/wazuh-docker/single-node/local_rules.xml
<!-- Local rules -->

<!-- Modify it at your will. -->
<!-- Copyright (C) 2015, Wazuh Inc. -->

<!-- Example -->
<group name="local,syslog,sshd,">

  <!--
  Dec 10 01:02:02 host sshd[1234]: Failed none for root from 1.1.1.1 port 1066 ssh2
  -->
  <rule id="100001" level="5">
    <if_sid>5716</if_sid>
    <srcip>1.1.1.1</srcip>
    <description>sshd: authentication failed from IP 1.1.1.1.</description>
    <group>authentication_failed,pci_dss_10.2.4,pci_dss_10.2.5,</group>
  </rule>
</group>

<group name="local,windows,recon,">
  <rule id="100030" level="9" frequency="3" timeframe="120">
    <if_matched_sid>60106</if_matched_sid>
    <field name="win.eventdata.commandLine" type="pcre2">(?!)(whoami|net user|net group|systeminfo|
    <same_source_ip />
    <description>Multiple reconnaissance commands executed in short window</description>
    <mitre><id>T1082</id><id>T1087</id></mitre>
    <group>recon,mitre_discovery,</group>
  </rule>

  <rule id="100031" level="12">
    <if_sid>91651</if_sid>
    <field name="win.eventdata.scriptBlockText" type="pcre2">(?!)(-enc|-EncodedCommand|FromBase64St
    <description>Base64-encoded PowerShell command execution detected</description>
  </rule>
</group>

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo
^X Exit      ^R Read File  ^_ Replace    ^U Paste      ^J Justify    ^_ Go To Line M-E Redo
```

Figure 66. Source: 170.png

Selected evidence screenshot previously retained for the project. Refer to the visible event/rule/endpoint identifiers and cross-check against raw logs; avoid claiming causation from proximity alone.

```
ubuntu@ubuntuuser:~/wazuh-docker/single-node$ docker exec -it single-node-wazuh.manager-1 /var/ossec/bin/wazuh-logtest
Starting wazuh-logtest v4.14.5
Type one log per line

test

**Phase 1: Completed pre-decoding.
    full event: 'test'

**Phase 2: Completed decoding.
    No decoder matched.

^Cubuntu@ubuntuuser:~/wazuh-docker/single-node$ docker restart single-node-wazuh.manager-1
single-node-wazuh.manager-1
ubuntu@ubuntuuser:~/wazuh-docker/single-node$ sleep 20
ubuntu@ubuntuuser:~/wazuh-docker/single-node$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS
PORTS
NAMES
9336661f9226   wazuh/wazuh-dashboard:4.14.5       "/entrypoint.sh"        2 weeks ago   Up 45 minutes
443/tcp, 0.0.0.0:443->5601/tcp, [::]:443->5601/tcp
single-node-wazuh.dashboard-1
b0c6d340abe1   wazuh/wazuh-manager:4.14.5         "/init"                 2 weeks ago   Up 32 seconds
0.0.0.0:1514-1515->1514-1515/tcp, [::]:1514-1515->1514-1515/tcp, 0.0.0.0:514->514/udp, [::]:514->514/udp, 0.0.0.0:55000->55000/tcp, [::]:55000->55000/tcp, 1516/tcp
single-node-wazuh.manager-1
4ec48a29610b   wazuh/wazuh-indexer:4.14.5         "/entrypoint.sh open..." 2 weeks ago   Up 45 minutes
0.0.0.0:9200->9200/tcp, [::]:9200->9200/tcp
single-node-wazuh.indexer-1
ubuntu@ubuntuuser:~/wazuh-docker/single-node$ docker exec -it single-node-wazuh.manager-1 grep -E 'rule id="10003' /var/ossec/etc/rules/local_rules.xml
<rule id="100030" level="9" frequency="3" timeframe="120">
  <rule id="100031" level="12">
  <rule id="100032" level="12">
ubuntu@ubuntuuser:~/wazuh-docker/single-node$ _
```

Figure 67. Source: 171.png

Selected evidence screenshot previously retained for the project. Refer to the visible event/rule/endpoint identifiers and cross-check against raw logs; avoid claiming causation from proximity alone.

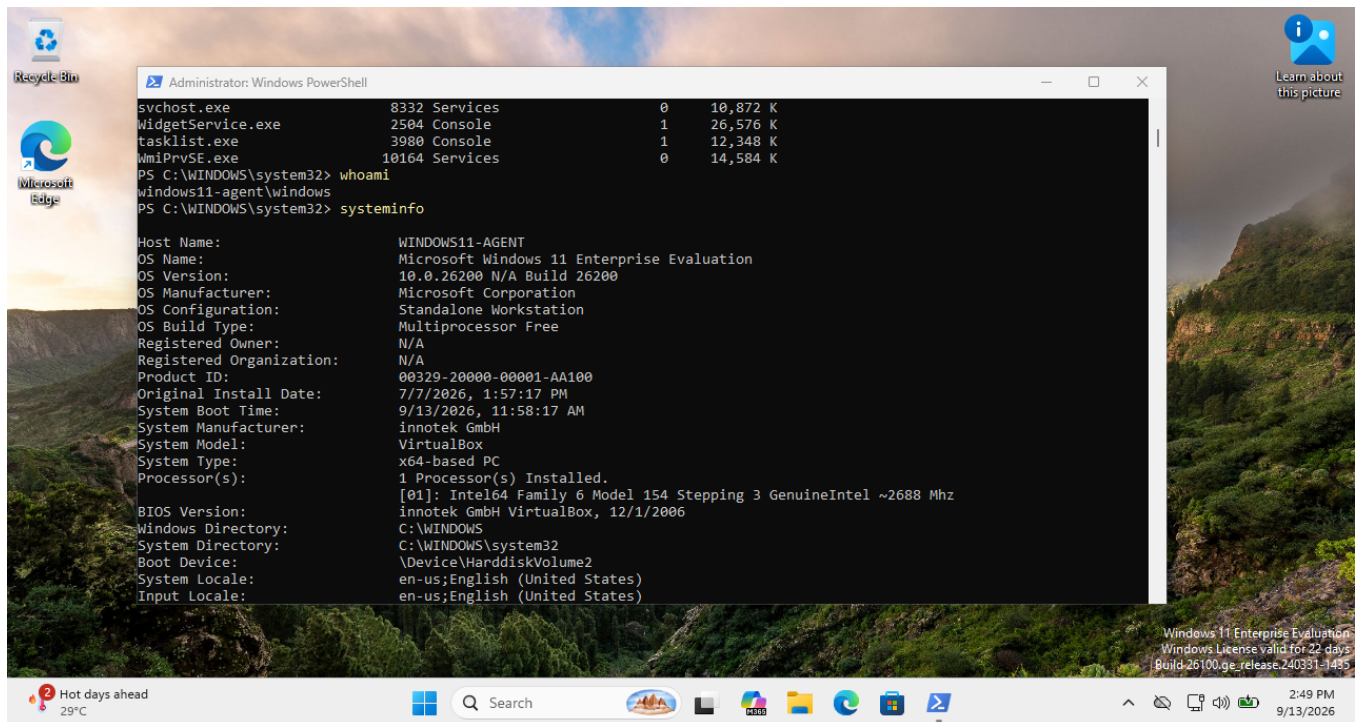
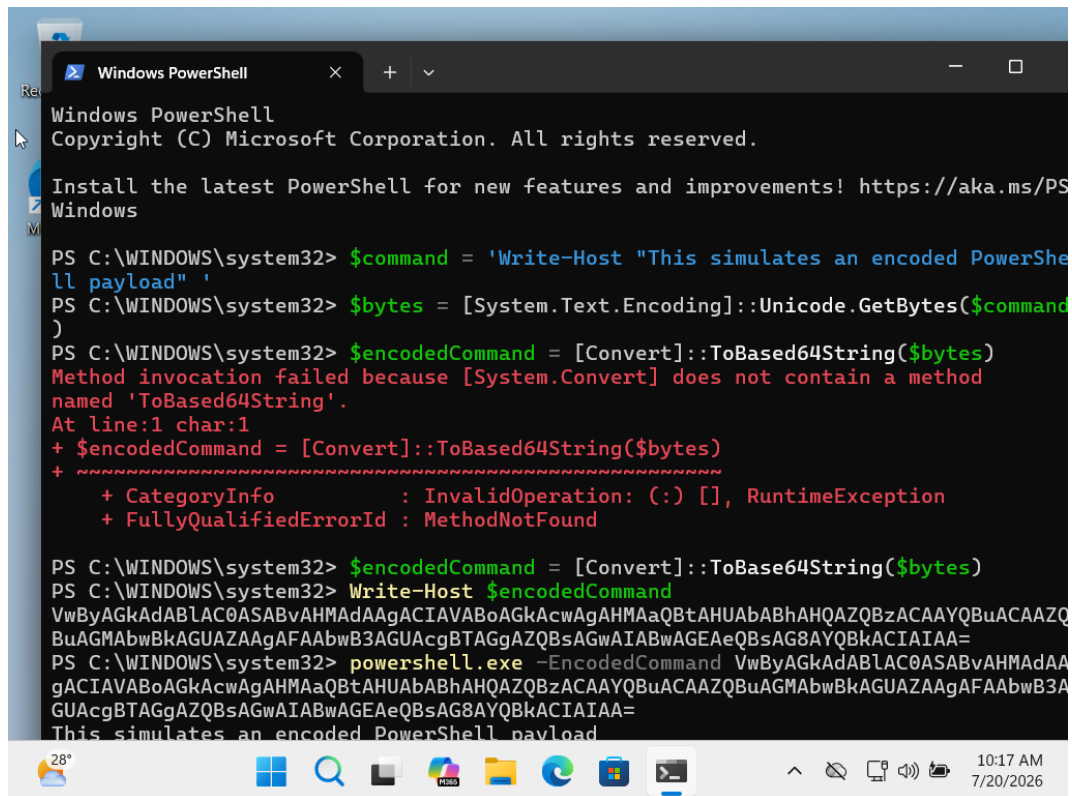


Figure 68. Source: powershell command.png

PowerShell test-command screenshot, including a failed attempt. It documents test execution and troubleshooting; do not label the failed attempt a successful detection.



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\WINDOWS\system32> $command = 'Write-Host "This simulates an encoded PowerShell payload" '
PS C:\WINDOWS\system32> $bytes = [System.Text.Encoding]::Unicode.GetBytes($command)
PS C:\WINDOWS\system32> $encodedCommand = [Convert]::ToBase64String($bytes)
Method invocation failed because [System.Convert] does not contain a method named 'ToBase64String'.
At line:1 char:1
+ $encodedCommand = [Convert]::ToBase64String($bytes)
+ ~~~~~
+ CategoryInfo          : InvalidOperation: (:) [], RuntimeException
+ FullyQualifiedErrorId : MethodNotFound

PS C:\WINDOWS\system32> $encodedCommand = [Convert]::ToBase64String($bytes)
PS C:\WINDOWS\system32> Write-Host $encodedCommand
VwByAGkAdABlAC0ASABvAHMAAAgACIAVABoAGkAcwAgAHMAAQbtAHUAbABhAHQAZQBzACAAYQBuACAAZQBuAGMAbwBkAGUAZAAGAFABwB3AGUAcgBTAGgAZQBzAGwAIABwAGEAeQBsAG8AYQBsACIAIAA=
PS C:\WINDOWS\system32> powershell.exe -EncodedCommand VwByAGkAdABlAC0ASABvAHMAAAgACIAVABoAGkAcwAgAHMAAQbtAHUAbABhAHQAZQBzACAAYQBuACAAZQBuAGMAbwBkAGUAZAAGAFABwB3AGUAcgBTAGgAZQBzAGwAIABwAGEAeQBsAG8AYQBsACIAIAA=
This simulates an encoded PowerShell payload
```

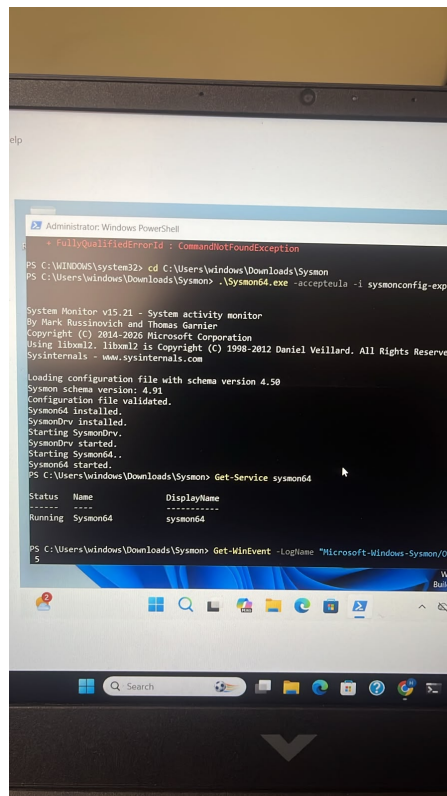
Figure 69. Source: ssh login failed.png

SSH authentication-failure screenshot from the controlled lab test. Pair it with Rule 5712 and Active Response logs to substantiate correlation and timed blocking.

```
status='connected'
agent@ubuntu-agent-01:~$ sudo tail -20 /var/log/auth.log
[sudo] password for agent:
Sep 12 08:04:39 ubuntu-agent-01 sshd[2720]: Failed password for invalid user wronguser from 192.168.1.9 port 50416 ssh2
Sep 12 08:04:40 ubuntu-agent-01 sshd[2720]: Connection closed by invalid user wronguser 192.168.1.9 port 50416 [preauth]
Sep 12 08:04:40 ubuntu-agent-01 sshd[2720]: PAM 2 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.1.9
Sep 12 08:04:49 ubuntu-agent-01 sshd[3106]: Invalid user wronguser from 192.168.1.9 port 36364
Sep 12 08:04:51 ubuntu-agent-01 sshd[3106]: pam_unix(sshd:auth): check pass; user unknown
Sep 12 08:04:51 ubuntu-agent-01 sshd[3106]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.1.9
Sep 12 08:04:53 ubuntu-agent-01 sshd[3106]: Failed password for invalid user wronguser from 192.168.1.9 port 36364 ssh2
Sep 12 08:04:56 ubuntu-agent-01 sshd[3106]: pam_unix(sshd:auth): check pass; user unknown
Sep 12 08:04:57 ubuntu-agent-01 sshd[3106]: Failed password for invalid user wronguser from 192.168.1.9 port 36364 ssh2
Sep 12 08:05:01 ubuntu-agent-01 sshd[3106]: pam_unix(sshd:auth): check pass; user unknown
Sep 12 08:05:03 ubuntu-agent-01 sshd[3106]: Failed password for invalid user wronguser from 192.168.1.9 port 36364 ssh2
Sep 12 08:05:05 ubuntu-agent-01 sshd[3106]: Connection closed by invalid user wronguser 192.168.1.9 port 36364 [preauth]
Sep 12 08:05:05 ubuntu-agent-01 sshd[3106]: PAM 2 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.1.9
Sep 12 08:17:01 ubuntu-agent-01 CRON[3165]: pam_unix(cron:session): session opened for user root(uid=0) by (uid=0)
Sep 12 08:17:01 ubuntu-agent-01 CRON[3165]: pam_unix(cron:session): session closed for user root
Sep 12 08:20:47 ubuntu-agent-01 sshd[3229]: Invalid user wronguser from 192.168.1.9 port 53638
Sep 12 08:20:50 ubuntu-agent-01 sshd[3229]: pam_unix(sshd:auth): check pass; user unknown
Sep 12 08:20:50 ubuntu-agent-01 sshd[3229]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.1.9
Sep 12 08:20:52 ubuntu-agent-01 sshd[3229]: Failed password for invalid user wronguser from 192.168.1.9 port 53638 ssh2
Sep 12 08:22:03 ubuntu-agent-01 sudo: agent : TTY=ttty1 ; PWD=/home/agent ; USER=root ; COMMAND=/usr/bin/tail -20 /var/log/auth.log
agent@ubuntu-agent-01:~$ _
```

Figure 70. Source: WhatsApp Image 2026-07-14 at 15.22.36.jpeg

Sysmon installation/service-status setup evidence from the Windows endpoint. It documents telemetry preparation rather than an alert outcome.



The image is a photograph of a laptop screen displaying a Windows PowerShell terminal window. The window title is "Administrator: Windows PowerShell". The terminal shows the execution of the Sysmon installation command and subsequent status checks. The output includes version information, configuration file validation, and the successful installation and starting of the Sysmon service. A table shows the service status as "Running".

```
Administrator: Windows PowerShell

PS C:\WINDOWS\system32> cd C:\Users\Windows\Downloads\Sysmon
PS C:\Users\Windows\Downloads\Sysmon> .\Sysmon64.exe -accepteula -i sysmonconfig-export

System Monitor v15.21 - System activity monitor
By Mark Russinovich and Thomas Garnier
Copyright (C) 2014-2026 Microsoft Corporation
Using libxml2. libxml2 is Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.
Sysinternals - www.sysinternals.com

Loading configuration file with schema version 4.50
Sysmon schema version: 4.51
Configuration file validated.
Sysmon64 installed.
SysmonDrv installed.
Starting SysmonDrv.
SysmonDrv started.
Starting Sysmon64.
Sysmon64 started.
PS C:\Users\Windows\Downloads\Sysmon> Get-Service sysmon64

Status Name DisplayName
-----
Running Sysmon64 sysmon64

PS C:\Users\Windows\Downloads\Sysmon> Get-WinEvent -LogName "Microsoft-Windows-Sysmon/Operational"
5
```